

Project Data Page:





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ASME TS Calc:	147



STEAM SUPERHEATER
A150NRJ10EP001

DESIGN CALCULATION

In Accordance with ASME Section VIII Division 2

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\STEAM SUPERHEATER.pvdb

Date of Analysis : Sep 16,2026 4:54pm

PV Elite 28 U2, April 2026

Vessel Design Summary:

ASME Code, Section VIII Division 2, 2025

Diameter Spec : 2100.000 x 2236.000 mm. OD
Vessel Design Length, Tangent to Tangent 11100.00 mm.

Distance of Bottom Tangent above Grade 7500.00 mm.
Specified Datum Line Distance -50.00 mm.

Shell Side Design Temperature 286 °C
Channel Side Design Temperature 286 °C

Shell Side Design Pressure 51.000 bars
Channel Side Design Pressure 191.000 bars

Shop Shell Side Test Pressure 22.163 bars
Shop Channel Side Test Pressure 21.930 bars

Wind Design Code ASCE/SEI 7-22
Earthquake Design Code ASCE/SEI 7-22

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-387 11	2	...	K11789	
Head	SA-387 11	2	...	K11789	
Nozzle	SA-336 F21	3	...	K31545	
Nozzle	SA-336 F11	3	...	K11572	
Nozzle Flg	SA-336 F21	3	...	K31545	
Nozzle Flg	SA-105	...	...	K03504	
Sup. Lugs	SA-387 11	2	...	K11789	
Tubes	SB-163	...	...	N08825	
Tubesheet	SA-336 F21	3	...	K31545	
Lug Bolting	SA-193 B7	...	<= 64	G41400	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
+Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (bars & mm.):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Dish bottom	51.000	1.03	52.000	1.5000	N/A	No
shell-2	51.000	1.03	52.000	1.5000	N/A	No
shell-3	51.000	1.03	52.000	1.5000	N/A	No
shell 4	51.000	1.03	52.000	1.5000	N/A	No
Channel-1	191.000	1.03	212.000	1.5000	N/A	No
Channel-2	191.000	1.03	212.000	1.5000	N/A	No
Top dish	191.000	1.03	203.000	1.5000	N/A	No

Element Types and Properties:

Element Type	"To" Elev mm.	Element Length mm.	Nominal Thickness mm.	Finished Thickness mm.	Reqd Thk Internal mm.	Reqd Thk External mm.	Long Eff	Circ Eff
Sphere	50.0	0.0	22.0	18.0	17.6	...	1.00	1.00
Cylinder	1550.0	1500.0	34.0	34.0	33.4	...	1.00	1.00
Cylinder	3550.0	2000.0	34.0	34.0	33.4	...	1.00	1.00
Cylinder	5050.0	1500.0	34.0	34.0	33.4	...	1.00	1.00



Cylinder	8150.0	3000.0	136.0	136.0	122.1	...	1.00	1.00
Cylinder	11150.0	3000.0	136.0	136.0	122.1	...	1.00	1.00
Sphere	11150.0	0.0	74.0	68.0	64.1	...	1.00	1.00

External Pressure Calculations:

From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
10	20	18	4.3497	1.03419	22.804
20	30	34	11.0554	1.03419	18.7361
30	40	34	11.0554	1.03419	18.7361
40	Tsht	34	11.0554	1.03419	18.7361
Tsht	50	34	No Calc	1.03419	No Calc
50	60	136	12.1518	1.03419	141.015
60	70	136	12.1518	1.03419	141.015
70	80	68	4.53423	1.03419	131.674

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm^4	Ring Inertia Available cm^4
10	20	No Calc	No Calc	No Calc	No Calc
20	30	5343.97	25399974	No Calc	No Calc
30	40	5343.97	25399974	No Calc	No Calc
40	Tsht	5343.97	25399974	No Calc	No Calc
Tsht	50	No Calc	No Calc	No Calc	No Calc
50	60	6350.03	25399974	No Calc	No Calc
60	70	6350.03	25399974	No Calc	No Calc
70	80	No Calc	No Calc	No Calc	No Calc

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
A2-24 "	WRC-297	0.751	Shell	Passed
B2-24 "	WRC-297	0.751	Shell	Passed
A1-24"	WRC-297	0.079	Shell	Passed
B1-24"	WRC-297	0.079	Shell	Passed

Loads for Foundation/Support Design:

Total Wind Shear on Support	3973. Kg
Total Earthquake Shear on Support	14899. Kg
Wind Moment on Support	1001. Kg-m.
Earthquake Moment on Support	34108. Kg-m.
Wind Load Case Test Moment on Support	330. Kg-m.
Test Case Wind Shear on Support	1311. Kg

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	99412.1 kg.
Shop Test - Fabricated + water (full)	137342.2 kg.
Shipping - Fab. + rem. intls.+ shipping app.	101477.7 kg.
Erected - Fab. + rem. intls.+ insul. (etc)	101477.7 kg.
Empty - Fab. + intls. + details + wghts.	101477.7 kg.
Operating - Empty + operating liquid (no ca)	101480.7 kg.
Field Test - Empty weight + test medium (full)	139407.8 kg.

STEAM SUPERHEATER

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PV Elite 28 U2

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Vessel Design Summary:

PV Elite® v28



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**Nozzle Calculation Summary (bars & mm.):**

Description	MAWP	Ext	MAPNC	Tbl 4.5.2	Weld	Pass/Fail
D1-3"	52	OK	...	...	OK	Passed
A2-24 "	52	OK	...	OK	OK	Passed
B2-24 "	52	OK	...	OK	OK	Passed
A1-24"	212	OK	...	OK	OK	Passed
B1-24"	212	OK	...	OK	OK	Passed
M1-24"	203	OK	...	...	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 52 Nozzle : A2-24 " [Shellside]
 Minimum MAWP Nozzles : 203 Nozzle : M1-24" [Tubeside]

VIII-2 Nozzle to Gross Structural Discontinuity Check:

- *Error: Nozzle A2-24 ", was too close to a gross structural discontinuity.*

Please see ASME VIII-2 Section 4.5.13.1 for more information.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm.	Layout Angle deg	Inside Radius mm.
10	D1-3"	0.000	0.000	39.600
20	A2-24 "	450.005	270.000	281.300
40	B2-24 "	4250.004	90.000	281.300
50	A1-24"	7100.000	90.000	266.300
60	B1-24"	10100.004	270.000	266.300
70	M1-24"	0.000	0.000	266.300

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Diameter	Pad Thk mm	Cut Length mm	Flg Class
D1-3"	3.000 in	Actual	LWN	4.620	20.574	...	...	319.67	...
A2-24 "	24.000 in	Actual	None	24.000	25.000	...	...	399.84	...
B2-24 "	24.000 in	Actual	None	24.000	25.000	...	...	399.84	...
A1-24"	24.000 in	Actual	None	24.000	40.000	...	...	527.91	...
B1-24"	24.000 in	Actual	None	24.000	40.000	...	...	527.91	...
M1-24"	24.000 in	Actual	WNF	24.000	40.000	...	...	491.05	1500

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm.):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
D1-3"	SA-336 F21	18.000	10.000	...	...	...
A2-24 "	SA-336 F11	34.000	10.000	...	...	...
B2-24 "	SA-336 F11	34.000	10.000	...	...	...
A1-24"	SA-336 F21	136.000	10.000	...	...	...
B1-24"	SA-336 F21	136.000	10.000	...	...	...
M1-24"	SA-336 F21	68.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm.	Layout Angle deg	Proj Outside mm.	Proj Inside mm.	Installed in Component
D1-3"	...	0.0	300.00	0.00	Dish bottom
A2-24 "	500.005	270.0	300.00	0.00	shell-2
B2-24 "	4300.004	90.0	300.00	0.00	shell 4
A1-24"	7150.000	90.0	300.00	0.00	Channel-1
B1-24"	10150.005	270.0	300.00	0.00	Channel-2
M1-24"	...	0.0	350.00	0.00	Top dish

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**Bill of Materials:**

QTY	DESCRIPTION	MATERIAL
1	SPHERICAL HEAD: 22.0mm. THK X 2064.0mm. ID X 0.0mm.	SA-387 11
1	SPHERICAL HEAD: 74.0mm. THK X 2100.0mm. ID X 0.0mm.	SA-387 11
1	INSULATION: 1050mm X 109mm THK	
2	INSULATION: 1500mm X 109mm THK	
1	INSULATION: 2000mm X 109mm THK	
1	TUBESHEET: 2236.0mm X 100.0mm THK	SA-336 F21
559	TUBES: 4500.0mm X 38.1mm DIA X 2.7mm THK	SB-163
2	INSULATION: 3000mm X 109mm THK	
2	CLADDING: 3000.000mm X 3.000mm THK	...
8	GUSSET PLATES: 400.0mm X 250.0mm X 450.0mm X 50.0mm	SA-387 11
1	TOP RING: 2100.0mm X 2600.0mm X 40.0mm	SA-387 11
1	BOTTOM RING: 2100mm X 2900mm X 50mm	SA-387 11
4	LUG BOLTS: 42.0mm TEMA Metric	SA-193 B7
1	INSULATION: 1118mm X 109mm THK	
1	CLADDING: 1118.000mm X 3.000mm THK	...
1	3.0" BLIND FLANGE(S)	SA-336 F21
1	CLASS 1500 GR 1.1, 24.0" BLIND FLANGE(S)	SA-105
1	CYLINDRICAL SEGMENT 34.0mm. THK X 2464.8mm. X 1500.0mm.	SA-387 11
1	CYLINDRICAL SEGMENT 34.0mm. THK X 2482.5mm. X 1500.0mm.	SA-387 11
1	CYLINDRICAL SEGMENT 34.0mm. THK X 1436.3mm. X 1500.0mm.	SA-387 11
1	CYLINDRICAL SEGMENT 34.0mm. THK X 2464.8mm. X 2000.0mm.	SA-387 11
1	CYLINDRICAL SEGMENT 34.0mm. THK X 2482.5mm. X 2000.0mm.	SA-387 11
1	CYLINDRICAL SEGMENT 34.0mm. THK X 1436.3mm. X 2000.0mm.	SA-387 11
1	CYLINDRICAL SEGMENT 34.0mm. THK X 2695.3mm. X 1500.0mm.	SA-387 11
1	CYLINDRICAL SEGMENT 34.0mm. THK X 3688.3mm. X 1500.0mm.	SA-387 11
1	CYLINDRICAL SEGMENT 34.0mm. THK X 0.0mm. X 1500.0mm.	SA-387 11
1	CYLINDRICAL SEGMENT 136.0mm. THK X 2382.3mm. X 3000.0mm.	SA-387 11
1	CYLINDRICAL SEGMENT 136.0mm. THK X 3770.5mm. X 3000.0mm.	SA-387 11
2	CYLINDRICAL SEGMENT 136.0mm. THK X 17.1mm. X 3000.0mm.	SA-387 11
1	CYLINDRICAL SEGMENT 136.0mm. THK X 3787.7mm. X 3000.0mm.	SA-387 11
1	CYLINDRICAL SEGMENT 136.0mm. THK X 2399.4mm. X 3000.0mm.	SA-387 11
1	NAMEPLATE	...

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Bill of Material:

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Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	PWHT	Thickness ratio	Gov Thk mm.	E*
Dish bottom	[9]	A	10	-43	Yes	0.263	18.000	1.00
shell-2	[20]	A	23	6	Yes	0.699	34.000	1.00
shell-3	[20]	A	23	6	Yes	0.699	34.000	1.00
shell 4	[20]	A	23	6	Yes	0.699	34.000	1.00
D1-3"	[1]	A	10	-43	Yes	0.263	18.000	1.00
A2-24 "	[1]	A	23	-30	Yes	0.263	34.000	1.00
B2-24 "	[1]	A	23	-30	Yes	0.263	34.000	1.00
Nozl. Bolting	[22]		-48		No	1.000	64.000	1.00
Tubesheet: SS	[13]	A	23	23	Yes	1.000	34.000	1.00
Warmest MDMT:			23	23				
Channel-1	[20][17] Impact Testing reqd				Yes			
Channel-2	[20][17] Impact Testing reqd				Yes			
Top dish	[9]	A	-104	-104	Yes	0.066	68.000	1.00
A1-24"	[1]				Yes			
B1-24"	[1]				Yes			
M1-24"	[1]	A	-104	-104	Yes	0.058	40.000	1.00
Nozzle Flg	[5]	A	-18	-104	Yes			
Nozl. Bolting	[22]		-48		No	1.000	64.000	1.00
Tubesheet: CS	[14]	A	42	42	Yes	1.000	100.000	1.00
Warmest MDMT:			42	42				
Exchanger Side	Computed MDMT		Required MDMT		Pass/Fail			
Shell	23.0		-10.0		Fail			

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.1.
- [5] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.3.
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per 3.11.2.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

Note: Lug supports should be modeled on an element of their own whose length is just slightly greater than the height of the lug assembly. The vessel does not have to be constructed with this extra element, just modeled this way. This is to insure there is proper mass distribution for these types of models.

Warning:

This is a vertical vessel on support lugs and the earthquake type selected is a building code. Building codes assume all of the mass is located above the point of support. This makes their use on lug supported vessels questionable. It is strongly suggested that based on the seismic zone and applicable parameters, an appropriate lateral (Gx) and (+-) vertical (Gy) acceleration be applied using the G loading option.

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.

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**Units used in this Analysis (bars):**

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm.
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0689	bars
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm.
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm.
Thickness	inches	25.4000	mm.

PV Elite Vessel Analysis Program: Input Data

STEAM SUPERHEATER

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Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	51	bars
Channel Side Design Pressure	191	bars
Shell Side Design Temperature	286.0	°C
Channel Side Design Temperature	286.0	°C
Radiography, Shell Side	None	
Radiography, Channel Side	None	
Service Type, Shell Side	None	
Service Type, Channel Side	None	
MDMT (CET), Shell Side	-10.0	°C
MDMT (CET), Tube Side	-10.0	°C
User defined MAWP, Shell Side	0	bars
User defined MAWP, Channel Side	0	bars
User defined MAPnc, Shell Side	0	bars
User defined MAPnc, Channel Side	0	bars
User defined Test Pres., Shell Side	0	bars
User defined Test Pres., Channel Side	0	bars

Projection of Nozzle from Vessel Top	0	mm.
Projection of Nozzle from Vessel Bottom	0	mm.
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	N	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	Hydrostatic
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

**Field Pressure Test:**

Type of Pressure Test	Hydrostatic
Pressure Test Position	Vertical
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+1.1WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+1.1WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+1.1WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+1.1WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-22	
Wind Load Reduction Scale Factor	1.100	
Basic Wind Speed	[V] 180	Km/hr
Surface Roughness Category	D: Flat, unobstructed	
Type of Surface	Moderately Smooth	
Base Elevation	7500	mm.
Percent Wind for Hydrottest	33.0	
Using User defined Wind Press. Vs Elev.	N	
Height of Hill or Escarpment	H or Hh	0 mm.
Distance Upwind of Crest	Lh	0 mm.
Distance from Crest to the Vessel	x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	
Site Elevation above Sea Level	0	mm.

Seismic Design Code	ASCE/SEI 7-22	
Importance Factor	1.500	
Force Modification Factor R	3.000	
Site Class	D	
Component Elevation Ratio	z/h	0.500
Amplification Factor	Ap	1.000
Force Factor		0.000
Consider Vertical Acceleration		Yes
Minimum Acceleration Multiplier		0.000
User Value of Sds (used if > 0)		0.000
User Value of Sd1 (used if > 0)		0.000
Component resonance duct. fact.	[CAR]	1.000
Component strength factor	[Rpo]	1.500
Overstrength factor	[Omega0]	2.000
Response Acceleration Factor	[SMS]	0.326
Response Acceleration Factor	[SM1]	0.139
Building Importance Factor	[Ie]	1.000
Site Latitude		0.000 deg
Site Longitude		0.000 deg

M.A.W.P + Static Head to Nozzle	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N



Perform Blast Load Analysis No
Material Database Year Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk) Yes
Using Metric Material Databases, ASME II D Yes
Using ASME VIII-1 Appendix 46 No
Calculate B31.3 type stress for Nozzles with Loads Yes
Reduce the MDMT due to lower membrane stress Yes
Consider Longitudinal Stress in MDMT Calculations Yes

Complete Listing of Vessel Elements and Details:

Element#	1/7	
Element From Node	10	
Element To Node	20	
Element Type	Spherical	
Description	Dish bottom	
Distance "FROM" to "TO"	0	mm.
Element Outside Diameter	2100	mm.
Element Thickness	18	mm.
Internal Corrosion Allowance	1.5	mm.
Nominal Thickness	22	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	51	bars
Design Temperature Internal Pressure	286	°C
Design External Pressure	1.0342	bars
Design Temperature External Pressure	286	°C
Effective Diameter Multiplier	1.23	
Material Name	SA-387 11	
Allowable Stress, Ambient	207	N./mm ²
Allowable Stress, Operating	165.12	N./mm ²
Allowable Stress, Hydrotest	294.49	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	0	mm.
Yield Stress, Operating	247.68	N./mm ²
MDMT Curve	A	
External Pressure Chart Name	CS-3	
UNS Number	K11789	
Class / Thickness / Grade	2 ::	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	10	
Detail Type	Insulation	
Detail ID	Ins: 10	
Dist. from "FROM" Node / Offset dist	-1050	mm.
Height/Length of Insulation	1050	mm.
Thickness of Insulation	110	mm.
Density	0.0002	kg./cm ³
Element From Node	10	
Detail Type	Nozzle	
Detail ID	D1-3"	
Dist. from "FROM" Node / Offset dist	0	mm.
Nozzle Diameter	3	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-336 F21	



Element#	2/7
Element From Node	20
Element To Node	30
Element Type	Cylinder
Description	shell-2
Distance "FROM" to "TO"	1500 mm.
Element Outside Diameter	2100 mm.
Element Thickness	34 mm.
Internal Corrosion Allowance	1.5 mm.
Nominal Thickness	34 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	51 bars
Design Temperature Internal Pressure	286 °C
Design External Pressure	1.0342 bars
Design Temperature External Pressure	286 °C
Effective Diameter Multiplier	1.23
Material Name	SA-387 11
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	20
Detail Type	Insulation
Detail ID	Ins: 20
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Insulation	1500 mm.
Thickness of Insulation	110 mm.
Density	0.0002 kg./cm ³
Element From Node	20
Detail Type	Nozzle
Detail ID	A2-24 "
Dist. from "FROM" Node / Offset dist	450 mm.
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	None
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	None
Nozzle Matl	SA-336 F11
Element From Node	20
Detail Type	Weight
Detail ID	Btm dish SF
Dist. from "FROM" Node / Offset dist	0 mm.
Miscellaneous Weight	970 Kg
Offset from Element Centerline	0 mm.

Element#	3/7
Element From Node	30
Element To Node	40
Element Type	Cylinder
Description	shell-3
Distance "FROM" to "TO"	2000 mm.
Element Outside Diameter	2100 mm.
Element Thickness	34 mm.
Internal Corrosion Allowance	1.5 mm.
Nominal Thickness	34 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	51 bars
Design Temperature Internal Pressure	286 °C
Design External Pressure	1.0342 bars



Design Temperature External Pressure	286 °C
Effective Diameter Multiplier	1.23
Material Name	SA-387 11
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	30
Detail Type	Insulation
Detail ID	Ins: 30
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Insulation	2000 mm.
Thickness of Insulation	110 mm.
Density	0.0002 kg./cm ³

Element#	4/7
Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	sheel 4
Distance "FROM" to "TO"	1500 mm.
Element Outside Diameter	2100 mm.
Element Thickness	34 mm.
Internal Corrosion Allowance	1.5 mm.
Nominal Thickness	34 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	51 bars
Design Temperature Internal Pressure	286 °C
Design External Pressure	1.0342 bars
Design Temperature External Pressure	286 °C
Effective Diameter Multiplier	1.23
Material Name	SA-387 11
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	40
Detail Type	Insulation
Detail ID	Ins: 40
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Insulation	1500 mm.
Thickness of Insulation	110 mm.
Density	0.0002 kg./cm ³

Element From Node	40
Detail Type	Nozzle
Detail ID	B2-24 "
Dist. from "FROM" Node / Offset dist	750 mm.
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	None
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	None
Nozzle Matl	SA-336 F11

Element#	5/7
Element From Node	50
Element To Node	60
Element Type	Cylinder
Description	Channel-1
Distance "FROM" to "TO"	3000 mm.
Element Outside Diameter	2236 mm.



Element Thickness	136	mm.
Internal Corrosion Allowance	1.5	mm.
Nominal Thickness	136	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	191	bars
Design Temperature Internal Pressure	286	°C
Design External Pressure	1.0342	bars
Design Temperature External Pressure	286	°C
Effective Diameter Multiplier	1.23	
Material Name	SA-387 11	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	50	
Detail Type	Insulation	
Detail ID	Ins: 50	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Insulation	3000	mm.
Thickness of Insulation	110	mm.
Density	0.0002	kg./cm ³

Element From Node	50	
Detail Type	Lining	
Detail ID	cld-2	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Lining	3000	mm.
Thickness of Lining	3	mm.
Density	0.00847	kg./cm ³

Element From Node	50	
Detail Type	Nozzle	
Detail ID	A1-24"	
Dist. from "FROM" Node / Offset dist	2000	mm.
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	None	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-336 F21	

Element From Node	50	
Detail Type	Lug	
Detail ID	Support Lugs	
Dist. from "FROM" Node / Offset dist	1000	mm.
Number of Lugs	4	
Dist. from OD to Lug Cntrline(dlug)	300	mm.
Height of Gusset Plates (hgp)	450	mm.
Force Bearing Width (wfb)	250	mm.
Weight of Lug	559.44	Kgf
Lug Start Angle (degrees)	0.0	

Element From Node	50	
Detail Type	Weight	
Detail ID	Internal dish+	
Dist. from "FROM" Node / Offset dist	-0.005233	mm.
Miscellaneous Weight	6000	Kgf
Offset from Element Centerline	0	mm.

Element From Node	50	
Detail Type	Weight	
Detail ID	Clip weight	
Dist. from "FROM" Node / Offset dist	-0.005233	mm.
Miscellaneous Weight	920	Kgf
Offset from Element Centerline	0	mm.



Element#	6/7
Element From Node	60
Element To Node	70
Element Type	Cylinder
Description	Channel-2
Distance "FROM" to "TO"	3000 mm.
Element Outside Diameter	2236 mm.
Element Thickness	136 mm.
Internal Corrosion Allowance	1.5 mm.
Nominal Thickness	136 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	191 bars
Design Temperature Internal Pressure	286 °C
Design External Pressure	1.0342 bars
Design Temperature External Pressure	286 °C
Effective Diameter Multiplier	1.23
Material Name	SA-387 11
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	60
Detail Type	Insulation
Detail ID	Ins: 60
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Insulation	3000 mm.
Thickness of Insulation	110 mm.
Density	0.0002 kg./cm ³
Element From Node	60
Detail Type	Lining
Detail ID	cld-1
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Lining	3000 mm.
Thickness of Lining	3 mm.
Density	0.00847 kg./cm ³
Element From Node	60
Detail Type	Nozzle
Detail ID	B1-24"
Dist. from "FROM" Node / Offset dist	2000 mm.
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	None
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	None
Nozzle Matl	SA-336 F21
Element From Node	60
Detail Type	Weight
Detail ID	Top dish SF
Dist. from "FROM" Node / Offset dist	-0.004651 mm.
Miscellaneous Weight	5700 Kg
Offset from Element Centerline	0 mm.
Element From Node	60
Detail Type	Weight
Detail ID	Lug weight
Dist. from "FROM" Node / Offset dist	-0.004651 mm.
Miscellaneous Weight	2700 Kg
Offset from Element Centerline	0 mm.



Element#	7/7
Element From Node	70
Element To Node	80
Element Type	Spherical
Description	Top dish
Distance "FROM" to "TO"	0 mm.
Element Outside Diameter	2236 mm.
Element Thickness	68 mm.
Internal Corrosion Allowance	1.5 mm.
Nominal Thickness	74 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	191 bars
Design Temperature Internal Pressure	286 °C
Design External Pressure	1.0342 bars
Design Temperature External Pressure	286 °C
Effective Diameter Multiplier	1.23
Material Name	SA-387 11
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	70
Detail Type	Insulation
Detail ID	Ins: 70
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Insulation	1118 mm.
Thickness of Insulation	110 mm.
Density	0.0002 kg./cm ³
Element From Node	70
Detail Type	Lining
Detail ID	clad
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Lining	1118 mm.
Thickness of Lining	3 mm.
Density	0.00847 kg./cm ³
Element From Node	70
Detail Type	Nozzle
Detail ID	M1-24"
Dist. from "FROM" Node / Offset dist	0 mm.
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	1500
Layout Angle	0.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	3706.5 Kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-336 F21

**XY Coordinate Calculations:**

From	To	X (Horiz.) mm.	Y (Vert.) mm.	DX (Horiz.) mm.	DY (Vert.) mm.
Dish bottom		...	...	...	...
shell-2		...	1500	...	1500
shell-3		...	3500	...	2000
shell 4		...	5000	...	1500
Channel-1		...	8100	...	3000
Channel-2		...	11100	...	3000
Top dish		...	11100	...	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd bars	Nominal Thickness mm.	Total Corr Allowance mm.	Element Diameter mm.	Allowable Stress(SE) N./mm ²	Element#
Dish bottom		51	22	1.5	2064	165.12	1
shell-2		51	34	1.5	2032	165.12	2
shell-3		51	34	1.5	2032	165.12	3
shell 4		51	34	1.5	2032	165.12	4
Channel-1		191	136	1.5	1964	165.12	5
Channel-2		191	136	1.5	1964	165.12	6
Top dish		191	74	1.5	2100	165.12	7

Element Required Thickness and MAWP :

From	To	Design Pressure bars	M.A.W.P. Corroded bars	M.A.P. New & Cold bars	Minimum Thickness mm.	Required Thickness mm.	Element#
Dish bottom		51	52.3034	71.5814	18	17.5856	1
shell-2		51	51.9128	68.1326	34	33.4196	2
shell-3		51	51.9128	68.1326	34	33.4196	3
shell 4		51	51.9128	68.1326	34	33.4196	4
Channel-1		191	211.636	268.47	136	122.114	5
Channel-2		191	211.636	268.47	136	122.114	6
Top dish		191	202.502	259.77	68	64.1125	7

Summary of Heat Exchanger Maximum Allowable Working Pressures :

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

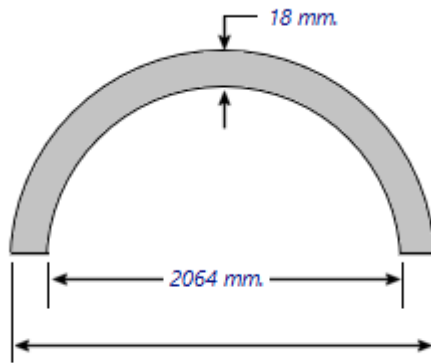
Shell Side MAWP = 17 bars, < design pressure: 51.000 bars
 Shell Side MAPnc = 32 bars
 Channel Side MAWP = 17 bars, < design pressure: 191.000 bars
 Channel Side MAPnc = 31 bars

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Dish bottom

Spherical Head Design Information:					
Design Pressure	P	51.000 bars	Int. Design Temperature		286 °C
Material		SA-387 11	Ext. Design Temperature		286 °C
Length		0.000 mm.	External Pressure Chart		CS-3
			UNS Number		K11789
Int. Corr. All.	c	1.5000 mm.	Allowable Stress (ope)	S	165.1 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	207.0 N./mm ²
Inside Diameter	Di	2064.000 mm.	Actual Stress		158.5 N./mm ²
Spec. Min. Thk t,e, or sa		18.0000 mm.	Spec. Nominal Thk.		22.000 mm.
Surface Area		69800.9 cm ²	SG of Contents		0.000
Element Volume		2302372.5 cm ³	Weight of Contents		0.0 Kgf
Empty Weight		1165.5 Kgf	Operating Weight		1165.5 Kgf
			Comp. All. Ope. Stress		75.3 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 25.800 mm.
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 12.900 mm.

Spherical Shell Calculation - Section 4.3.5.1

Computed Minimum Required Thickness [t]:
 $= D/2 \cdot (\exp(0.5P/(S \cdot E)) - 1) + ci + co$
 $= 2067.0/2 \cdot (\exp(0.5 \cdot 51.0/(165.119 \cdot 1.0)) - 1) + 1.5 + 0.0$
 $= 16.0856 + 1.5 + 0.0 = 17.5856 \text{ mm.}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= 2 \cdot S \cdot E \cdot \ln((2 \cdot t + D)/D) - P_{liq}$
 $= 2 \cdot 165.1 \cdot 1.0 \cdot \ln((2 \cdot 16.5 + 2067.0)/2067.0) - 0.0$
 $= 52.303 - 0.0 = 52.3034 \text{ bars}$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= 2 \cdot S \cdot E \cdot \ln((2 \cdot t + D)/D)$
 $= 2 \cdot 206.996 \cdot 1.0 \cdot \ln((2 \cdot 18.0 + 2064.0)/2064.0)$
 $= 71.5814 \text{ bars}$



Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D^2 / (E (D_o^2 - D^2))$$

$$= 51.0 \cdot 2067.0^2 / (1.0 (2100.0^2 - 2067.0^2))$$

$$= 158.467 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= 0.5 \cdot P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 0.5 \cdot 51.0 / (1.0 \cdot \ln((2 \cdot 16.5 + 2067.0) / 2067.0))$$

$$= 161.004 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

For Heads formed from Segments $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.582 %
 For one Piece doubly Curved Heads $100 \cdot \ln(D_b / (D_f - 2t))$ 36.767 %

Where D_b is the blank diameter for one piece doubly curved surfaces [D_b]:

$$= \text{square root}(4 \cdot \text{Head Surface Area} / 3.14159)$$

$$= \text{square root}(4 \cdot 69800.9 / 3.14159)$$

$$= 2981.162 \text{ mm.}$$

MDMT Calculations in the Spherical Portion:

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : -43 °C
 User entered Minimum Design Temperature : -10 °C
 The Curve for this Material is : A
 From Figure 3.8M- with PWHT, the MDMT is : 10 °C
 Temperature reduction per Graph 3.13 : 53 °C
 The computed stress ratio is : 0.263

The computed thickness ratio [sr]:

$$= (t \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (4.337 \cdot 1.0) / (18.0 - 1.5 - 0.0)$$

$$= 0.263$$

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	17.334 bars
Governing Thk.	18.000 mm.	Min. Des. Mtl. Temp.	-10 °C
Stress/Thk. Ratio	0.263	Basic MDMT	10 °C
		Temperature Difference	53 °C
		Computed Min. Temp.	-43 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 25.800 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 12.900 mm.

EXTERNAL PRESSURE RESULTS for: Dish bottom

Spherical Head From 10 to 20 Ext. Chart: CS-2 at 286 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 0.075 \cdot E_y (t / R_o)$$

$$= 0.075 \cdot 0.18730 \text{E}+09 (16.5 / 1050.0)$$

$$= 220.759 \text{ N./mm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= F_{he} / S_y$$

$$= 220.8 / 247.7$$



$$= 0.891$$

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 247.7/517.1$$

$$= 0.479$$

Ratio [AE]:

$$= F_{he} / E$$

$$= 220.8/18730E+09$$

$$= 0.001179$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.479) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.160$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$

$$= 247.7 (1 + 0.002) / \ln(1 + 0.002)^{0.160318}$$

$$= 672.232 \text{ N./mm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.479)$$

$$= 0.313$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (517.1 \cdot \exp(0.313)) / (0.313^{0.313})$$

$$= 1016.803 \text{ N./mm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4791.5 - 0.5 \cdot 0.4792.5 - 0.4793.5$$

$$= 0.342$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	258	-1.764	0.311E-05	0.262E-05	0.775E-05	-0.580E-05	0.122E+08	0.201E-01
2	129	-4.571	0.825E-07	0.825E-07	0.169E-05	-0.169E-05	0.143E+09	0.274E-03
3	193	-3.167	0.690E-06	0.684E-06	0.412E-05	-0.406E-05	0.510E+08	0.262E-02
4	161	-3.869	0.265E-06	0.265E-06	0.276E-05	-0.275E-05	0.933E+08	0.554E-03
5	145	-4.220	0.153E-06	0.153E-06	0.219E-05	-0.218E-05	0.119E+09	0.465E-04
6	137	-4.396	0.113E-06	0.113E-06	0.193E-05	-0.193E-05	0.131E+09	0.132E-03
7	141	-4.308	0.132E-06	0.132E-06	0.206E-05	-0.205E-05	0.125E+09	0.477E-04
8	143	-4.264	0.142E-06	0.142E-06	0.212E-05	-0.212E-05	0.122E+09	0.195E-05
9	144	-4.242	0.147E-06	0.147E-06	0.215E-05	-0.215E-05	0.120E+09	0.219E-04
10	143	-4.253	0.145E-06	0.145E-06	0.214E-05	-0.213E-05	0.121E+09	0.990E-05
11	143	-4.259	0.143E-06	0.143E-06	0.213E-05	-0.213E-05	0.121E+09	0.395E-05
12	143	-4.261	0.143E-06	0.143E-06	0.212E-05	-0.212E-05	0.122E+09	0.993E-06
13	143	-4.263	0.142E-06	0.142E-06	0.212E-05	-0.212E-05	0.122E+09	0.482E-06
14	143	-4.262	0.143E-06	0.142E-06	0.212E-05	-0.212E-05	0.122E+09	0.255E-06
15	143	-4.262	0.142E-06	0.142E-06	0.212E-05	-0.212E-05	0.122E+09	0.114E-06
16	143	-4.262	0.143E-06	0.142E-06	0.212E-05	-0.212E-05	0.122E+09	0.706E-07
17	143	-4.262	0.143E-06	0.142E-06	0.212E-05	-0.212E-05	0.122E+09	0.215E-07

Factor of Safety [FS]:

$$\text{Since } F_{ic} > 0.55 F_y \text{ and } F_{ic} < F_y$$

$$= 2.407 - 0.741 \cdot F_{ic} / F_y$$

$$= 2.407 - 0.741 \cdot 143.5/247.68$$

$$= 1.978$$

Allowable Circumferential Compressive Stress [Fha]:

$$= F_{ic}/FS$$

STEAM SUPERHEATER

A150NRJ10EP001

PV Elite 28 U2

Licensee: SL User

FileName : STEAM SUPERHEATER

Element: Dish bottom

PV Elite

v28



HEXAGON

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Step: 4 4:54pm Sep 16, 2026

= 143.504/1.978
= 72.562 N./mm²

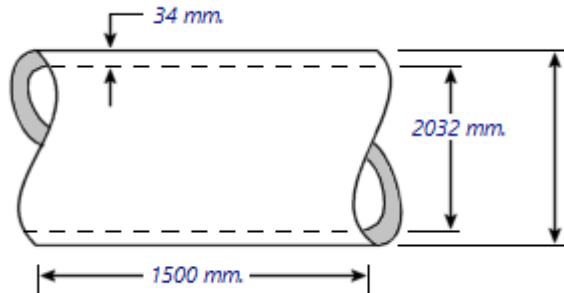
Allowable External Pressure at the given Thickness [Pa]:

= $2 \cdot F_{ha}[t/R_o]$
= $2 \cdot 72.562[16.5/1050.0]$
= 22.804 bars

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: shell-2

Cylindrical Shell Design Information:					
Design Pressure	P	51.000 bars	Int. Design Temperature		286 °C
Material		SA-387 11	Ext. Design Temperature		286 °C
Length		1500.000 mm.	External Pressure Chart		CS-3
			UNS Number		K11789
Int. Corr. All.	c	1.5000 mm.	Allowable Stress (ope)	S	165.1 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	207.0 N./mm ²
Inside Diameter	Di	2032.000 mm.	Actual Stress		159.7 N./mm ²
Spec. Min. Thk t,e, or sa		34.0000 mm.	Spec. Nominal Thk.		34.000 mm.
Surface Area		98960.2 cm ²	SG of Contents		0.000
Element Volume		2968293.5 cm ³	Weight of Contents		0.0 Kgf
Empty Weight		2565.5 Kgf	Operating Weight		2565.5 Kgf
			Comp. All. Ope. Stress		85.0 N./mm ²
Radiography Information:					
...			...		
Circ. Joint Efficiency		100 %	Long. Joint Eff.	E or v	100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 2035.0 \cdot (\exp(51.0/(165.119 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 31.9196 + 1.5 + 0.0 = 33.4196 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 165.119 \cdot 1.0 \cdot \ln((2 \cdot 32.5 + 2035.0) / 2035.0) - 0.0$$

$$= 51.913 - 0.0 = 51.9128 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 206.996 \cdot 1.0 \cdot \ln((2 \cdot 34.0 + 2032.0) / 2032.0)$$

$$= 68.1326 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 51.0 \cdot 2035.0 / (1.0 (2100.0 - 2035.0))$$

$$= 159.6787 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 51.0 / (1.0 \cdot \ln((2 \cdot 32.5 + 2035.0) / 2035.0))$$

$$= 162.2157 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

$$\text{Cylinders formed from Plate} \quad (50 \cdot t_{\text{nom}} / R_f) \cdot (1 - R_f / R_o) \quad 1.646 \%$$

Minimum Design Metal Temperature Results:

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	: 6 °C
User entered Minimum Design Temperature	: -10 °C
The Curve for this Material is	: A
From Figure 3.8M- with PWHT, the MDMT is	: 23 °C
Temperature reduction per Graph 3.13	: 17 °C
The computed stress ratio is	: 0.699

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{\text{nom}} - c_i - c_{\text{ext}})$$

$$= (22.715 \cdot 1.0) / (34.0 - 1.5 - 0.0)$$

$$= 0.699$$

Warning: Impact Testing Is Mandatory!

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	17.334 bars
Governing Thk.	34.000 mm.	Min. Des. Mtl. Temp.	-10 °C
Stress/Thk. Ratio	0.699	Basic MDMT	23 °C
		Temperature Difference	17 °C
		Computed Min. Temp.	6 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: shell-2**Cylindrical Shell From 20 to 30 Ext. Chart: CS-2 at 286 °C****Maximum External Pressure per ASME Division 2 Section 4.4:**

Determine Parameter [Mx]:

$$= L / \sqrt{R_o \cdot t}$$

$$= 5343.97 / \sqrt{1050.0 \cdot 32.5}$$

$$= 28.929$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 13 \text{ and } M_x < 2 \cdot (D_o/t)^{0.94}, \text{ Ch} = 1.12 \cdot M_x^{-1.058}$$

$$= 1.12 \cdot 28.929^{-1.058}$$

$$= 0.0319$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot \text{Ch} \cdot E_y(t / D_o)$$

$$= 1.6 \cdot 0.032 \cdot 0.18730 \text{E}+09(32.5/2100.0)$$

$$= 147.734 \text{ N./mm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= F_{he} / F_y$$

$$= 147.7/247.7$$

$$= 0.596$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{ts}$$

$$= 247.7/517.1$$

$$= 0.479$$



Ratio [AE]:

$$= F_{he} / E$$

$$= 147.7 / 1.8730E+09$$

$$= 0.000789$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.479) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.160$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$

$$= 247.7 (1 + 0.002) / \ln(1 + 0.002)^{0.160318}$$

$$= 672.232 \text{ N./mm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f / (m2f^2 - R)$$

$$= 0.6 (1.0 - 0.479)$$

$$= 0.313$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (517.1 \cdot \exp(0.313)) / (0.313^{0.313})$$

$$= 1016.803 \text{ N./mm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4791.5 - 0.5 \cdot 0.4792.5 - 0.4793.5$$

$$= 0.342$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	258	-1.764	0.311E-05	0.262E-05	0.775E-05	-0.580E-05	0.122E+08	0.205E-01
2	129	-4.571	0.825E-07	0.825E-07	0.169E-05	-0.169E-05	0.143E+09	0.116E-03
3	64	-5.975	0.219E-08	0.219E-08	0.368E-06	-0.368E-06	0.186E+09	0.441E-03
4	96	-5.273	0.183E-07	0.183E-07	0.897E-06	-0.897E-06	0.175E+09	0.236E-03
5	113	-4.922	0.410E-07	0.410E-07	0.126E-05	-0.126E-05	0.162E+09	0.917E-04
6	121	-4.747	0.588E-07	0.588E-07	0.146E-05	-0.146E-05	0.153E+09	0.165E-05
7	117	-4.834	0.493E-07	0.493E-07	0.136E-05	-0.136E-05	0.158E+09	0.473E-04
8	119	-4.791	0.539E-07	0.539E-07	0.141E-05	-0.141E-05	0.156E+09	0.234E-04
9	120	-4.769	0.563E-07	0.563E-07	0.144E-05	-0.144E-05	0.155E+09	0.110E-04
10	120	-4.758	0.576E-07	0.575E-07	0.145E-05	-0.145E-05	0.154E+09	0.474E-05
11	120	-4.752	0.582E-07	0.582E-07	0.146E-05	-0.146E-05	0.154E+09	0.155E-05
12	121	-4.749	0.585E-07	0.585E-07	0.146E-05	-0.146E-05	0.153E+09	0.463E-07
13	121	-4.751	0.584E-07	0.583E-07	0.146E-05	-0.146E-05	0.154E+09	0.754E-06
14	121	-4.750	0.584E-07	0.584E-07	0.146E-05	-0.146E-05	0.154E+09	0.354E-06
15	121	-4.750	0.585E-07	0.585E-07	0.146E-05	-0.146E-05	0.154E+09	0.154E-06
16	121	-4.750	0.585E-07	0.585E-07	0.146E-05	-0.146E-05	0.154E+09	0.538E-07
17	121	-4.750	0.585E-07	0.585E-07	0.146E-05	-0.146E-05	0.153E+09	0.379E-08

Factor of Safety [FS]:

$$\text{Since } F_{ic} [121.07] < 0.55 \cdot 247.68, FS = 2$$

Allowable Circumferential Compressive Stress [Fha]:

$$= F_{ic} / FS$$

$$= 121.071 / 2.0$$

$$= 60.535 \text{ N./mm}^2$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot F_{ha} (t / D_o)$$

$$= 2 \cdot 605.32 (32.5 / 2100.0)$$

$$= 18.736 \text{ bars}$$

STEAM SUPERHEATER

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FileName : STEAM SUPERHEATER

Element: shell-2

PV Elite

v28



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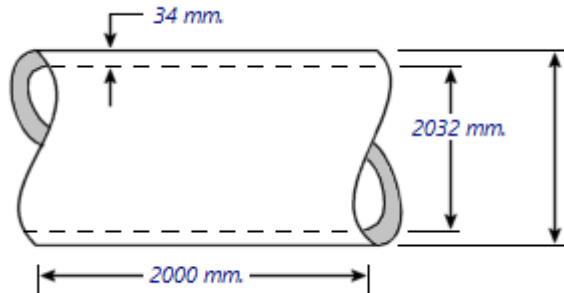
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Step: 5 4:54pm Sep 16, 2026

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: shell-3

Cylindrical Shell Design Information:					
Design Pressure	P	51.000 bars	Int. Design Temperature		286 °C
Material		SA-387 11	Ext. Design Temperature		286 °C
Length		2000.000 mm.	External Pressure Chart		CS-3
			UNS Number		K11789
Int. Corr. All.	c	1.5000 mm.	Allowable Stress (ope)	S	165.1 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	207.0 N./mm ²
Inside Diameter	Di	2032.000 mm.	Actual Stress		159.7 N./mm ²
Spec. Min. Thk t,e, or sa		34.0000 mm.	Spec. Nominal Thk.		34.000 mm.
Surface Area		131946.9 cm ²	SG of Contents		0.000
Element Volume		5212168.0 cm ³	Weight of Contents		0.0 Kg
Empty Weight		3420.7 Kg	Operating Weight		3420.7 Kg
			Comp. All. Ope. Stress		85.0 N./mm ²
Radiography Information:					
...			...		
Circ. Joint Efficiency		100 %	Long. Joint Eff.	E or v	100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 2035.0 \cdot (\exp(51.0/(165.119 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 31.9196 + 1.5 + 0.0 = 33.4196 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D)/D) - P_{liq}$$

$$= 165.119 \cdot 1.0 \cdot \ln((2 \cdot 32.5 + 2035.0)/2035.0) - 0.0$$

$$= 51.913 - 0.0 = 51.9128 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D)/D)$$

$$= 206.996 \cdot 1.0 \cdot \ln((2 \cdot 34.0 + 2032.0)/2032.0)$$

$$= 68.1326 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E(D_o - D))$$

$$= 51.0 \cdot 2035.0 / (1.0(2100.0 - 2035.0))$$

$$= 159.6787 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D)/D))$$



$$= 51.0 / (1.0 \cdot \ln((2 \cdot 32.5 + 2035.0) / 2035.0))$$

$$= 162.2157 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

$$\text{Cylinders formed from Plate} \quad (50 \cdot t_{\text{nom}} / R_f) \cdot (1 - R_f / R_o) \quad 1.646 \%$$

Minimum Design Metal Temperature Results:

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	: 6 °C
User entered Minimum Design Temperature	: -10 °C
The Curve for this Material is	: A
From Figure 3.8M- with PWHT, the MDMT is	: 23 °C
Temperature reduction per Graph 3.13	: 17 °C
The computed stress ratio is	: 0.699

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{\text{nom}} - c_i - c_{\text{ext}})$$

$$= (22.715 \cdot 1.0) / (34.0 - 1.5 - 0.0)$$

$$= 0.699$$

Warning: Impact Testing Is Mandatory!

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	17.334 bars
Governing Thk.	34.000 mm.	Min. Des. Mtl. Temp.	-10 °C
Stress/Thk. Ratio	0.699	Basic MDMT	23 °C
		Temperature Difference	17 °C
		Computed Min. Temp.	6 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: shell-3**Cylindrical Shell From 30 to 40 Ext. Chart: CS-2 at 286 °C****Maximum External Pressure per ASME Division 2 Section 4.4:**

Determine Parameter [Mx]:

$$= L / \sqrt{R_o \cdot t}$$

$$= 5343.97 / \sqrt{1050.0 \cdot 32.5}$$

$$= 28.929$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 13 \text{ and } M_x < 2 \cdot (D_o/t)^{0.94}, \text{ Ch} = 1.12 \cdot M_x^{-1.058}$$

$$= 1.12 \cdot 28.929^{-1.058}$$

$$= 0.0319$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot \text{Ch} \cdot E_y(t / D_o)$$

$$= 1.6 \cdot 0.032 \cdot 0.18730 \text{E}+09(32.5/2100.0)$$

$$= 147.734 \text{ N./mm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= F_{he} / F_y$$

$$= 147.7/247.7$$

$$= 0.596$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 247.7/517.1$$

$$= 0.479$$



Ratio [AE]:

$$= F_{he} / E$$

$$= 147.7 / 1.8730E+09$$

$$= 0.000789$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.479) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.160$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$

$$= 247.7 (1 + 0.002) / \ln(1 + 0.002)^{0.160318}$$

$$= 672.232 \text{ N./mm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f / (m2f^2 - R)$$

$$= 0.6 / (1.0 - 0.479)$$

$$= 0.313$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (517.1 \cdot \exp(0.313)) / (0.313^{0.313})$$

$$= 1016.803 \text{ N./mm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4791.5 - 0.5 \cdot 0.4792.5 - 0.4793.5$$

$$= 0.342$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	258	-1.764	0.311E-05	0.262E-05	0.775E-05	-0.580E-05	0.122E+08	0.205E-01
2	129	-4.571	0.825E-07	0.825E-07	0.169E-05	-0.169E-05	0.143E+09	0.116E-03
3	64	-5.975	0.219E-08	0.219E-08	0.368E-06	-0.368E-06	0.186E+09	0.441E-03
4	96	-5.273	0.183E-07	0.183E-07	0.897E-06	-0.897E-06	0.175E+09	0.236E-03
5	113	-4.922	0.410E-07	0.410E-07	0.126E-05	-0.126E-05	0.162E+09	0.917E-04
6	121	-4.747	0.588E-07	0.588E-07	0.146E-05	-0.146E-05	0.153E+09	0.165E-05
7	117	-4.834	0.493E-07	0.493E-07	0.136E-05	-0.136E-05	0.158E+09	0.473E-04
8	119	-4.791	0.539E-07	0.539E-07	0.141E-05	-0.141E-05	0.156E+09	0.234E-04
9	120	-4.769	0.563E-07	0.563E-07	0.144E-05	-0.144E-05	0.155E+09	0.110E-04
10	120	-4.758	0.576E-07	0.575E-07	0.145E-05	-0.145E-05	0.154E+09	0.474E-05
11	120	-4.752	0.582E-07	0.582E-07	0.146E-05	-0.146E-05	0.154E+09	0.155E-05
12	121	-4.749	0.585E-07	0.585E-07	0.146E-05	-0.146E-05	0.153E+09	0.463E-07
13	121	-4.751	0.584E-07	0.583E-07	0.146E-05	-0.146E-05	0.154E+09	0.754E-06
14	121	-4.750	0.584E-07	0.584E-07	0.146E-05	-0.146E-05	0.154E+09	0.354E-06
15	121	-4.750	0.585E-07	0.585E-07	0.146E-05	-0.146E-05	0.154E+09	0.154E-06
16	121	-4.750	0.585E-07	0.585E-07	0.146E-05	-0.146E-05	0.154E+09	0.538E-07
17	121	-4.750	0.585E-07	0.585E-07	0.146E-05	-0.146E-05	0.153E+09	0.379E-08

Factor of Safety [FS]:

$$\text{Since } F_{ic} [121.07] < 0.55 \cdot 247.68, FS = 2$$

Allowable Circumferential Compressive Stress [Fha]:

$$= F_{ic} / FS$$

$$= 121.071 / 2.0$$

$$= 60.535 \text{ N./mm}^2$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot F_{ha} (t / D_o)$$

$$= 2 \cdot 605.32 (32.5 / 2100.0)$$

$$= 18.736 \text{ bars}$$

STEAM SUPERHEATER

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PV Elite 28 U2

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FileName : STEAM SUPERHEATER

Element: shell-3

PV Elite®

v28



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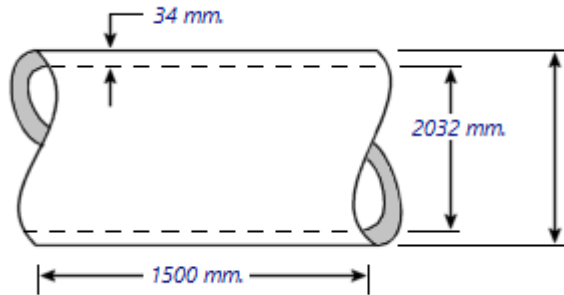
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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: sheel 4

Cylindrical Shell Design Information:					
Design Pressure	P	51.000 bars	Int. Design Temperature		286 °C
Material		SA-387 11	Ext. Design Temperature		286 °C
Length		1500.000 mm.	External Pressure Chart		CS-3
			UNS Number		K11789
Int. Corr. All.	c	1.5000 mm.	Allowable Stress (ope)	S	165.1 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	207.0 N./mm ²
Inside Diameter	Di	2032.000 mm.	Actual Stress		159.7 N./mm ²
Spec. Min. Thk t,e, or sa		34.0000 mm.	Spec. Nominal Thk.		34.000 mm.
Surface Area		98960.2 cm ²	SG of Contents		0.000
Element Volume		3909126.0 cm ³	Weight of Contents		0.0 Kg
Empty Weight		2565.5 Kg	Operating Weight		2565.5 Kg
			Comp. All. Ope. Stress		85.0 N./mm ²
Radiography Information:					
...			...		
Circ. Joint Efficiency		100 %	Long. Joint Eff.	E or v	100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 2035.0 \cdot (\exp(51.0/(165.119 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 31.9196 + 1.5 + 0.0 = 33.4196 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 165.119 \cdot 1.0 \cdot \ln((2 \cdot 32.5 + 2035.0) / 2035.0) - 0.0$$

$$= 51.913 - 0.0 = 51.9128 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 206.996 \cdot 1.0 \cdot \ln((2 \cdot 34.0 + 2032.0) / 2032.0)$$

$$= 68.1326 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 51.0 \cdot 2035.0 / (1.0 (2100.0 - 2035.0))$$

$$= 159.6787 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 51.0 / (1.0 \cdot \ln((2 \cdot 32.5 + 2035.0) / 2035.0))$$

$$= 162.2157 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

$$\text{Cylinders formed from Plate} \quad (50 \cdot t_{\text{nom}} / R_f) \cdot (1 - R_f / R_o) \quad 1.646 \%$$

Minimum Design Metal Temperature Results:

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	: 6 °C
User entered Minimum Design Temperature	: -10 °C
The Curve for this Material is	: A
From Figure 3.8M- with PWHT, the MDMT is	: 23 °C
Temperature reduction per Graph 3.13	: 17 °C
The computed stress ratio is	: 0.699

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{\text{nom}} - c_i - c_{\text{ext}})$$

$$= (22.715 \cdot 1.0) / (34.0 - 1.5 - 0.0)$$

$$= 0.699$$

Warning: Impact Testing Is Mandatory!

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	17.334 bars
Governing Thk.	34.000 mm.	Min. Des. Mtl. Temp.	-10 °C
Stress/Thk. Ratio	0.699	Basic MDMT	23 °C
		Temperature Difference	17 °C
		Computed Min. Temp.	6 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: sheel 4**Cylindrical Shell From 40 to Tsht Ext. Chart: CS-2 at 286 °C****Maximum External Pressure per ASME Division 2 Section 4.4:**

Determine Parameter [Mx]:

$$= L / \sqrt{R_o \cdot t}$$

$$= 5343.97 / \sqrt{1050.0 \cdot 32.5}$$

$$= 28.929$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 13 \text{ and } M_x < 2 \cdot (D_o/t)^{0.94}, \text{ Ch} = 1.12 \cdot M_x^{-1.058}$$

$$= 1.12 \cdot 28.929^{-1.058}$$

$$= 0.0319$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot \text{Ch} \cdot E_y(t / D_o)$$

$$= 1.6 \cdot 0.032 \cdot 0.18730 \text{E}+09(32.5/2100.0)$$

$$= 147.734 \text{ N./mm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= F_{he} / F_y$$

$$= 147.7/247.7$$

$$= 0.596$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 247.7/517.1$$

$$= 0.479$$



Ratio [AE]:

$$= F_{he} / E$$

$$= 147.7 / 1.8730E+09$$

$$= 0.000789$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.479) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.160$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$

$$= 247.7 (1 + 0.002) / \ln(1 + 0.002)^{0.160318}$$

$$= 672.232 \text{ N./mm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f / (m2f^2 - R)$$

$$= 0.6 / (1.0 - 0.479)$$

$$= 0.313$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (517.1 \cdot \exp(0.313)) / (0.313^{0.313})$$

$$= 1016.803 \text{ N./mm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4791.5 - 0.5 \cdot 0.4792.5 - 0.4793.5$$

$$= 0.342$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	258	-1.764	0.311E-05	0.262E-05	0.775E-05	-0.580E-05	0.122E+08	0.205E-01
2	129	-4.571	0.825E-07	0.825E-07	0.169E-05	-0.169E-05	0.143E+09	0.116E-03
3	64	-5.975	0.219E-08	0.219E-08	0.368E-06	-0.368E-06	0.186E+09	0.441E-03
4	96	-5.273	0.183E-07	0.183E-07	0.897E-06	-0.897E-06	0.175E+09	0.236E-03
5	113	-4.922	0.410E-07	0.410E-07	0.126E-05	-0.126E-05	0.162E+09	0.917E-04
6	121	-4.747	0.588E-07	0.588E-07	0.146E-05	-0.146E-05	0.153E+09	0.165E-05
7	117	-4.834	0.493E-07	0.493E-07	0.136E-05	-0.136E-05	0.158E+09	0.473E-04
8	119	-4.791	0.539E-07	0.539E-07	0.141E-05	-0.141E-05	0.156E+09	0.234E-04
9	120	-4.769	0.563E-07	0.563E-07	0.144E-05	-0.144E-05	0.155E+09	0.110E-04
10	120	-4.758	0.576E-07	0.575E-07	0.145E-05	-0.145E-05	0.154E+09	0.474E-05
11	120	-4.752	0.582E-07	0.582E-07	0.146E-05	-0.146E-05	0.154E+09	0.155E-05
12	121	-4.749	0.585E-07	0.585E-07	0.146E-05	-0.146E-05	0.153E+09	0.463E-07
13	121	-4.751	0.584E-07	0.583E-07	0.146E-05	-0.146E-05	0.154E+09	0.754E-06
14	121	-4.750	0.584E-07	0.584E-07	0.146E-05	-0.146E-05	0.154E+09	0.354E-06
15	121	-4.750	0.585E-07	0.585E-07	0.146E-05	-0.146E-05	0.154E+09	0.154E-06
16	121	-4.750	0.585E-07	0.585E-07	0.146E-05	-0.146E-05	0.154E+09	0.538E-07
17	121	-4.750	0.585E-07	0.585E-07	0.146E-05	-0.146E-05	0.153E+09	0.379E-08

Factor of Safety [FS]:

$$\text{Since } F_{ic} [121.07] < 0.55 \cdot 247.68, FS = 2$$

Allowable Circumferential Compressive Stress [Fha]:

$$= F_{ic} / FS$$

$$= 121.071 / 2.0$$

$$= 60.535 \text{ N./mm}^2$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot F_{ha} (t / D_o)$$

$$= 2 \cdot 605.32 (32.5 / 2100.0)$$

$$= 18.736 \text{ bars}$$

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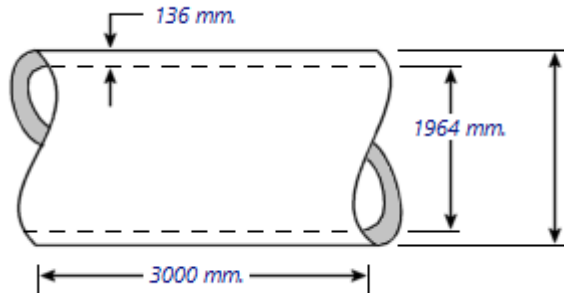
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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Channel-1

Cylindrical Shell Design Information:					
Design Pressure	P	191.000 bars	Int. Design Temperature		286 °C
Material		SA-387 11	Ext. Design Temperature		286 °C
Length		3000.000 mm.	External Pressure Chart		CS-3
			UNS Number		K11789
Int. Corr. All.	c	1.5000 mm.	Allowable Stress (ope)	S	165.1 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	207.0 N./mm ²
Inside Diameter	Di	1964.000 mm.	Actual Stress		139.7 N./mm ²
Spec. Min. Thk t,e, or sa		136.0000 mm.	Spec. Nominal Thk.		136.000 mm.
Surface Area		210738.0 cm ²	SG of Contents		0.000
Element Volume		9090167.0 cm ³	Weight of Contents		0.00 Kg
Empty Weight		20862.2 Kg	Operating Weight		20862.2 Kg
			Comp. All. Ope. Stress		103.7 N./mm ²
Radiography Information:					
...			...		
Circ. Joint Efficiency		100 %	Long. Joint Eff.	E or v	100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 1967.0 \cdot (\exp(191.0/(165.119 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 120.6141 + 1.5 + 0.0 = 122.1141 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 165.119 \cdot 1.0 \cdot \ln((2 \cdot 134.5 + 1967.0) / 1967.0) - 0.0$$

$$= 211.636 - 0.0 = 211.6357 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 206.996 \cdot 1.0 \cdot \ln((2 \cdot 136.0 + 1964.0) / 1964.0)$$

$$= 268.4698 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 191.0 \cdot 1967.0 / (1.0 (2236.0 - 1967.0))$$

$$= 139.6724 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 191.0 / (1.0 \cdot \ln((2 \cdot 134.5 + 1967.0) / 1967.0))$$

$$= 149.0190 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

$$\text{Cylinders formed from Plate} \quad (50 \cdot t_{\text{nom}} / R_f) \cdot (1 - R_f / R_o) \quad 6.476 \%$$

Minimum Design Metal Temperature Results:*Warning:*

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm.. This component must be impact tested.

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	17.152 bars
Governing Thk.	136.000 mm.	Min. Des. Mtl. Temp.	-10 °C
Stress/Thk. Ratio	-1.000	Basic MDMT	-17 °C
		Temperature Difference	0 °C
		Computed Min. Temp.	-17 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Channel-1**Cylindrical Shell From 50 to 60 Ext. Chart: CS-2 at 286 °C****Maximum External Pressure per ASME Division 2 Section 4.4:**

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$

$$= 6350.03 / \sqrt{(1118.0 \cdot 134.5)}$$

$$= 16.375$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 13 \text{ and } M_x < 2 \cdot (D_o/t)^{0.94}, \text{ Ch} = 1.12 \cdot M_x^{-1.058}$$

$$= 1.12 \cdot 16.375^{-1.058}$$

$$= 0.0582$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot \text{Ch} \cdot E_y (t / D_o)$$

$$= 1.6 \cdot 0.0582 \cdot 0.18730 \text{E}+09 (134.5 / 2236.0)$$

$$= 1048.419 \text{ N./mm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= F_{he} / F_y$$

$$= 1048.4 / 247.7$$

$$= 4.233$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 247.7 / 517.1$$

$$= 0.479$$

Ratio [AE]:

$$= F_{he} / E$$

$$= 1048.4 / 0.18730 \text{E}+09$$

$$= 0.005597$$

Factor [m1]:

$$= (\ln(r) + (\text{epsp} - \text{epsys})) / \ln(\ln(1 + \text{epsp}) / \ln(1 + \text{epsys}))$$

$$= (\ln(0.479) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.160$$



Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys}) m_1$$

$$= 247.7 (1 + 0.002) / \ln(1 + 0.002) 0.160318$$

$$= 672.232 \text{ N./mm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m_2 f(m_2^2 - R)$$

$$= 0.6 (1.0 - 0.479)$$

$$= 0.313$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m_2)) / (m_2 m_2)$$

$$= (517.1 \cdot \exp(0.313)) / (0.313 0.313)$$

$$= 1016.803 \text{ N./mm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R_{1.5} - 0.5 \cdot R_{2.5} - R_{3.5}$$

$$= 1.5 \cdot 0.479_{1.5} - 0.5 \cdot 0.479_{2.5} - 0.479_{3.5}$$

$$= 0.342$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	258	-1.764	0.311E-05	0.262E-05	0.775E-05	-0.580E-05	0.122E+08	0.157E-01
2	129	-4.571	0.825E-07	0.825E-07	0.169E-05	-0.169E-05	0.143E+09	0.469E-02
3	193	-3.167	0.690E-06	0.684E-06	0.412E-05	-0.406E-05	0.510E+08	0.180E-02
4	226	-2.466	0.155E-05	0.149E-05	0.578E-05	-0.544E-05	0.256E+08	0.324E-02
5	210	-2.816	0.105E-05	0.103E-05	0.491E-05	-0.477E-05	0.364E+08	0.176E-03
6	202	-2.992	0.854E-06	0.844E-06	0.450E-05	-0.442E-05	0.432E+08	0.918E-03
7	206	-2.904	0.948E-06	0.934E-06	0.470E-05	-0.460E-05	0.396E+08	0.401E-03
8	208	-2.860	0.997E-06	0.981E-06	0.481E-05	-0.469E-05	0.380E+08	0.120E-03
9	209	-2.838	0.102E-05	0.101E-05	0.486E-05	-0.473E-05	0.372E+08	0.258E-04
10	208	-2.849	0.101E-05	0.994E-06	0.483E-05	-0.471E-05	0.376E+08	0.478E-04
11	208	-2.844	0.102E-05	0.100E-05	0.484E-05	-0.472E-05	0.374E+08	0.111E-04
12	208	-2.841	0.102E-05	0.100E-05	0.485E-05	-0.472E-05	0.373E+08	0.731E-05
13	208	-2.843	0.102E-05	0.102E-05	0.485E-05	-0.472E-05	0.373E+08	0.191E-05
14	208	-2.842	0.102E-05	0.100E-05	0.485E-05	-0.472E-05	0.373E+08	0.269E-05
15	208	-2.842	0.102E-05	0.100E-05	0.485E-05	-0.472E-05	0.373E+08	0.389E-06
16	208	-2.842	0.102E-05	0.100E-05	0.485E-05	-0.472E-05	0.373E+08	0.763E-06
17	208	-2.842	0.102E-05	0.100E-05	0.485E-05	-0.472E-05	0.373E+08	0.187E-06

Factor of Safety [FS]:

$$\text{Since } F_{ic} > 0.55 F_y \text{ and } F_{ic} < F_y$$

$$= 2.407 - 0.741 \cdot F_{ic} / F_y$$

$$= 2.407 - 0.741 \cdot 208.89 / 247.68$$

$$= 1.782$$

Allowable Circumferential Compressive Stress [Fha]:

$$= F_{ic} / FS$$

$$= 208.894 / 1.782$$

$$= 117.223 \text{ N./mm}^2$$

Allowable External Pressure at the given Thickness [Pa]:

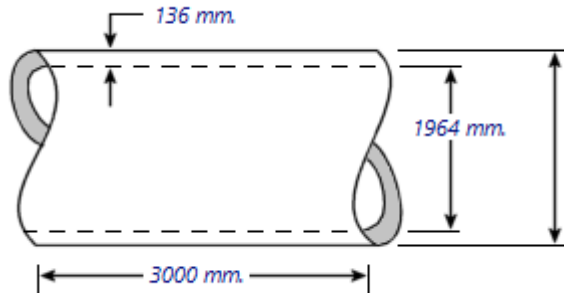
$$= 2 \cdot F_{ha} (t / D_o)$$

$$= 2 \cdot 1172.157 (134.5 / 2236.0)$$

$$= 141.015 \text{ bars}$$

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Channel-2

Cylindrical Shell Design Information:					
Design Pressure	P	191.000 bars	Int. Design Temperature		286 °C
Material		SA-387 11	Ext. Design Temperature		286 °C
Length		3000.000 mm.	External Pressure Chart		CS-3
			UNS Number		K11789
Int. Corr. All.	c	1.5000 mm.	Allowable Stress (ope)	S	165.1 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	207.0 N./mm ²
Inside Diameter	Di	1964.000 mm.	Actual Stress		139.7 N./mm ²
Spec. Min. Thk t,e, or sa		136.0000 mm.	Spec. Nominal Thk.		136.000 mm.
Surface Area		210738.0 cm ²	SG of Contents		0.000
Element Volume		9090167.0 cm ³	Weight of Contents		0.00 Kgf
Empty Weight		20862.2 Kgf	Operating Weight		20862.2 Kgf
			Comp. All. Ope. Stress		103.7 N./mm ²
Radiography Information:					
...			...		
Circ. Joint Efficiency		100 %	Long. Joint Eff.	E or v	100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 1967.0 \cdot (\exp(191.0/(165.119 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 120.6141 + 1.5 + 0.0 = 122.1141 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 165.119 \cdot 1.0 \cdot \ln((2 \cdot 134.5 + 1967.0) / 1967.0) - 0.0$$

$$= 211.636 - 0.0 = 211.6357 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 206.996 \cdot 1.0 \cdot \ln((2 \cdot 136.0 + 1964.0) / 1964.0)$$

$$= 268.4698 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 191.0 \cdot 1967.0 / (1.0 (2236.0 - 1967.0))$$

$$= 139.6724 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 191.0 / (1.0 \cdot \ln((2 \cdot 134.5 + 1967.0) / 1967.0))$$

$$= 149.0190 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

$$\text{Cylinders formed from Plate} \quad (50 \cdot t_{\text{nom}} / R_f) \cdot (1 - R_f / R_o) \quad 6.476 \%$$

Minimum Design Metal Temperature Results:*Warning:*

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm.. This component must be impact tested.

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	17.152 bars
Governing Thk.	136.000 mm.	Min. Des. Mtl. Temp.	-10 °C
Stress/Thk. Ratio	-1.000	Basic MDMT	-17 °C
		Temperature Difference	0 °C
		Computed Min. Temp.	-17 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Channel-2**Cylindrical Shell From 60 to 70 Ext. Chart: CS-2 at 286 °C****Maximum External Pressure per ASME Division 2 Section 4.4:**

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$

$$= 6350.031 / \sqrt{(1118.0 \cdot 134.5)}$$

$$= 16.375$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 13 \text{ and } M_x < 2 \cdot (D_o/t)^{0.94}, \text{ Ch} = 1.12 \cdot M_x - 1.058$$

$$= 1.12 \cdot 16.375 - 1.058$$

$$= 0.0582$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot \text{Ch} \cdot E_y (t / D_o)$$

$$= 1.6 \cdot 0.058 \cdot 0.18730 \text{E}+09 (134.5 / 2236.0)$$

$$= 1048.419 \text{ N./mm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= F_{he} / F_y$$

$$= 1048.4 / 247.7$$

$$= 4.233$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 247.7 / 517.1$$

$$= 0.479$$

Ratio [AE]:

$$= F_{he} / E$$

$$= 1048.4 / 0.18730 \text{E}+09$$

$$= 0.005597$$

Factor [m1]:

$$= (\ln(r) + (\text{epsp} - \text{epsys})) / \ln(\ln(1 + \text{epsp}) / \ln(1 + \text{epsys}))$$

$$= (\ln(0.479) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.160$$



Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys}) m_1$$

$$= 247.7 (1 + 0.002) / \ln(1 + 0.002) 0.160318$$

$$= 672.232 \text{ N./mm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m_2 f(m_2^2 - R)$$

$$= 0.6 (1.0 - 0.479)$$

$$= 0.313$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m_2)) / (m_2 m_2)$$

$$= (517.1 \cdot \exp(0.313)) / (0.313 0.313)$$

$$= 1016.803 \text{ N./mm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R_{1.5} - 0.5 \cdot R_{2.5} - R_{3.5}$$

$$= 1.5 \cdot 0.479_{1.5} - 0.5 \cdot 0.479_{2.5} - 0.479_{3.5}$$

$$= 0.342$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	258	-1.764	0.311E-05	0.262E-05	0.775E-05	-0.580E-05	0.122E+08	0.157E-01
2	129	-4.571	0.825E-07	0.825E-07	0.169E-05	-0.169E-05	0.143E+09	0.469E-02
3	193	-3.167	0.690E-06	0.684E-06	0.412E-05	-0.406E-05	0.510E+08	0.180E-02
4	226	-2.466	0.155E-05	0.149E-05	0.578E-05	-0.544E-05	0.256E+08	0.324E-02
5	210	-2.816	0.105E-05	0.103E-05	0.491E-05	-0.477E-05	0.364E+08	0.176E-03
6	202	-2.992	0.854E-06	0.844E-06	0.450E-05	-0.442E-05	0.432E+08	0.918E-03
7	206	-2.904	0.948E-06	0.934E-06	0.470E-05	-0.460E-05	0.396E+08	0.401E-03
8	208	-2.860	0.997E-06	0.981E-06	0.481E-05	-0.469E-05	0.380E+08	0.120E-03
9	209	-2.838	0.102E-05	0.101E-05	0.486E-05	-0.473E-05	0.372E+08	0.258E-04
10	208	-2.849	0.101E-05	0.994E-06	0.483E-05	-0.471E-05	0.376E+08	0.478E-04
11	208	-2.844	0.102E-05	0.100E-05	0.484E-05	-0.472E-05	0.374E+08	0.111E-04
12	208	-2.841	0.102E-05	0.100E-05	0.485E-05	-0.472E-05	0.373E+08	0.731E-05
13	208	-2.843	0.102E-05	0.100E-05	0.485E-05	-0.472E-05	0.373E+08	0.191E-05
14	208	-2.842	0.102E-05	0.100E-05	0.485E-05	-0.472E-05	0.373E+08	0.269E-05
15	208	-2.842	0.102E-05	0.100E-05	0.485E-05	-0.472E-05	0.373E+08	0.390E-06
16	208	-2.842	0.102E-05	0.100E-05	0.485E-05	-0.472E-05	0.373E+08	0.762E-06
17	208	-2.842	0.102E-05	0.100E-05	0.485E-05	-0.472E-05	0.373E+08	0.186E-06

Factor of Safety [FS]:

$$\text{Since } F_{ic} > 0.55 F_y \text{ and } F_{ic} < F_y$$

$$= 2.407 - 0.741 \cdot F_{ic} / F_y$$

$$= 2.407 - 0.741 \cdot 208.89 / 247.68$$

$$= 1.782$$

Allowable Circumferential Compressive Stress [Fha]:

$$= F_{ic} / FS$$

$$= 208.894 / 1.782$$

$$= 117.223 \text{ N./mm}^2$$

Allowable External Pressure at the given Thickness [Pa]:

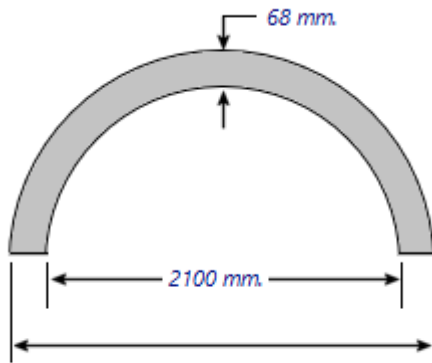
$$= 2 \cdot F_{ha} (t / D_o)$$

$$= 2 \cdot 1172.157 (134.5 / 2236.0)$$

$$= 141.015 \text{ bars}$$

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Top dish

Spherical Head Design Information:					
Design Pressure	P	191.000 bars	Int. Design Temperature		286 °C
Material		SA-387 11	Ext. Design Temperature		286 °C
Length		0.000 mm.	External Pressure Chart		CS-3
			UNS Number		K11789
Int. Corr. All.	c	1.5000 mm.	Allowable Stress (ope)	S	165.1 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	207.0 N./mm ²
Inside Diameter	Di	2100.000 mm.	Actual Stress		146.4 N./mm ²
Spec. Min. Thk t _e , or s _a		68.0000 mm.	Spec. Nominal Thk.		74.000 mm.
Surface Area		79380.3 cm ²	SG of Contents		0.000
Element Volume		2424958.0 cm ³	Weight of Contents		0.0 Kgf
Empty Weight		4259.6 Kgf	Operating Weight		4259.6 Kgf
			Comp. All. Ope. Stress		94.1 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency		Long. Joint Eff.	E or v
100 %		100 %	

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 26.250 mm.
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 13.125 mm.

Spherical Shell Calculation - Section 4.3.5.1

Computed Minimum Required Thickness [t]:

$$= D/2 \left(\exp(0.5P/(S \cdot E)) - 1 \right) + c_i + c_o$$

$$= 2103.0/2 \left(\exp(0.5 \cdot 191.0/(165.119 \cdot 1.0)) - 1 \right) + 1.5 + 0.0$$

$$= 62.6125 + 1.5 + 0.0 = 64.1125 \text{ mm.}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= 2 \cdot S \cdot E \cdot \ln((2 \cdot t + D)/D) - P_{liq}$$

$$= 2 \cdot 165.1 \cdot 1.0 \cdot \ln((2 \cdot 66.5 + 2103.0)/2103.0) - 0.0$$

$$= 202.502 - 0.0 = 202.5023 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= 2 \cdot S \cdot E \cdot \ln((2 \cdot t + D)/D)$$

$$= 2 \cdot 206.996 \cdot 1.0 \cdot \ln((2 \cdot 68.0 + 2100.0)/2100.0)$$

$$= 259.7701 \text{ bars}$$



Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D^2 / (E (D_o^2 - D^2))$$

$$= 191.0 \cdot 2103.0^2 / (1.0 (2236.0^2 - 2103.0^2))$$

$$= 146.385 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= 0.5 \cdot P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 0.5 \cdot 191.0 / (1.0 \cdot \ln((2 \cdot 66.5 + 2103.0) / 2103.0))$$

$$= 155.740 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

For Heads formed from Segments $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 5.106 %

For one Piece doubly Curved Heads $100 \cdot \ln(D_b / (D_f - 2t))$ 41.468 %

Where D_b is the blank diameter for one piece doubly curved surfaces [D_b]:

$$= \text{square root}(4 \cdot \text{Head Surface Area} / 3.14159)$$

$$= \text{square root}(4 \cdot 79380.3 / 3.14159)$$

$$= 3179.152 \text{ mm.}$$

MDMT Calculations in the Spherical Portion:

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : -104 °C

User entered Minimum Design Temperature : -10 °C

The Curve for this Material is : A

As the Stress Ratio < 0.240, the MDMT is : -104 °C

The computed stress ratio is : 0.066

The computed thickness ratio [sr]:

$$= (t \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (4.366 \cdot 1.0) / (68.0 - 1.5 - 0.0)$$

$$= 0.066$$

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	17.152 bars
Governing Thk.	68.000 mm.	Min. Des. Mtl. Temp.	-10 °C
Stress/Thk. Ratio	0.066	Basic MDMT	-104 °C
		Temperature Difference	0 °C
		Computed Min. Temp.	-104 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 26.250 mm.

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 13.125 mm.

EXTERNAL PRESSURE RESULTS for: Top dish

Spherical Head From 70 to 80 Ext. Chart: CS-2 at 286 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 0.075 \cdot E_y(t / R_o)$$

$$= 0.075 \cdot 0.18730 \text{E}+09(66.5 / 1118.0)$$

$$= 835.612 \text{ N./mm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= F_{he} / S_y$$

$$= 835.6 / 247.7$$



$$= 3.374$$

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 247.7/517.1$$

$$= 0.479$$

Ratio [AE]:

$$= F_{he} / E$$

$$= 835.6/.18730E+09$$

$$= 0.004461$$

Factor [m1]:

$$= (\ln(r) + (\text{epsp} - \text{epsys})) / \ln(\ln(1 + \text{epsp}) / \ln(1 + \text{epsys}))$$

$$= (\ln(0.479) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.160$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \text{epsys}) / \ln(1 + \text{epsys})^{m1}$$

$$= 247.7 (1 + 0.002) / \ln(1 + 0.002) 0.160318$$

$$= 672.232 \text{ N./mm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.479)$$

$$= 0.313$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (517.1 \cdot \exp(0.313)) / (0.313^{0.313})$$

$$= 1016.803 \text{ N./mm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4791.5 - 0.5 \cdot 0.4792.5 - 0.4793.5$$

$$= 0.342$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	258	-1.764	0.311E-05	0.262E-05	0.775E-05	-0.580E-05	0.122E+08	0.168E-01
2	129	-4.571	0.825E-07	0.825E-07	0.169E-05	-0.169E-05	0.143E+09	0.356E-02
3	193	-3.167	0.690E-06	0.684E-06	0.412E-05	-0.406E-05	0.510E+08	0.659E-03
4	226	-2.466	0.155E-05	0.149E-05	0.578E-05	-0.544E-05	0.256E+08	0.437E-02
5	210	-2.816	0.105E-05	0.103E-05	0.491E-05	-0.477E-05	0.364E+08	0.131E-02
6	202	-2.992	0.854E-06	0.844E-06	0.450E-05	-0.442E-05	0.432E+08	0.218E-03
7	197	-3.080	0.769E-06	0.761E-06	0.431E-05	-0.424E-05	0.469E+08	0.244E-03
8	199	-3.036	0.810E-06	0.802E-06	0.440E-05	-0.433E-05	0.450E+08	0.194E-04
9	200	-3.014	0.832E-06	0.823E-06	0.445E-05	-0.437E-05	0.441E+08	0.979E-04
10	200	-3.025	0.821E-06	0.812E-06	0.443E-05	-0.435E-05	0.446E+08	0.389E-04
11	200	-3.030	0.816E-06	0.807E-06	0.442E-05	-0.434E-05	0.448E+08	0.965E-05
12	200	-3.033	0.813E-06	0.804E-06	0.441E-05	-0.434E-05	0.449E+08	0.489E-05
13	200	-3.032	0.815E-06	0.805E-06	0.441E-05	-0.434E-05	0.448E+08	0.237E-05
14	200	-3.032	0.814E-06	0.805E-06	0.441E-05	-0.434E-05	0.449E+08	0.126E-05
15	200	-3.032	0.814E-06	0.805E-06	0.441E-05	-0.434E-05	0.449E+08	0.558E-06
16	200	-3.032	0.814E-06	0.805E-06	0.441E-05	-0.434E-05	0.449E+08	0.350E-06
17	200	-3.032	0.814E-06	0.805E-06	0.441E-05	-0.434E-05	0.449E+08	0.104E-06

Factor of Safety [FS]:

$$\text{Since } F_{ic} > 0.55 F_y \text{ and } F_{ic} < F_y$$

$$= 2.407 - 0.741 \cdot F_{ic} / F_y$$

$$= 2.407 - 0.741 \cdot 200.15/247.68$$

$$= 1.808$$

Allowable Circumferential Compressive Stress [Fha]:

$$= F_{ic}/FS$$

STEAM SUPERHEATER

A150NRJ10EP001

PV Elite 28 U2

Licensee: SL User

FileName : STEAM SUPERHEATER

Element: Top dish

PV Elite® v28



HEXAGON

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Step: 10 4:54pm Sep 16, 2026

= 200.152/1.808
= 110.692 N./mm²

Allowable External Pressure at the given Thickness [Pa]:

= 2 • Fha[t/Ro]
= 2 • 110.692[66.5/1118.0]
= 131.674 bars

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**Hydrostatic Test Pressure Results:**

Hydrotest Pressure Calculation [Shell Side]:

= Test Factor • ShellSide MAWP(St/S)

= 1.25 • 17.334(1.023)

= 22.163 bars

Hydrotest Pressure Calculation [Channel Side]:

= Test Factor • Channel Side MAWP(St/S)

= 1.25 • 17.152(1.023)

= 21.930 bars

Horizontal Test performed per: per VIII-2, Section 8.2

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
Dish bottom	64.7	294.5	0.220	22.37
shell-2	67.9	294.5	0.231	22.36
shell-3	67.9	294.5	0.231	22.36
shell 4	67.9	294.5	0.231	22.36
Channel-1	17.1	294.5	0.058	22.12
Channel-2	17.1	294.5	0.058	22.12
Top dish	17.6	294.5	0.060	22.14

Theoretical Maximum Allowable test pressure Shellside is : 96.733 bars

Theoretical Maximum Allowable test pressure Tubeside : 369.370 bars

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
D1-3"	Nozzle	207.00	169.56	1.221
A2-24 "	Nozzle	207.00	165.12	1.254
B2-24 "	Nozzle	207.00	165.12	1.254
A1-24"	Nozzle	207.00	169.56	1.221
B1-24"	Nozzle	207.00	169.56	1.221
M1-24"	Nozzle	207.00	169.56	1.221

Minimum

1.221

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Dish bottom	207.00	165.12	1.254
shell-2	207.00	165.12	1.254
shell-3	207.00	165.12	1.254
shell 4	207.00	165.12	1.254
Channel-1	207.00	165.12	1.254
Channel-2	207.00	165.12	1.254
Top dish	207.00	165.12	1.254

Minimum

1.254

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	161.00	157.40	1.023



Tubesheet Material	207.00	120.60	1.716
--------------------	--------	--------	-------

Minimum 1.023

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
D1-3"	5.18	294.49	0.018
A2-24 "	26.13	294.50	0.089
B2-24 "	26.13	294.50	0.089
A1-24"	15.73	294.50	0.053
B1-24"	15.73	294.50	0.053
M1-24"	15.74	294.50	0.053

Elements Suitable for Test Pressure.

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**Input Echo, Tubesheet Number 1, Description: Tube sheet****Shell Data:****Main Shell Description: sheel 4**

Shell Test Pressure		22.16 bars
Shell Thickness	ts	34.0000 mm.
Shell Internal Corrosion Allowance	cas	0.0000 mm.
Shell External Corrosion Allowance	casext	0.0000 mm.
Inside Diameter of Shell	Ds	2032.000 mm.
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-387 11

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Shell Material UNS Number		K11789
Shell Allowable Stress at Temperature	Ss	207.00 N./mm ²
Shell Allowable Stress at Ambient		207.00 N./mm ²

Channel Data:**Channel Description: Channel-1**

Channel Type:		Cylinder
Channel Test Pressure		21.93 bars
Channel Thickness	tc	136.0000 mm.
Channel Internal Corrosion Allowance	cac	0.0000 mm.
Channel External Corrosion Allowance	cacext	0.0000 mm.
Inside Diameter of Channel	Dc	1964.000 mm.
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-387 11

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Channel Material UNS Number		K11789
Channel Allowable Stress at Temperature	Sc	207.00 N./mm ²
Channel Allowable Stress at Ambient		207.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	559
Tube Wall Thickness	et	2.7000 mm.
Tube Outside Diameter	D	38.1000 mm.
Total Straight Tube Length	Lt	4500.00 mm.
Straight Tube Length (bet. inner tubsht faces) L		4400.00 mm.
Design Temperature of the Tubes		21.11 °C
Tube Material		SB-163
Tube Material UNS Number		N08825
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		161.00 N./mm ²
Tube Allowable Stress At Ambient		161.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	240.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	75.0000 mm.
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	2.1100 mm.
Groove Weld Leg	ag	2.1100 mm.
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		4.21.2
Is the Tube Inset		No

Radius to Outermost Tube Hole Center	ro	921.615 mm.
Largest Center-to-Center Tube Distance	U1	800.0000 mm.
Length of Expanded Portion of Tube	ltx	0.0000 mm.
Tube-side pass partition groove depth	hg	0.0000 mm.
Area of Tube Layout	Ap	25623.5 cm ²
Perimeter of Tube Layout	Cp	5952.6 mm.

Tubesheet Data:



Tubesheet TYPE: U-tube, Integral both sides, Conf. a

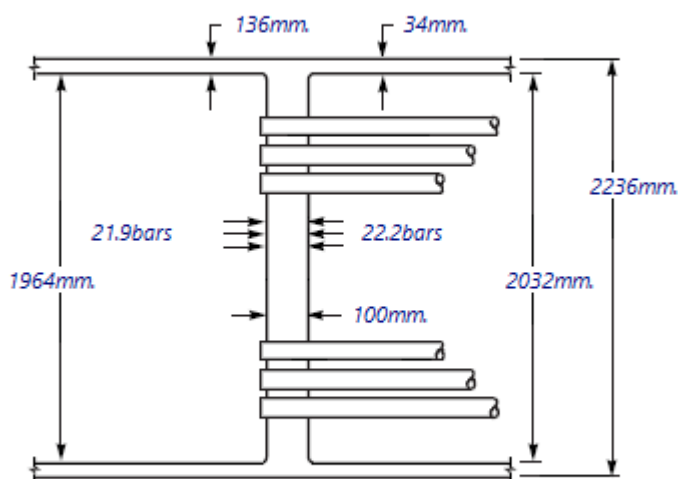
Tubesheet Design Metal Temperature	T	21.11 °C
Tubesheet Material		SA-336 F21
Tubesheet Material UNS Number		K31545
Tubesheet Allowable Stress at Temperature	S	207.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	207.00 N./mm ²
Thickness of Tubesheet	h	100.0000 mm.
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000 mm.
Tubesheet Corr. Allowance (Channel side)	Catc	0.0000 mm.
Tubesheet Outside Diameter	A	2236.000 mm.

Area of the Untubed Lanes	AL	0.0 cm ²
---------------------------	----	---------------------

Junction Stress Reduction option	Perform Plastic Calculation
Perform Differential Pressure Design	No
Run Multiple Load Cases	Yes

Tubesheet Integral with	Both
Tubesheet Extended as Flange	No

Tubesheet Integral w/Shell and Channel



Configuration a:

ASME Division 2 Tubesheet Results per Part 4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Material Group C

Elastic Mod. at Design Temperature	21.1 °C	0.20408E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20408E+09 KPa.

Channel - TM-1 Material Group C

Elastic Mod. at Design Temperature	21.1 °C	0.20408E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20408E+09 KPa.

Tubes - TM-4 N08825

Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.19305E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19305E+09 KPa.



TubeSheet - TM-1 Material Group D

Elastic Mod. at Design Temperature	21.1 °C	0.21097E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.21097E+09 KPa.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i$ Equation 4.3.1
 $= 0.5 \cdot 32.7 \cdot (\exp(21.93/(161.0 \cdot 1.0)) - 1) + 0$
 $= 0.2242 + 0$
 $= 0.2242 \text{ mm.}$

Tube Required Thickness under External Pressure (Shellside pressure) :

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$

$$= 5321.615 / \sqrt{(19.05 \cdot 2.7)}$$

$$= 742.017$$

Compute the Factor [Ch]:

Since $M_x > 2 \cdot (D_o/t)^{0.94}$, $Ch = 0.55 / (D_o/t)$
 $= 0.55 / (38.1/2.7)$
 $= 0.0390$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y(t / D_o)$$

$$= 1.6 \cdot 0.039 \cdot 0.19305E+09(2.7/38.1)$$

$$= 853.203 \text{ N./mm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / F_y$$

$$= 853.2/240.0$$

$$= 3.555$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 240.0/586.1$$

$$= 0.409$$

Ratio [AE]:

$$= Fhe / E$$

$$= 853.2/0.19305E+09$$

$$= 0.004419$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{ps} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{ps}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.409) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.194$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$

$$= 240.0(1 + 0.002) / \ln(1 + 0.002)^{0.194347}$$

$$= 804.806 \text{ N./mm}^2$$

Curve Fitting Exponent for, Stainless Steel and Nickel Base Alloys [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.75(1.0 - 0.409)$$

$$= 0.443$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (586.1 \cdot \exp(0.443)) / (0.443^{0.443})$$

$$= 1309.018 \text{ N./mm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:



$$\begin{aligned}
 &= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5 \\
 &= 1.5 \cdot 0.4091.5 - 0.5 \cdot 0.4092.5 - 0.4093.5 \\
 &= 0.295
 \end{aligned}$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	293	-0.963	0.485E-05	0.122E-05	0.131E-04	0.502E-05	0.404E+07	0.680E-01
2	146	-3.828	0.274E-06	0.274E-06	0.549E-05	-0.547E-05	0.922E+08	0.283E-02
3	219	-2.395	0.147E-05	0.141E-05	0.914E-05	-0.842E-05	0.243E+08	0.463E-02
4	183	-3.112	0.691E-06	0.685E-06	0.727E-05	-0.715E-05	0.497E+08	0.731E-03
5	201	-2.754	0.103E-05	0.101E-05	0.819E-05	-0.790E-05	0.351E+08	0.131E-02
6	192	-2.933	0.846E-06	0.835E-06	0.773E-05	-0.754E-05	0.419E+08	0.172E-03
7	187	-3.022	0.766E-06	0.757E-06	0.750E-05	-0.734E-05	0.456E+08	0.306E-03
8	190	-2.978	0.805E-06	0.795E-06	0.761E-05	-0.744E-05	0.437E+08	0.739E-04
9	191	-2.955	0.826E-06	0.815E-06	0.767E-05	-0.749E-05	0.428E+08	0.472E-04
10	190	-2.966	0.815E-06	0.805E-06	0.764E-05	-0.747E-05	0.433E+08	0.138E-04
11	190	-2.961	0.820E-06	0.810E-06	0.765E-05	-0.748E-05	0.430E+08	0.166E-04
12	190	-2.964	0.818E-06	0.807E-06	0.765E-05	-0.747E-05	0.431E+08	0.134E-05
13	190	-2.965	0.817E-06	0.806E-06	0.764E-05	-0.747E-05	0.432E+08	0.625E-05
14	190	-2.964	0.817E-06	0.807E-06	0.764E-05	-0.747E-05	0.432E+08	0.246E-05
15	190	-2.964	0.818E-06	0.807E-06	0.765E-05	-0.747E-05	0.432E+08	0.557E-06
16	190	-2.964	0.818E-06	0.807E-06	0.765E-05	-0.747E-05	0.431E+08	0.392E-06
17	190	-2.964	0.818E-06	0.807E-06	0.765E-05	-0.747E-05	0.432E+08	0.823E-07

Factor of Safety [FS]:

$$\begin{aligned}
 &\text{Since } F_{ic} > 0.55 F_y \text{ and } F_{ic} < F_y \\
 &= 2.407 - 0.741 \cdot F_{ic} / F_y \\
 &= 2.407 - 0.741 \cdot 190.72/240.0 \\
 &= 1.818
 \end{aligned}$$

Allowable Circumferential Compressive Stress [Fha]:

$$\begin{aligned}
 &= F_{ic}/FS \\
 &= 190.719/1.818 \\
 &= 104.897 \text{ N./mm}^2
 \end{aligned}$$

Allowable External Pressure at the given Thickness [Pa]:

$$\begin{aligned}
 &= 2 \cdot F_{ha} (t / D_o) \\
 &= 2 \cdot 1048.912 (2.7/38.1) \\
 &= 148.665 \text{ bars}
 \end{aligned}$$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.9376 mm.
Allowable Internal Pressure at Corroded thickness	246.05 bars
Required Internal Design Pressure	21.93 bars
Allowable External Pressure at Corroded thickness	148.66 bars
Required External Design Pressure	22.16 bars
Required Thickness due to Shell Side pressure	0.9376 mm.

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$\begin{aligned}
 &= 2 \cdot r_o + d_t \\
 &= 2 \cdot 921.6149 + 38.1 \\
 &= 1881.330 \text{ mm.}
 \end{aligned}$$

Basic ligament efficiency for shear [μ]:



$$\begin{aligned}
 &= (p - dt) / p \\
 &= (75.0 - 38.1) / 75.0 \\
 &= 0.492
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$$\begin{aligned}
 &= D_s / D_o \text{ (Configurations a, b, c)} \\
 &= 2032.0 / 1881.33 \\
 &= 1.0801
 \end{aligned}$$

Ratio [pc]:

$$\begin{aligned}
 &= D_c / D_o \text{ (Configurations a, e, f)} \\
 &= 1964.0 / 1881.33 \\
 &= 1.0439
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$$\begin{aligned}
 &= l_{tx} / h \\
 &= 0 / 100.0 \\
 &= 0.000 \text{ (must be } 0 \leq p \leq 1)
 \end{aligned}$$

Effective tube hole Diameter [d*]:

$$\begin{aligned}
 &= \max(dt - 2t_t (E_t T / E) \cdot (StT / S) \cdot (p); dt - 2t_t) \\
 &= \max(38.1 - 2 \cdot 2.7 (0.19305E+09 / 2.1097E+09) \cdot (160 / 206) \cdot (0); 38.1 - 2 \cdot 2.7) \\
 &= 38.1000 \text{ mm.}
 \end{aligned}$$

Effective tube pitch [p*]:

$$\begin{aligned}
 &= p / \sqrt{(1 - 4 \cdot \min(AI; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2))} \\
 &= 75.0 / \sqrt{(1 - 4 \cdot \min(0; 4 \cdot 1881.33 \cdot 75.0) / (\pi \cdot 1881.33^2))} \\
 &= 75.0000 \text{ mm.}
 \end{aligned}$$

Effective ligament efficiency for bending [μ^*]:

$$\begin{aligned}
 &= (p^* - d^*) / p^* \\
 &= (75.0 - 38.1) / 75.0 \\
 &= 0.4920
 \end{aligned}$$

 $h/p = 1.333333$; $\mu^* = 0.492000$; $E^* = 118457968 \text{ KPa.}$ $E^*/E = 0.561479$; $v^* = 0.292221$ **ASME Section VIII-2 Part 4.18.7.4 Step 4:**Shell coefficient [β_s]:

$$\begin{aligned}
 &= ((12(1 - v_s^2))^{1/4}) / \sqrt{(D_s + t_s)t_s} \\
 &= ((12(1 - 0.3^2))^{1/4}) / \sqrt{(2032.0 + 34.0)34.0} \\
 &= 0.0069 \text{ 1/mm.}
 \end{aligned}$$

Shell coefficient [k_s]:

$$\begin{aligned}
 &= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - v_s^2)) \\
 &= 0.007 \cdot 2040801 \cdot 34.0^3 / (6(1 - 0.3^2)) \\
 &= 100761800.0000 \text{ bars} \cdot \text{mm.}^2
 \end{aligned}$$

Shell coefficient [λ_s]:

$$\begin{aligned}
 &= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2) \\
 &= 6(2032.0 + 34.0) \cdot 0.10076E+09 / 100.0^3 \cdot (1 + 100.0 \cdot 0.007 + (100.0 \cdot 0.007)^2 / 2) \\
 &= 2399541.2 \text{ bars}
 \end{aligned}$$

Shell coefficient [δ_s]:

$$\begin{aligned}
 &= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot v_s / (2(D_s + t_s))) \\
 &= 2032.0^2 / (4 \cdot 0.20408E+09 \cdot 34.0) \cdot (1 - 2032.0 \cdot 0.3 / (2(2032.0 + 34.0))) \\
 &= 0.12681E+00 \text{ mm./N./mm}^2
 \end{aligned}$$

Parameter [w_s]:

$$\begin{aligned}
 &= p_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s) \\
 &= 1.0801 \cdot 0.10076E+09 \cdot 0.0069 \cdot 0.126812 (1 + 100.0 \cdot 0.0069) \\
 &= 15959.5127 \text{ mm.}^2
 \end{aligned}$$

Channel coefficient [β_c]:

$$= \left(\left(12(1 - v_c^2) \right)^{1/4} / \sqrt{((D_c + t_c)t_c)} \right) \\ = \left(\left(12(1 - 0.3^2) \right)^{1/4} / \sqrt{((1964.0 + 136.0)136.0)} \right) \\ = 0.0034 \text{ 1/mm.}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - v_s^2)) \\ = 0.003 \cdot 2040801 \cdot 136.0^3 / (6(1 - 0.3^2)) \\ = 3198168832.0000 \text{ bars} \cdot \text{mm.}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + \\ h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta) \\ = 6(1964.0 + 136.0 / \cos(0)) \cdot 0.31982E+1 / 100.0^3 \cdot (1.0 + \\ 100.0 \cdot 0.003 \cdot (1.0 + 1.0) / 2 + (100.0 \cdot 0.003)^2 / 2 \cdot 1.0) \\ = 56335404.0 \text{ bars}$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c))) \\ = 1964.0^2 / (4 \cdot 0.20408E+09 \cdot 136.0) \cdot (1 - 1964.0 \cdot 0.3 / (2(1964.0 + 136.0))) \\ = 0.29868E-01 \text{ mm./N./mm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

Parameter [ω_c]:

$$= \rho_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + CqDelta \cdot h \cdot \beta_c^2) - \\ \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)] \\ = 1.044 \cdot 0.31982E+1 [0.029868(1.0 + 0.003 \cdot 1.0 \cdot 100.0) - \\ 0(1.0 / 100.0 + 1.0 \cdot 0.003 / 2)] \\ = 45461.8164 \text{ mm.}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o \\ = 2236.0 / 1881.33 \\ = 1.1885$$

Coefficient [F]:

$$= (1 - v_s) / E_s \cdot (\lambda_s + \lambda_c + E \cdot \ln(K)) \\ = (1 - 0.29) / 1.1846E+09 \cdot (2399541 + 56335404 + \\ 0.21097E+09 \cdot \ln(1.19)) \\ = 35.3115$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= D_o^2 / 16 \cdot [(\rho_s - 1) \cdot (\rho_s^2 + 1) \cdot P_s - (\rho_c - 1) \cdot (\rho_c^2 + 1) \cdot P_t] \\ = 1881.33^2 / 16 \cdot [(1.08 - 1) \cdot (1.08^2 + 1) \cdot 22.163 - \\ (1.044 - 1) \cdot (1.044^2 + 1) \cdot 21.93] \\ = 405234.6875 \text{ bars} \cdot \text{mm.}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= Mts + \omega_c \cdot P_t - \omega_s \cdot P_s \\ = 405234.688 + 45461.816 \cdot 21.93 - 15959.513 \cdot 22.163 \\ = 1048478.6250 \text{ bars} \cdot \text{mm.}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= (M_s - D_o^2 / 32 \cdot F(P_s - P_t)) / (1 + F) \\ = (1048478 - 1881.33^2 / 32 \cdot 35.311(22.16 - 21.93)) / (1 + 35.31) \\ = 3725.9094 \text{ bars} \cdot \text{mm.}^2$$



Maximum bending moment acting on center of tubesheet [Mo]:

$$\begin{aligned}
 &= M_p + D_o^2/64 \cdot (3 + \nu) \cdot (P_s - P_t) \\
 &= 3725.91 + 1881.33^2/64 \cdot (3 + 0.292) \cdot (22.16 - 21.93) \\
 &= 46295.7461 \text{ bars}\cdot\text{mm}^2
 \end{aligned}$$

Maximum bending moment acting on tubesheet [M]:

$$\begin{aligned}
 &= \max(|M_p|; |M_o|) \\
 &= \max(|3725.909|; |46295.746|) \\
 &= 46295.7461 \text{ bars}\cdot\text{mm}^2
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$\begin{aligned}
 &= 6 \cdot M / (\mu \cdot (h - h_g')^2) \\
 &= 6 \cdot 46295.746 / (0.492 \cdot (100.0 - 0)^2) \\
 &= 5.6462 \text{ N./mm}^2
 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$\begin{aligned}
 &= 2 \cdot \beta T \cdot S_y \\
 &= 2 \cdot 0.95 \cdot 310.0 \\
 &= 588.99 \text{ N./mm}^2
 \end{aligned}$$

Tubesheet bending stress at final thickness [σ]:

$$\begin{aligned}
 &= 6 \cdot M / (\mu \cdot (h - h_g')^2) \\
 &= 6 \cdot 25846.502 / (0.492 \cdot (7.3157 - 0)^2) \\
 &= 588.9891 \text{ N./mm}^2
 \end{aligned}$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$\begin{aligned}
 &= h + C_{ats} + C_{atc} \\
 &= 7.3157 + 0 + 0 \\
 &= 7.3157 \text{ mm.}
 \end{aligned}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$\begin{aligned}
 &= \max(H_{reqB}; H_{reqS}) \\
 &= \max(7.3157; 0.0868) \\
 &= 7.3157 \text{ mm.}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [| $P_s - P_t$ |]:

$$\begin{aligned}
 &= |P_s - P_t| \\
 &= |22.16 - 21.93| \\
 &= 0.234 \text{ bars}
 \end{aligned}$$

Shear Stress check [τ_{limit}]:

$$\begin{aligned}
 &= 4 \cdot \mu \cdot h/D_o \cdot 0.8 \cdot \beta T \cdot S_y \\
 &= 4 \cdot 0.492 \cdot 100.0/1881.33 \cdot 0.8 \cdot 0.95 \cdot 309.994 \\
 &= 24.645 \text{ N./mm}^2
 \end{aligned}$$

| $P_s - P_t$ | is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$\begin{aligned}
 &= 1/(4 \cdot \mu \cdot (0.8 \cdot \beta T \cdot S_y)) \cdot (4 \cdot A_p/C_p) \cdot |P_s - P_t| + C_{ats} + C_{atc} \\
 &= 1/(4 \cdot 0.492 \cdot 235.596) \cdot (4 \cdot 25623.5/5952.57) \cdot |22.16 - 21.93| + 0 \\
 &= 0.0868 \text{ mm.}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Shell thickness of: 34.000mm.

Min. Shell len. adjacent to the tubesheet is: 473.123 mm.

Note: For a given Channel thickness of: 136.000 mm.

Min. Channel len. adjacent to the tubesheet is: 930.278mm.

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$



$$= \pi (38.1 - 2.7) 2.7$$

$$= 3.003 \text{ cm}^2$$

Tube Weld Size Results per VIII-2 4.21.2:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 2.7 (38.1 - 2.7) 227.996$$

$$= 6980.961 \text{ Kgf}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 2.11 (38.1 + 0.67 \cdot 2.11) 228.0$$

$$= 5176.0366 \text{ Kgf}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 227.996 / 227.996$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 2.11 (38.1 + 0.67 \cdot 2.11) 228.0$$

$$= 3349.2002 \text{ Kgf}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{(0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d} - 0.75 \cdot d_o \right)$$

$$= 2 \left(\sqrt{(0.75 \cdot 38.1)^2 + 1.07 \cdot 2.7 \cdot (38.1 - 2.7) 1.0 \cdot 1.0} - 0.75 \cdot 38.1 \right)$$

$$= 3.473 \text{ mm.}$$

Minimum Required Fillet Weld Leg [afr]: 1.7367 mm.

Minimum Required Groove Weld Leg [agr]: 1.7367 mm.

Tube-Tubesheet Jt allowable, 6980.96 is $\geq$ tube strength 6980.96 Kgf**Note:** This tube-tubesheet joint is a Full Strength joint**Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):**

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	5.6 $\leq$	589.0 N./mm ²	Ok
Tubesheet shear stress	0.2 $\leq$	235.6 N./mm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm.)	Required	Actual	P/F
Tubesheet Thickness :	7.316	100.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.737	2.110	Ok
Tube-Tubesheet Groove Weld Leg :	1.737	2.110	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025**Results for 3 Load Cases:**

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Stresses Bend Allwd Shear Allwd	Case Type	Pass/Fail
D1uc	70.557 ...	290	589 19 236	Ps+Pt D1	Ok
D2uc	70.770 ...	289	589 19 236	Ps+Pt D2	Ok
D3uc	7.316 ...	6	589 ... 236	Ps+Pt D3	Ok



Max: 70.7704 ... mm. 0.492 0.082 (Str. Ratio)

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures
derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

Case#	Shell	Channel	Shell Stress	Shell Allwd	Channel Stress	Channel Allwd	Pass Fail
D1uc	...	...	...	442	...	442	Ok
D2uc	...	...	...	442	...	442	Ok
D3uc	...	...	...	442	...	442	Ok
Max	...	...	...	...	...	...	...

Note:

Because there was no net pressure differential across the tubesheet due to the specified pressures psd,min and ptd,min; load case D4 was not processed as the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm.)	Required	Actual	P/F
Tubesheet thickness :	70.770	100.000	Ok
Tube thickness :	0.938	2.700	Ok
Tube-tubesheet fillet weld leg :	1.737	2.110	Ok
Tube-tubesheet groove weld leg :	1.737	2.110	Ok

Given Shell Thickness: 34.0000 mm.

Given Channel Thickness: 136.0000 mm.

Min Shell length of thk, (34.000) adj. to tubesheet: 473.123 mm.

Min Channel length of thk, (136.000) adj. to tubesheet: 930.278 mm.

Note: This is a full strength Tube to Tubesheet Joint.

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**External Pressure Calculation Results Summary:****ASME Code, Section VIII Division 2, 2025****External Pressure Calculations:**

From	To	Section Length mm.	Outside Diameter mm.	Corroded Thickness mm.	Factor A	Factor B N./mm ²
10	20	No Calc	2100	16.5	No Calc	No Calc
20	30	5343.97	2100	32.5	No Calc	No Calc
30	40	5343.97	2100	32.5	No Calc	No Calc
40	Tsht	5343.97	2100	32.5	No Calc	No Calc
Tsht	50	No Calc	...	32.5	No Calc	No Calc
50	60	6350.03	2236	134.5	No Calc	No Calc
60	70	6350.03	2236	134.5	No Calc	No Calc
70	80	No Calc	2236	66.5	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
10	20	18	4.3497	1.03419	22.804
20	30	34	11.0554	1.03419	18.7361
30	40	34	11.0554	1.03419	18.7361
40	Tsht	34	11.0554	1.03419	18.7361
Tsht	50	34	No Calc	1.03419	No Calc
50	60	136	12.1518	1.03419	141.015
60	70	136	12.1518	1.03419	141.015
70	80	68	4.53423	1.03419	131.674

Minimum

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External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	5343.97	25399974	No Calc	No Calc
30	40	5343.97	25399974	No Calc	No Calc
40	Tsht	5343.97	25399974	No Calc	No Calc
Tsht	50	No Calc	No Calc	No Calc	No Calc
50	60	6350.03	25399974	No Calc	No Calc
60	70	6350.03	25399974	No Calc	No Calc
70	80	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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**Element and Detail Weights:**

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	1165.51	2302372	1087.6	2312426	116.551
20	30	2565.55	2968293	2454.14	2982670	256.555
30	40	3420.73	5212168	3272.19	5231337	342.073
40	50	2565.55	3909126	2454.14	3923503	256.555
50	60	20862.2	9090167	20646.8	9117958	2086.22
60	70	20862.2	9090167	20646.8	9117958	2086.22
70	80	4259.6	2424958	4178.95	2435365	425.96
Total		55701	34997256.00	54740	35121220.00	5570



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm.	Y Offset, Dtl. Cent. mm.	Z Offset Dtl. Cent. mm.	Description
10	Insl	153.745	...	-525	...	Ins: 10
10	Nozl	33.8456	...	-1173	...	D1-3"
20	Insl	229.385	...	750	...	Ins: 20
20	Nozl	306.935	-1166	450.005	...	A2-24 "
20	Wght	970	...	...	...	Btm dish SF
30	Insl	305.846	...	1000	...	Ins: 30
40	Insl	229.385	...	750	...	Ins: 40
40	Nozl	306.935	1166	750.004	...	B2-24 "
50	Insl	487.001	...	1500	...	Ins: 50
50	Lini	470.183	...	1500	...	cld-2
50	Nozl	1136.09	1132	2000	...	A1-24"
50	Lugs	2461.54	...	1225	...	Support Lugs
50	Wght	6000	...	-0.0052334	...	Internal dish+
50	Wght	920	...	-0.0052334	...	Clip weight
60	Insl	487.001	...	1500	...	Ins: 60
60	Lini	470.183	...	1500	...	cld-1
60	Nozl	1136.09	-1132	2000	...	B1-24"
60	Wght	5700	...	-0.0046512	...	Top dish SF
60	Wght	2700	...	-0.0046512	...	Lug weight
70	Insl	173.234	...	559	...	Ins: 70
70	Lini	175.723	...	559	...	clad
70	Nozl	4077.15	...	1191.01	...	M1-24"
40	FTsh	2804.43	...	1550	...	Tube sheet
40	Tube	8471.6	...	-750	...	Tubes

Total Weight of Each Detail Type:

Insulation	2065.6
Cladding & Overlay	1116.1
Nozzles	6997.0
Lugs	2461.5
Weights	16290.0
Exchanger Components	11276.0
Liquid in Tubes	3.1

Sum of the Detail Weights 40209.4 kg.

Weight Summation Results: (kg.)



	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	61271.4	61271.4	61271.4	61271.4	61271.4	61271.4
Cladding	1116.1	1116.1	1116.1	1116.1	1116.1	1116.1
Nozzles	6997.0	6997.0	6997.0	6997.0	6997.0	6997.0
Lugs	2461.5	2461.5	2461.5	2461.5	2461.5	2461.5
Wld Weights	16290.0	16290.0	16290.0	16290.0	16290.0	16290.0
Exchanger	11276.0	11276.0	11276.0	11276.0	11276.0	11276.0
Insulation	...	...	2065.6	2065.6	2065.6	2065.6
Tube Ope Lqd	...	...	...	...	...	3.1
Test Liquid	...	34844.3	...	...	...	...
Tube Tst Lqd	...	3085.8	...	...	...	...
Totals	99412.1	137342.2	101477.7	101477.7	101477.7	101480.7

Field Installation Options:**Miscellaneous Weight Percent: 10.0 %**

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	99412.1 kg.
Shop Test Wt.	- Fabricated weight + water (full)	137342.2 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	101477.7 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	101477.7 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	101477.7 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	101480.7 kg.
Field Test Wt.	- Empty weight + test medium (full)	139407.8 kg.

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	47750.4 kg.
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	56410.6 kg.
Mass of the Upper 1/3 of the Vertical Vessel (empty)	48131.3 kg.

Exchanger Tube Data

Volume of Exchanger tubes :	3087706.2 cm ³
Weight of Ope Liq in tubes :	3.1 kg.
Weight of Water in tubes :	3085.8 kg.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	69800.9	66917.4
20	30	98960.2	95755.8
30	40	131947	127674
40	50	98960.2	95755.8
50	60	210738	185103
60	70	210738	185103
70	80	79380.3	69272.1

Total	900524.5	825580.7 cm ²
Total	969.3	888.6 ft ²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kg.	Total. Ele. Oper. Wgt. kg.	Total. Ele. Hydro. Wgt. kg.	Total Dtl. Offset Mom. Kg-m.	Oper. Wgt. No Liquid kg.
10	20	1469.66	1469.66	3770.62	...	1469.66

20	30	4328.42	4328.42	7294.9	...	4328.42
30	40	4068.65	4068.65	9277.64	...	4068.65
40	50	14634.4	14637.5	21627	...	14634.4
50	Lugs	10653.9	10653.9	13682.1	...	10653.9
Lugs	60	21307.8	21307.8	27364.2	...	21307.8
60	70	33441.6	33441.6	42526.3	...	33441.6
70	80	9111.67	9111.67	11535.1	...	9111.67

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kg.	Cumulative Oper. Wgt. kg.	Cumulative Hydro. Wgt. kg.
10	20	...	...	...
20	30	-1469.66	-1469.66	-3770.62
30	40	-5798.08	-5798.08	-11065.5
40	50	-9866.73	-9866.73	-20343.2
50	Lugs	-24501.2	-24504.3	-41970.2
Lugs	60	63861.1	63861.1	81425.6
60	70	42553.3	42553.3	54061.4
70	80	9111.67	9111.67	11535.1

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. Kg-m.	Cumulative Oper. Mom. Kg-m.	Cumulative Hydro. Mom. Kg-m.
10	20	...	...	...
20	30	...	...	...
30	40	...	...	...
40	50	...	...	...
50	Lugs	...	...	...
Lugs	60	...	...	...
60	70	...	...	...
70	80	...	...	...

Nozzle Flange MAWP Results: (bars & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press.	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
M1-24"	202.39	255.10	286	1500	GR 1.1	...	...	...	...

Shellside Flange Rating

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 202.386 bars
Lowest Flange Pressure Rating was (Amb)[TubeSide]: 255.100 bars

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Warning:

There are nozzles in this model, but no flange MAWPs were de-rated. Be sure this is what you intended. There is a check box in the nozzle dialog that instructs PV Elite to perform the de-rating for each nozzle flange. See ASME VIII-2, 4.16.12 for more information.

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 38.1732 Hz.

The Natural Frequency for the Vessel (Ope...) is 38.1729 Hz.

The Natural Frequency for the Vessel (Filled) is 28.2228 Hz.

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-22
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 180 Km/hr
Surface Roughness Category	D: Flat, unobstructed
Type of Surface	Moderately Smooth
Base Elevation	7500 mm.
Percent Wind for Hydrotest	33.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm.
Distance Upwind of Crest Lh	0 mm.
Distance from Crest to the Vessel x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q)/(1 + 1.7 \cdot gV \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1+1.7 \cdot 3.4 \cdot 0.161 \cdot 0.945)/(1+1.7 \cdot 3.4 \cdot 0.161)))$
 $= \min(0.85, 0.901)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	38.173 Hz
Natural Frequency of Vessel (Empty)	38.173 Hz
Natural Frequency of Vessel (Test)	28.223 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.572
Structure Height to Diameter ratio	5.295
Height to top of Structure	12327.999 mm.

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:
Exposure Category: D from Table 26.11-1
 $\alpha = 11.5$; $Zg = 589788$ mm.

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 667.697 + 7500.0 = 8167.697$ mm.
 $= 26.797$ ft. Imperial Units

Velocity Pressure coefficient evaluated at height z $[Kz]$:
Because z (26.797 ft.) > 15 ft.
 $= 2.41(z / Zg)^{2/\alpha}$
 $= 2.41(26.797/1935.0)^{2/11.5}$
 $= 1.145$

Type of Hill: No Hill

Wind Directionality Factor $[Kd]$:
 $= 1$, per Table 26.6-1

As there is No Hill Present: [Kzt]:

$K1 = 0, K2 = 0, K3 = 0$

Topographical Factor [Kzt]:

$= (1 + K1 \cdot K2 \cdot K3)^2$
 $= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$
 $= 1.0$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$
 $= \max(16, 0.00256 \cdot 1.145 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 111.85^2)$
 $= 36.7 \text{ psf } [179.034] \text{ Kgs/m}^2$

Force on the first element [F]:

$= qz \cdot G \cdot Cf \cdot \text{WindArea}$
 $= 36.669 \cdot 0.85 \cdot 0.572 \cdot 27.985$
 $= 498.6 \text{ lbs. } [226.1] \text{ Kgf}$

Element	Hgt (z) mm.	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
Dish bottom	8167.7	0.000	0.000	0.000	1.145	1.000	179.034
shell-2	9282.0	0.000	0.000	0.000	1.171	1.000	183.061
shell-3	11032.0	0.000	0.000	0.000	1.206	1.000	188.643
shell 4	12782.0	0.000	0.000	0.000	1.238	1.000	193.536
Channel-1	15132.0	0.000	0.000	0.000	1.275	1.000	199.301
Channel-2	18132.0	0.000	0.000	0.000	1.315	1.000	205.670
Top dish	20153.2	0.000	0.000	0.000	1.340	1.000	209.485

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

Cf - Correction factor for natural frequency
D - Average outer insulated diameter of vessel, mm.
Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
Dr - Average outer insulated of top half of vessel, mm.
f - Natural frequency of vibration (Hertz)
f1 - Natural frequency of bare vessel based on a unit value of $(D/L^2)(10^4)$
L - Total height of structure mm.
Lc - Total length of conical section(s) of vessel, mm.
tb - Uncorroded plate thickness at bottom of vessel, mm.
V30 - Design Wind Speed provided by user, Km/hr
Vc - Critical wind velocity, Km/hr
Vw - Maximum wind speed at top of structure, Km/hr
W - Total corroded weight of structure, Kgf
Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. Kgf
Z - Maximum amplitude of vibration at top of vessel, mm.
Dl - Logarithmic decrement (taken as 0.03 for Welded Structures)
Vp - Vib. Chance, $\leq 0.320\text{E-}06$ (High); $0.320\text{E-}06 < 0.400\text{E-}06$ (Probable)
P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5
:: $0.0/11100.0 = 0.0$

:: $[D / L^2] \cdot 10^4 < 8$ (English Units, ft.)
:: $[7.78/36.42^2] \cdot 10^4 = 58.694$ [Geometry Violation]

Compute the vibration possibility. If $Vp > 0.400\text{E-}06$ no chance. [Vp]:

$= W / (L \cdot Dr^2)$
 $= 100424 / (11100.0 \cdot 2320.848^2)$
 $= 0.16797\text{E-}05$



Since Vp is > 0.400E-06 no further vibration analysis is required !

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm.	Pressure Kgs/m ²	Force Kgf
Btm dish SF	0.00	7500.00	179.03	0.00
Internal dish	0.00	12599.99	193.03	0.00
Clip weight	0.00	12599.99	193.03	0.00
Top dish SF	0.00	15600.00	200.29	0.00
Lug weight	0.00	15600.00	200.29	0.00

The Natural Frequency for the Vessel (Ope...) is 38.1729 Hz.

Wind Load Calculation:

From	To	Wind Height mm.	Wind Diameter mm.	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	8167.7	2853.6	25998.9	179.034	248.759
20	30	9282	2853.6	42804	183.061	418.761
30	40	11032	2853.6	57072	188.643	575.375
40	50	12782	2853.6	42804	193.536	442.724
50	60	15132	3020.88	90626.4	199.301	965.274
60	70	18132	3020.88	90626.4	205.67	996.119
70	80	20153.2	3020.88	29136.4	209.485	326.193

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code		ASCE/SEI 7-22	
Importance Factor		1.500	
Force Modification Factor R		3.000	
Site Class		D	
Component Elevation Ratio	z/h	0.500	
Amplification Factor	Ap	1.000	
Force Factor		0.000	
Consider Vertical Acceleration		Yes	
Minimum Acceleration Multiplier		0.000	
User Value of Sds (used if > 0)		0.000	
User Value of Sd1 (used if > 0)		0.000	
Component resonance duct. fact.	[CAR]	1.000	
Component strength factor	[Rpo]	1.500	
Overstrength factor	[Omega0]	2.000	
Response Acceleration Factor	[SMS]	0.326	
Response Acceleration Factor	[SM1]	0.139	
Building Importance Factor	[Ie]	1.000	
Site Latitude		0.000	deg
Site Longitude		0.000	deg

Seismic Analysis Results:

$$SDS = 2/3 * Sms = 2/3 * 0.326 = 0.217$$

$$SD1 = 2/3 * Sm1 = 2/3 * 0.139 = 0.093$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
= Ct • hnx where Ct = 0.020, x = 0.75 and hn = Structural Height (ft.)
= 0.020(39.86220.75)
= 0.317 seconds

The Coefficient Cu from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
= (1/Natural Frequency) = (1/38.173)
= 0.026

Check the Value of T which is the smaller of Cu•Ta and T:
= Minimum Value of (1.7 • 0.317, 0.026) per 12.8.2
= 0.026

Loads on Building/Structure Supported Equipment:

Factor [a1]:
= min(1/Ta, 2.5)
= min(1/0.026, 2.5)
= 2.500

Factor [a2]:
= max(0, 1 - (0.4/Ta)²)
= max(0, 1 - (0.4/0.026)²)
= 0.000

Factor for force amplification factor [Hf]:
= 1 + (a1 • zh) + (a2(zh)¹⁰)
= 1 + (2.5 • 0.5) + (0.0(0.5)¹⁰)
= 2.250

Structure ductility reduction factor [Rp]:
= max(1.3, $\sqrt{(1.1 \cdot R / (Ie \cdot \omega O))}$)
= max(1.3, $\sqrt{(1.1 \cdot 3.0 / (1.0 \cdot 2.0))}$)
= 1.300



Seismic Force per equation 13.3-1, [Fp]:

$$= 0.4 \cdot SDS \cdot I_p \cdot W_p (H_f / R_\mu) (CAR / RPO)$$

$$= 0.4 \cdot 0.217 \cdot 1.5 \cdot 218296 (2.25/1.3) (1.0/1.5)$$

$$= 14898.584 \text{ Kgf}$$

Maximum value of Fp per equation 13.3-2:

$$= 1.6 \cdot Sds \cdot I \cdot W$$

$$= 1.6 \cdot 0.217 \cdot 1.5 \cdot 99019 = 51648.42 \text{ Kgf}$$

Minimum value of Fp per equation 13.3-3:

$$= 0.3 \cdot 0.22 \cdot 1.5 \cdot 99019$$

$$= 9684.080 \text{ Kgf}$$

Total Base Shear V = Fp, [V]:

$$= 14898.584 \text{ Kgf}$$

Vertical load per 12.4-4, [Ev]:

$$= 0.2 \cdot Sds \cdot W$$

$$= 0.2 \cdot 0.217 \cdot 99019$$

$$= 4304.0 \text{ Kgf}$$

Distribute the Base shear force to each element according to the equations

$F_x = C_{vx} \cdot V$ (eqn. 12.8-11) and the vertical distribution factor

$C_{vx} = W_x \cdot h_x^k / (\text{Sum of } W_i \cdot h_i^k)$ and k is an exponent which is related to the period of Vibration.

In this case, the value of k was 1.0

The Natural Frequency for the Vessel (Ope...) is 38.1729 Hz.

Earthquake Load Calculation:

From	To	Earthquake Height mm.	Earthquake Weight Kgf	Element Ope Load Kgf	Element Emp Load Kgf	Element Vertical Load Kgf
10	20	-0.01524	1469.66	0.000468	0.000468	-63.8811
20	30	750	4328.42	67.8324	67.8315	-252.023
30	40	2500	4068.65	212.538	212.535	-428.874
40	50	4250	14637.5	1299.88	1299.59	-1065.12
50	Lugs	6000	10653.9	1335.69	1335.67	-1528.21
Lugs	60	7100	21307.8	3161.13	3161.09	2775.83
60	70	9600	33441.6	6708.19	6708.1	1849.65
70	80	11100	9111.67	2113.33	2113.3	396.054

The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm.	Cumulative Wind Shear Kgf	Earthquake Shear Kgf	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	6592.3	...	...	...	...
20	30	5350	248.759	0.000468	122.467	0.0002304
30	40	3600	667.52	67.8328	809.689	50.8762
40	50	1850	1242.89	280.371	2720.14	399.087
50	Lugs	500	1685.62	1580.25	4916.57	1794.58
Lugs	60	1000	3973.2	14898.6	1000.66	34107.8
60	70	3500	1322.31	8821.51	2642.81	17504
70	80	5521.16	326.193	2113.33	170.013	1101.47

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
Wind: 1.100; Earthquake: 1.000.

Note:

Review the Vessel Design Summary for the cumulative shear force and bending moment on the support.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear Kg	Centroid Deflection mm.	Elem. End Deflection mm.	Elem. Ang. Rotation
10	20	...	0.02927	0.02927	0.85647E-05
20	30	248.759	0.022764	0.02927	0.85647E-05
30	40	667.52	0.0092956	0.016576	0.77255E-05
40	50	1242.89	0.0010939	0.0037759	0.42253E-05
50	Lugs	1685.62	-0.00011711	...	...
Lugs	60	3973.2	0.00047117	0.0016102	0.75432E-06
60	70	1322.31	0.0038178	0.0062877	0.11236E-05
70	80	326.193	0.0062878	0.0062878	0.11236E-05

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed Km/hr	2nd Crit. Wind Speed Km/hr
10	20	1955.46	12221.6
20	30	1955.46	12221.6
30	40	1955.46	12221.6
40	50	1955.46	12221.6
50	60	2070.09	12938.1
60	70	2070.09	12938.1
70	80	2070.09	12938.1

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
Allowable deflection : 55.500 Actual deflection : 0.029 mm.

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Longitudinal Stress Constants:

From	To	Metal Area New cm ²	Metal Area Corroded cm ²	Section Modulus New mm. ³	Section Modulus Corroded mm. ³
10	20	1177.34	1080.01	60759856	55816316
20	30	2206.78	2110.95	112165024	107447672
30	40	2206.78	2110.95	112165024	107447672
40	50	2206.78	2110.95	112165024	107447672
50	60	8972.39	8879.77	444255328	440254592
60	70	8972.39	8879.77	444255328	440254592
70	80	4631.46	4532.43	243630656	238740800

**Longitudinal Allowable Stresses:**

From	To	Tensile N./mm ²	Hydrotest Tensile N./mm ²	Compressive N./mm ²	Hydrotest Compressive N./mm ²
10	20	165.119	294.495	-75.3382	-113.866
20	30	165.119	294.495	-84.9728	-118.011
30	40	165.119	294.495	-84.9728	-118.011
40	50	165.119	294.495	-84.9728	-118.011
50	Lugs	165.119	294.495	-103.706	-126.314
Lugs	60	165.119	294.495	-103.706	-126.314
60	70	165.119	294.495	-103.706	-126.314
70	80	165.119	294.495	-94.0802	-122.006

Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure N./mm ²	Longitudinal Stress External Pressure N./mm ²	Longitudinal Stress Hydrotest Pressure N./mm ²
10	20	158.467	-3.31686	68.8664
20	30	78.5843	-1.69698	34.151
30	40	78.5843	-1.69698	34.151
40	50	78.5843	-1.69698	34.151
50	60	65.3665	-0.45736	7.50506
60	70	65.3665	-0.45736	7.50506
70	80	146.385	-0.89604	16.8072

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty N./mm ²	Wght. Str. Operating N./mm ²	Wght. Str. Hydrotest N./mm ²	Wght. Str. Emp. Mom. N./mm ²	Wght. Str. Opr. Mom. N./mm ²
10	20	...	...	...	...	...
20	30	0.068276	0.068276	0.17517	...	...
30	40	0.26936	0.26936	0.51407	...	...
40	50	0.45838	0.45838	0.94508	...	...
50	Lugs	0.27059	0.27063	0.46352	...	...
Lugs	60	-0.70528	-0.70528	-0.70528	...	...
60	70	-0.46996	-0.46996	-0.46996	...	...
70	80	-0.19715	-0.19715	-0.19715	...	...

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. N./mm ²	Bend. Str. Oper. Wind N./mm ²	Bend. Str. Oper. Equ. N./mm ²	Bend. Str. Hyd. Wind N./mm ²	Bend. Str. Hyd. Equ. N./mm ²
10	20	...	...	...	...	...
20	30	...	0.011177	0.21029E-07	0.0036886	...
30	40	...	0.073899	0.0046434	0.024387	...
40	50	...	0.24826	0.036424	0.081927	...
50	Lugs	...	0.10952	0.039974	0.03614	...
Lugs	60	...	0.02229	0.75975	0.0073555	...
60	70	...	0.058868	0.3899	0.019427	...
70	80	...	0.0069835	0.045245	0.0023046	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case N./mm ²	Vortex Shedding Empty Case N./mm ²	Vortex Shedding Test Case N./mm ²	Earthquake Empty Case N./mm ²
10	20	...	...	...	...
20	30	...	...	...	0.21028E-07
30	40	...	...	...	0.0046434
40	50	...	...	...	0.036424
50	Lugs	...	...	...	0.039969
Lugs	60	...	...	...	0.75975

60	70	...	...	...	0.38989
70	80	...	...	...	0.045244

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind N./mm ²	Longitudinal Stress Y Forces Seismic N./mm ²
10	20	...	-0.0058006
20	30	...	-0.011708
30	40	...	-0.019924
40	50	...	-0.049482
50	Lugs	...	-0.016878
Lugs	60	...	0.030656
60	70	...	0.020428
70	80	...	0.0085694

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded N./mm ²	Earthquake For/Mom Corroded N./mm ²	Wind For/Mom No Corrosion N./mm ²	Earthquake For/Mom No Corrosion N./mm ²
10	20	...	...	...	...
20	30	...	...	...	...
30	40	...	...	...	...
40	50	...	...	...	...
50	Lugs	...	...	...	...
Lugs	60	...	...	...	...
60	70	...	...	...	...
70	80	...	...	...	...

Bending Stresses due to Blast Load:

From	To	Bending Stress N./mm ²
10	20	...
20	30	...
30	40	...
40	50	...
50	Lugs	...
Lugs	60	...
60	70	...
70	80	...

Stress Combination Load Cases for Vertical Vessels:

Load Case Definition Key

IP = Longitudinal Stress due to Internal Pressure
EP = Longitudinal Stress due to External Pressure
HP = Longitudinal Stress due to Hydrotest Pressure
NP = No Pressure
EW = Longitudinal Stress due to Weight (No Liquid)
OW = Longitudinal Stress due to Weight (Operating)
HW = Longitudinal Stress due to Weight (Hydrotest)
WI = Bending Stress due to Wind Moment (Operating)
EQ = Bending Stress due to Earthquake Moment (Operating)
EE = Bending Stress due to Earthquake Moment (Empty)
HI = Bending Stress due to Wind Moment (Hydrotest)
HE = Bending Stress due to Earthquake Moment (Hydrotest)
WE = Bending Stress due to Wind Moment (Empty) (no CA)
WF = Bending Stress due to Wind Moment (Filled) (no CA)
CW = Longitudinal Stress due to Weight (Empty) (no CA)
VO = Bending Stress due to Vortex Shedding Loads (Ope)
VE = Bending Stress due to Vortex Shedding Loads (Emp)
VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
FW = Axial Stress due to Vertical Forces for the Wind Case
FS = Axial Stress due to Vertical Forces for the Seismic Case
BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

For Division 2, the overall stress intensity is computed in accordance with the Section VIII Division II Example problem manual PTB-3, example 4.3.6, stress due to combined loads. The stresses are combined per equation 4.3.44 to determine the overall stress intensity. The unity check is from Section 4.4.12 equations 4.4.85 and 4.4.90.

An asterisk (*) in the final column denotes overstress.

Note: Performing Combined Stress Analysis per VIII-2 paragraph 4.3.10.2.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	0.00	165.12	0.00	75.34	0.0000	-0.0000	...
20	0.08	165.12		169.93	0.0005		...
30	0.34	165.12		169.93	0.0021		0.0017
40	0.71	165.12		169.93	0.0043		0.0029
50	0.38	165.12		198.54	0.0023		0.0015
50	0.73	165.12	-0.73	198.54	0.0044	0.0037	0.0038
60	0.53	165.12	-0.53	198.54	0.0032	0.0027	0.0025
70	0.20	165.12	-0.20	110.69	0.0012	0.0018	0.0048

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	0.01	165.12	-0.01	72.56	0.0000	0.0001	0.0028
20	0.06	165.12		169.93	0.0003		...



30	0.25	165.12		169.93	0.0015		0.0016
40	0.45	165.12		169.93	0.0027		0.0026
50	0.29	165.12		198.54	0.0018		0.0014
50	1.43	165.12	-1.43	198.54	0.0087	0.0072	0.0038
60	0.84	165.12	-0.84	198.54	0.0051	0.0042	0.0024
70	0.23	165.12	-0.23	110.69	0.0014	0.0021	0.0076

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	0.00	165.12	0.00	72.56	0.0000	-0.0000	0.0032
20	0.08	165.12		169.93	0.0005		...
30	0.34	165.12		169.93	0.0021		0.0017
40	0.71	165.12		169.93	0.0043		0.0029
50	0.38	165.12		198.54	0.0023		0.0015
50	0.73	165.12	-0.73	198.54	0.0044	0.0037	0.0038
60	0.53	165.12	-0.53	198.54	0.0032	0.0027	0.0025
70	0.20	165.12	-0.20	110.69	0.0012	0.0018	0.0048

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	0.01	165.12	-0.01	72.56	0.0000	0.0001	0.0028
20	0.06	165.12		169.93	0.0003		...
30	0.25	165.12		169.93	0.0015		0.0016
40	0.45	165.12		169.93	0.0027		0.0026
50	0.29	165.12		198.54	0.0018		0.0014
50	1.43	165.12	-1.43	198.54	0.0087	0.0072	0.0038
60	0.84	165.12	-0.84	198.54	0.0051	0.0042	0.0024
70	0.23	165.12	-0.23	110.69	0.0014	0.0021	0.0076

Analysis of Load Case 5 : NP+HW+HI

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	0.00	294.49	0.00	113.87	0.0000	-0.0000	0.0032
20	0.18	294.49		118.01	0.0006		0.0011
30	0.54	294.49		118.01	0.0018		0.0031
40	1.03	294.49		118.01	0.0035		0.0058
50	0.50	294.49		126.31	0.0017		0.0025
50	0.71	294.49	-0.71	126.31	0.0024	0.0056	0.0038
60	0.49	294.49	-0.49	126.31	0.0017	0.0039	0.0025
70	0.20	294.49	-0.20	122.01	0.0007	0.0016	0.0044

Analysis of Load Case 6 : NP+HW+HE

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	0.00	294.49	0.00	113.87	0.0000	-0.0000	0.0027
20	0.18	294.49		118.01	0.0006		0.0011
30	0.51	294.49		118.01	0.0017		0.0031
40	0.95	294.49		118.01	0.0032		0.0058
50	0.46	294.49		126.31	0.0016		0.0025
50	0.71	294.49	-0.71	126.31	0.0024	0.0056	0.0038
60	0.47	294.49	-0.47	126.31	0.0016	0.0037	0.0025
70	0.20	294.49	-0.20	122.01	0.0007	0.0016	0.0042

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	158.47	165.12		72.56	0.9597		0.0027
20	138.29	165.12		169.93	0.8375		...
30	138.29	165.12		169.93	0.8375		0.0017
40	138.29	165.12		169.93	0.8375		0.0029
50	121.04	165.12		198.54	0.7330		0.0015



Stress due to Combined Loads:

Step: 23

4:54pm

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50	121.07	165.12	198.54	0.7332	0.0038
60	121.06	165.12	198.54	0.7332	0.0025
70	146.29	165.12	94.08	0.8860	...

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	158.46	165.12		75.34	0.9597		...
20	138.29	165.12		169.93	0.8375		...
30	138.29	165.12		169.93	0.8375		0.0016
40	138.29	165.12		169.93	0.8375		0.0026
50	121.03	165.12		198.54	0.7330		0.0014
50	121.10	165.12		198.54	0.7334		0.0038
60	121.08	165.12		198.54	0.7333		0.0024
70	146.31	165.12		94.08	0.8861		...

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	3.32	165.12	-3.32	72.56	0.0201	0.0457	0.0454
20	1.64	165.12	-1.64	169.93	0.0099	0.0097	0.0564
30	1.50	165.12	-1.50	169.93	0.0091	0.0088	0.0593
40	1.49	165.12	-1.49	169.93	0.0090	0.0087	0.0617
50	0.30	165.12	-0.30	198.54	0.0018	0.0015	0.0089
50	1.18	165.12	-1.18	198.54	0.0072	0.0060	0.0095
60	0.99	165.12	-0.99	198.54	0.0060	0.0050	0.0092
70	1.10	165.12	-1.10	148.58	0.0067	0.0074	...

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	3.32	165.12	-3.32	72.56	0.0201	0.0458	0.0454
20	1.64	165.12	-1.64	169.93	0.0099	0.0097	...
30	1.45	165.12	-1.45	169.93	0.0088	0.0085	0.0590
40	1.32	165.12	-1.32	169.93	0.0080	0.0078	0.0610
50	0.24	165.12	-0.24	198.54	0.0015	0.0012	0.0087
50	1.89	165.12	-1.89	198.54	0.0115	0.0095	0.0108
60	1.30	165.12	-1.30	198.54	0.0079	0.0065	0.0096
70	1.13	165.12	-1.13	148.58	0.0068	0.0076	0.0026

Analysis of Load Case 11 : HP+HW+HI

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	66.87	294.49		113.87	0.2271		...
20	58.84	294.49		118.01	0.1998		0.0011
30	58.84	294.49		118.01	0.1998		0.0031
40	58.85	294.49		118.01	0.1998		0.0058
50	14.78	294.49		126.31	0.0502		0.0025
50	14.87	294.49		126.31	0.0505		0.0038
60	14.85	294.49		126.31	0.0504		0.0025
70	17.15	294.49		122.01	0.0582		...

Analysis of Load Case 12 : HP+HW+HE

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	66.87	294.49		113.87	0.2271		...
20	58.84	294.49		118.01	0.1998		0.0011
30	58.84	294.49		118.01	0.1998		0.0031
40	58.85	294.49		118.01	0.1998		0.0058
50	14.78	294.49		126.31	0.0502		0.0025
50	14.87	294.49		126.31	0.0505		0.0038
60	14.85	294.49		126.31	0.0504		0.0025
70	17.15	294.49		122.01	0.0582		...



Analysis of Load Case 13 : IP+WE+EW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	158.47	165.12		75.34	0.9597		...
20	138.29	165.12		169.93	0.8375		...
30	138.29	165.12		169.93	0.8375		0.0017
40	138.29	165.12		169.93	0.8375		0.0029
50	121.03	165.12		198.54	0.7330		0.0015
50	121.07	165.12		198.54	0.7332		0.0038
60	121.06	165.12		198.54	0.7332		0.0025
70	146.29	165.12		94.08	0.8859		...

Analysis of Load Case 14 : IP+WF+CW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	158.47	165.12		75.34	0.9597		...
20	138.29	165.12		169.93	0.8375		...
30	138.29	165.12		169.93	0.8375		0.0016
40	138.29	165.12		169.93	0.8375		0.0027
50	121.03	165.12		198.54	0.7330		0.0014
50	121.07	165.12		198.54	0.7332		0.0038
60	121.06	165.12		198.54	0.7332		0.0025
70	146.29	165.12		94.08	0.8860		...

Analysis of Load Case 15 : IP+VO+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	158.47	165.12		75.34	0.9597		...
20	138.29	165.12		169.93	0.8375		...
30	138.29	165.12		169.93	0.8375		0.0017
40	138.29	165.12		169.93	0.8375		0.0029
50	121.03	165.12		198.54	0.7330		0.0015
50	121.07	165.12		198.54	0.7332		0.0038
60	121.06	165.12		198.54	0.7332		0.0025
70	146.29	165.12		94.08	0.8859		...

Analysis of Load Case 16 : IP+VE+EW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	158.47	165.12		75.34	0.9597		...
20	138.29	165.12		169.93	0.8375		...
30	138.29	165.12		169.93	0.8375		0.0017
40	138.29	165.12		169.93	0.8375		0.0029
50	121.03	165.12		198.54	0.7330		0.0015
50	121.07	165.12		198.54	0.7332		0.0038
60	121.06	165.12		198.54	0.7332		0.0025
70	146.29	165.12		94.08	0.8859		...

Analysis of Load Case 17 : NP+VO+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	0.00	165.12	0.00	75.34	0.0000	-0.0000	...
20	0.07	165.12		169.93	0.0004		...
30	0.27	165.12		169.93	0.0016		0.0017
40	0.46	165.12		169.93	0.0028		0.0029
50	0.27	165.12		198.54	0.0016		0.0015
50	0.71	165.12	-0.71	198.54	0.0043	0.0036	0.0038
60	0.47	165.12	-0.47	198.54	0.0028	0.0024	0.0025
70	0.20	165.12	-0.20	110.69	0.0012	0.0018	0.0042

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
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Node	Intensity	Intensity	Stress	Stress Fxa	Ratio	Ratio	Check
10	158.46	165.12		72.56	0.9597		0.0027
20	138.29	165.12		169.93	0.8375		...
30	138.29	165.12		169.93	0.8375		0.0016
40	138.29	165.12		169.93	0.8375		0.0026
50	121.03	165.12		198.54	0.7330		0.0014
50	121.07	165.12		198.54	0.7332		0.0036
60	121.06	165.12		198.54	0.7332		0.0024
70	146.29	165.12		94.08	0.8860		...

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	3.32	165.12	-3.32	72.56	0.0201	0.0458	0.0454
20	1.64	165.12	-1.64	169.93	0.0099	0.0097	...
30	1.45	165.12	-1.45	169.93	0.0088	0.0085	0.0038
40	1.29	165.12	-1.29	169.93	0.0078	0.0076	0.0057
50	0.20	165.12	-0.20	198.54	0.0012	0.0010	0.0014
50	1.13	165.12	-1.13	198.54	0.0069	0.0057	0.0036
60	0.91	165.12	-0.91	198.54	0.0055	0.0046	0.0024
70	1.08	165.12	-1.08	110.69	0.0066	0.0098	0.0079

Analysis of Load Case 20 : BL+IP+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	158.47	165.12		75.34	0.9597		...
20	138.29	165.12		169.93	0.8375		...
30	138.29	165.12		169.93	0.8375		0.0017
40	138.29	165.12		169.93	0.8375		0.0029
50	121.03	165.12		198.54	0.7330		0.0015
50	121.07	165.12		198.54	0.7332		0.0038
60	121.06	165.12		198.54	0.7332		0.0025
70	146.29	165.12		94.08	0.8859		...

Absolute Maximum of the all of the Stress Ratio's

0.9597

Governing Element: Dish bottom

Governing Load Case 7 : IP+OW+WI+FW+BW

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
Dish bottom	158.5	...	...	...	158.5	+ Tens
Dish bottom	158.5	...	...	...	...	- Comp
shell-2	78.6	0.1	...	...	78.7	+ Tens
shell-2	78.6	0.1	...	...	...	- Comp
shell-3	78.6	0.3	...	0.1	78.9	+ Tens
shell-3	78.6	0.3	...	-0.1	...	- Comp
sheel 4	78.6	0.5	...	0.2	79.3	+ Tens
sheel 4	78.6	0.5	...	-0.2	...	- Comp
Channel-1	65.4	0.3	...	0.1	65.7	+ Tens
Channel-1	65.4	0.3	...	-0.1	...	- Comp
Channel-1	65.4	-0.7	...	...	64.7	+ Tens
Channel-1	65.4	-0.7	...	...	...	- Comp
Channel-2	65.4	-0.5	...	0.1	65.0	+ Tens
Channel-2	65.4	-0.5	...	-0.1	...	- Comp
Top dish	146.4	-0.2	...	...	146.2	+ Tens



Top dish	146.4	-0.2	...	...	...	- Comp
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Additional Results per Section 4.4, see 4.4.15**Nomenclature :**

L - Section Length mm.

Fxa - Allowable comp mem stress due to compression with lamc ≤ 0.15 , N./mm²Fba - Allowable all comp mem stress due to bending moment, N./mm²Fva - Allowable shear stress when only shear is present, N./mm²Fha - Allowable hoop comp memb stress under ext pressure alone, N./mm²Fxa - Allowable axial comp w/hoop compression and lamc ≤ 0.15 , N./mm²Fhxa - Allowable hoop comp w/axial compression and lamc ≤ 0.15 , N./mm²Faha - Allowable axial comp w/hoop compression and lamc > 0.15 , N./mm²Fbha - Allowable axial comp from bending w/hoop compression, N./mm²Fhba - Allowable hoop comp w/long compression due to moment, N./mm²**Allowable Stress Results for Case 1 : NP+EW+1.1WI+FW+BW**

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10							6895		6895	6895	6895
20	0.396	5343	169	169	103	60	6895	169	6895	6895	6895
30	0.396	5343	169	169	103	60	6895	169	6895	6895	6895
40	0.396	5343	169	169	103	60	6895	169	6895	6895	6895
50	0.419	6350	198	198	151	117	6895	198	6895	6895	6895
60	0.419	6350	198	198	151	117	6895	198	6895	6895	6895
70						110	6895		6895	6895	6895

Allowable Stress Results for Case 2 : NP+EW+EE+FS+BS

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10						72	6895		6895	6895	6895
20	0.396	5343	169	169	103	60	6895	169	6895	6895	6895
30	0.396	5343	169	169	103	60	6895	169	6895	6895	6895
40	0.396	5343	169	169	103	60	6895	169	6895	6895	6895
50	0.419	6350	198	198	151	117	6895	198	6895	6895	6895
60	0.419	6350	198	198	151	117	6895	198	6895	6895	6895
70						110	6895		6895	6895	6895

Allowable Stress Results for Case 3 : NP+OW+1.1WI+FW+BW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10						72	6895		6895	6895	6895
20	0.396	5343	169	169	103	60	6895	169	6895	6895	6895
30	0.396	5343	169	169	103	60	6895	169	6895	6895	6895
40	0.396	5343	169	169	103	60	6895	169	6895	6895	6895
50	0.419	6350	198	198	151	117	6895	198	6895	6895	6895
60	0.419	6350	198	198	151	117	6895	198	6895	6895	6895
70						110	6895		6895	6895	6895

Allowable Stress Results for Case 4 : NP+OW+EQ+FS+BS

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10						72	6895		6895	6895	6895
20	0.396	5343	169	169	103	60	6895	169	6895	6895	6895
30	0.396	5343	169	169	103	60	6895	169	6895	6895	6895
40	0.396	5343	169	169	103	60	6895	169	6895	6895	6895
50	0.419	6350	198	198	151	117	6895	198	6895	6895	6895
60	0.419	6350	198	198	151	117	6895	198	6895	6895	6895
70						110	6895		6895	6895	6895

Allowable Stress Results for Case 5 : NP+HW+HI

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10						74	6895		6895	6895	6895
20	0.384	5343	172	172	103	62	6895	172	6895	6895	6895
30	0.384	5343	172	172	103	62	6895	172	6895	6895	6895
40	0.384	5343	172	172	103	62	6895	172	6895	6895	6895
50	0.403	6350	199	199	151	117	6895	199	6895	6895	6895
60	0.403	6350	199	199	151	117	6895	199	6895	6895	6895
70						111	6895		6895	6895	6895

Allowable Stress Results for Case 6 : NP+HW+HE



From Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10					74	6895		6895	6895	6895
20 0.384	5343	172	172	103	62	6895	172	6895	6895	6895
30 0.384	5343	172	172	103	62	6895	172	6895	6895	6895
40 0.384	5343	172	172	103	62	6895	172	6895	6895	6895
50 0.403	6350	199	199	151	117	6895	199	6895	6895	6895
60 0.403	6350	199	199	151	117	6895	199	6895	6895	6895
70					111	6895		6895	6895	6895

Allowable Stress Results for Case 7 : IP+OW+1.1WI+FW+BW

From Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10					72	6895		6895	6895	6895
20 0.396	5343	169	169	103	60	6895	169	6895	6895	6895
30 0.396	5343	169	169	103	60	6895	169	6895	6895	6895
40 0.396	5343	169	169	103	60	6895	169	6895	6895	6895
50 0.419	6350	198	198	151	117	6895	198	6895	6895	6895
60 0.419	6350	198	198	151	117	6895	198	6895	6895	6895
70						6895		6895	6895	6895

Allowable Stress Results for Case 8 : IP+OW+EQ+FS+BS

From Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10						6895		6895	6895	6895
20 0.396	5343	169	169	103	60	6895	169	6895	6895	6895
30 0.396	5343	169	169	103	60	6895	169	6895	6895	6895
40 0.396	5343	169	169	103	60	6895	169	6895	6895	6895
50 0.419	6350	198	198	151	117	6895	198	6895	6895	6895
60 0.419	6350	198	198	151	117	6895	198	6895	6895	6895
70						6895		6895	6895	6895

Allowable Stress Results for Case 9 : EP+OW+1.1WI+FW+BW

From Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10					72	6895		6895	6895	6895
20 0.396	5343	169	169	103	60	31	63	29		60
30 0.396	5343	169	169	103	60	35	63	33	1	60
40 0.396	5343	169	169	103	60	38	63	36	4	60
50 0.419	6350	198	198	151	117	176	143	164	3	116
60 0.419	6350	198	198	151	117	139	145	129	7	116
70					148	6895		6895	6895	6895

Allowable Stress Results for Case 10 : EP+OW+EQ+FS+BS

From Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10					72	6895		6895	6895	6895
20 0.396	5343	169	169	103	60	31	63	29	6895	6895
30 0.396	5343	169	169	103	60	34	63	32		60
40 0.396	5343	169	169	103	60	38	63	35		60
50 0.419	6350	198	198	151	117	171	143	160	87	99
60 0.419	6350	198	198	151	117	135	145	126	49	109
70					148	6895		6895	6895	6895

Allowable Stress Results for Case 11 : HP+HW+HI

From Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10						6895		6895	6895	6895
20 0.384	5343	172	172	103	62	6895	172	6895	6895	6895
30 0.384	5343	172	172	103	62	6895	172	6895	6895	6895
40 0.384	5343	172	172	103	62	6895	172	6895	6895	6895
50 0.403	6350	199	199	151	117	6895	199	6895	6895	6895
60 0.403	6350	199	199	151	117	6895	199	6895	6895	6895
70						6895		6895	6895	6895

Allowable Stress Results for Case 12 : HP+HW+HE

From Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10						6895		6895	6895	6895
20 0.384	5343	172	172	103	62	6895	172	6895	6895	6895
30 0.384	5343	172	172	103	62	6895	172	6895	6895	6895
40 0.384	5343	172	172	103	62	6895	172	6895	6895	6895
50 0.403	6350	199	199	151	117	6895	199	6895	6895	6895
60 0.403	6350	199	199	151	117	6895	199	6895	6895	6895
70						6895		6895	6895	6895



Allowable Stress Results for Case 13 : IP+WE+EW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10							6895		6895	6895	6895
20	0.396	5343	169	169	103	60	6895	169	6895	6895	6895
30	0.396	5343	169	169	103	60	6895	169	6895	6895	6895
40	0.396	5343	169	169	103	60	6895	169	6895	6895	6895
50	0.419	6350	198	198	151	117	6895	198	6895	6895	6895
60	0.419	6350	198	198	151	117	6895	198	6895	6895	6895
70							6895		6895	6895	6895

Allowable Stress Results for Case 14 : IP+WF+CW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10							6895		6895	6895	6895
20	0.396	5343	169	169	103	60	6895	169	6895	6895	6895
30	0.396	5343	169	169	103	60	6895	169	6895	6895	6895
40	0.396	5343	169	169	103	60	6895	169	6895	6895	6895
50	0.419	6350	198	198	151	117	6895	198	6895	6895	6895
60	0.419	6350	198	198	151	117	6895	198	6895	6895	6895
70							6895		6895	6895	6895

Allowable Stress Results for Case 15 : IP+VO+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10							6895		6895	6895	6895
20	0.396	5343	169	169	103	60	6895	169	6895	6895	6895
30	0.396	5343	169	169	103	60	6895	169	6895	6895	6895
40	0.396	5343	169	169	103	60	6895	169	6895	6895	6895
50	0.419	6350	198	198	151	117	6895	198	6895	6895	6895
60	0.419	6350	198	198	151	117	6895	198	6895	6895	6895
70							6895		6895	6895	6895

Allowable Stress Results for Case 16 : IP+VE+EW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10							6895		6895	6895	6895
20	0.396	5343	169	169	103	60	6895	169	6895	6895	6895
30	0.396	5343	169	169	103	60	6895	169	6895	6895	6895
40	0.396	5343	169	169	103	60	6895	169	6895	6895	6895
50	0.419	6350	198	198	151	117	6895	198	6895	6895	6895
60	0.419	6350	198	198	151	117	6895	198	6895	6895	6895
70							6895		6895	6895	6895

Allowable Stress Results for Case 17 : NP+VO+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10							6895		6895	6895	6895
20	0.396	5343	169	169	103	60	6895	169	6895	6895	6895
30	0.396	5343	169	169	103	60	6895	169	6895	6895	6895
40	0.396	5343	169	169	103	60	6895	169	6895	6895	6895
50	0.419	6350	198	198	151	117	6895	198	6895	6895	6895
60	0.419	6350	198	198	151	117	6895	198	6895	6895	6895
70							110	6895	6895	6895	6895

Allowable Stress Results for Case 18 : FS+BS+IP+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10						72	6895		6895	6895	6895
20	0.396	5343	169	169	103	60	6895	169	6895	6895	6895
30	0.396	5343	169	169	103	60	6895	169	6895	6895	6895
40	0.396	5343	169	169	103	60	6895	169	6895	6895	6895
50	0.419	6350	198	198	151	117	6895	198	6895	6895	6895
60	0.419	6350	198	198	151	117	6895	198	6895	6895	6895
70							6895		6895	6895	6895

Allowable Stress Results for Case 19 : FS+BS+EP+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10						72	6895		6895	6895	6895
20	0.396	5343	169	169	103	60	31	63	29	6895	6895
30	0.396	5343	169	169	103	60	34	63	32	6895	6895
40	0.396	5343	169	169	103	60	38	63	35	6895	6895
50	0.419	6350	198	198	151	117	171	143	160	6895	6895



Stress due to Combined Loads:

Step: 23 4:54pm Sep 16, 2026

60	0.419	6350	198	198	151	117	135	145	126	6895	6895
70						110	6895		6895	6895	6895

Allowable Stress Results for Case 20 : BL+IP+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10							6895		6895	6895	6895
20	0.396	5343	169	169	103	60	6895	169	6895	6895	6895
30	0.396	5343	169	169	103	60	6895	169	6895	6895	6895
40	0.396	5343	169	169	103	60	6895	169	6895	6895	6895
50	0.419	6350	198	198	151	117	6895	198	6895	6895	6895
60	0.419	6350	198	198	151	117	6895	198	6895	6895	6895
70							6895		6895	6895	6895

Note:

Load cases involving Wind and Seismic are assumed to cause a net section bending moment. These load case components are: EQ, EE, WI, WE, VO, VE, VF, HE, HI, WF. Static bending moments due to the weights of nozzles etc. are considered to be local in nature and will not cause a net section bending moment. This is an important consideration when computing the unity check per section 4.4.12 equations 4.4.86, 4.4.87, 4.4.90, 4.4.91 and 4.4.92.

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Shop/Field Installation Options :

Insulation is installed in the Shop.
Lining is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Insulation	5683.572 mm.
Center of Gravity of the Lining	8660.343 mm.
Center of Gravity of the Nozzles	10155.117 mm.
Center of Gravity of the Lugs	6325.000 mm.
Center of Gravity of the Added Weights (Operating)	6343.274 mm.
Center of Gravity of the Added Weights (Empty)	6343.274 mm.
Center of Gravity of the Tubesheet(s)	5050.000 mm.
Center of Gravity of the Tubes	2750.000 mm.
Center of Gravity of Bare Shell New and Cold	7330.716 mm.
Center of Gravity of Bare Shell Corroded	7362.258 mm.
Vessel CG in the Operating Condition	6892.236 mm.
Vessel CG in the Fabricated (Shop/Empty) Condition	6878.212 mm.
Vessel CG in the Test Condition	6539.437 mm.

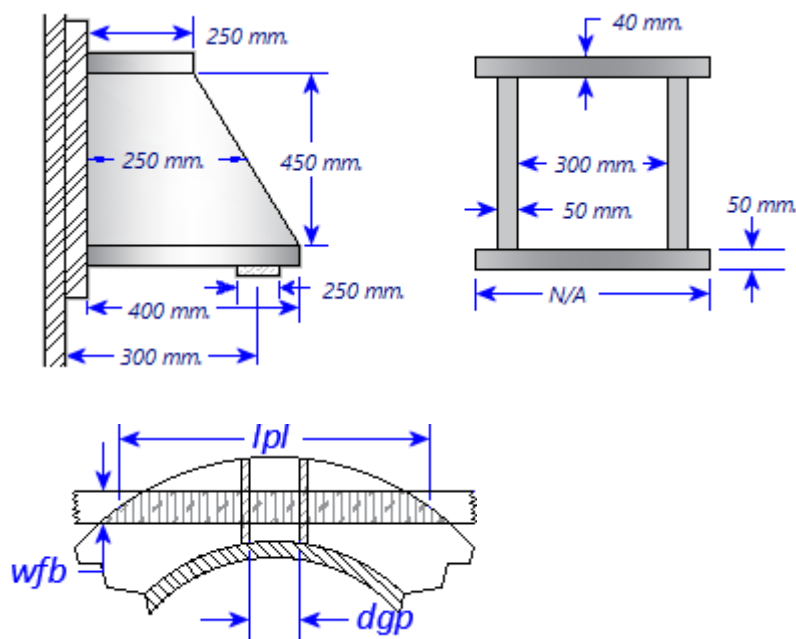
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**Support Lug Calculations: Operating Case****Input Data: Support Lugs**

Type of Geometry : Gussets with Continuous Top Ring

Number of Support Lugs	Nlug	4
Distance from Vessel OD to Lug Contact Point	Dlug	300.0000 mm.
Lug Support Force Bearing Width	Wfb	250.0000 mm.
Lug Material		SA-387 11
Lug Yield Stress		247.68 N./mm ²
Radial Width of bottom Support Lug Plate	Wpl	400.0000 mm.
Effective Force Bearing Length	Lpl	550.0000 mm.
Thickness of bottom Support Lug Plate	Tpl	50.0000 mm.
Distance between Gussets	Dgp	300.0000 mm.
Mean Width of Gusset Plate	Wgp	250.0000 mm.
Height of Gusset Plate	Hgp	450.0000 mm.
Thickness of Gusset Plate	Tgp	50.0000 mm.
Radial Width of Reinforcing Ring	Wtp	250.0000 mm.
Thickness of Top Plate	Ttp	40.0000 mm.

Bolt Material	SA-193 B7
Bolt Allowable Stress at Design Temperature	172.00 N./mm ²
Thread Series	TEMA Metric
Bolt Diameter	42.00 mm.

**Results for Support Lugs:****Description: Support Lugs**

Overturning Moment at Support Lug	34107.8 Kg-m.
Weight Load at the top of one Lug	25830.8 Kgf

Force on one Lug, Operating Condition [Flug]:

$$= (W/Nlug + Mlug / (Rlug \cdot Nlug / 2))$$

$$= (103323/4 + 34108 / (1418.0 \cdot 4/2))$$

$$= 37857.30 \text{ Kgf}$$

Ring Girder Calculations Per Bednar pp.157-159:

Force P acting in the plane of the ring [P]:

$$= Flug \cdot Dlug / (Hgp + Tpl)$$



$$= 37857 \cdot 300.0 / (450.0 + 50.0)$$

$$= 22714.38 \text{ Kg}$$

Angle $[\alpha]$:

$$= \pi / N_{lug}$$

$$= \pi / 4$$

$$= 45.0 \text{ deg}$$

Maximum bending moment (at the load points) $[M1]$:

$$= 0.5 \cdot P(OD/2 + W_{tp}/2) \cdot (1/\alpha - \cos(\alpha)/\sin(\alpha))$$

$$= 0.5 \cdot 22714.38 \cdot 1243.0(1/0.79 - \cos(45.0)/\sin(45.0))$$

$$= 3857.40 \text{ Kg-m.}$$

Tangential (axial) thrust induced in the ring $[T1]$:

(Top ring in compression, bottom ring in tension)

$$= 0.5 \cdot P \cdot \cos(\alpha) / \sin(\alpha)$$

$$= 0.5 \cdot 22714.38 \cdot \cos(45.0) / \sin(45.0)$$

$$= 11357.19 \text{ Kg}$$

Resulting combined stress at the load point $[S1]$:

$$= T1 / (T_{tp} \cdot W_{tp}) + M1 / (T_{tp} \cdot W_{tp}^2 / 6)$$

$$= 11357 / (40.0 \cdot 250.0) + 3857.4 / (40.0 \cdot 250.0^2 / 6)$$

$$= 101.93 \text{ N./mm}^2$$

Bending moment at the midpoints between the loads $[M2]$:

$$= 0.5 \cdot P(OD/2 + W_{tp}/2) \cdot (1/\sin(\alpha) - 1/\alpha)$$

$$= 0.5 \cdot 22714.38 \cdot 1243.0(1/\sin(45.0) - 1/0.79)$$

$$= 1990.17 \text{ Kg-m.}$$

Axial thrust $[T2]$:

$$= 0.5 \cdot P / \sin(\alpha)$$

$$= 0.5 \cdot 22714.38 / \sin(45.0)$$

$$= 16061.50 \text{ Kg}$$

Combined maximum stress $[S2]$:

$$= T2 / (W_{tp} \cdot T_{tp}) + M2 / (T_{tp} \cdot W_{tp}^2 / 6)$$

$$= 16061 / (250.0 \cdot 40.0) + 1990.17 / (40.0 \cdot 250.0^2 / 6)$$

$$= 62.59 \text{ N./mm}^2$$

Ring Allowable Stress $[Sa]$:

$$= 2/3 \cdot Y_{lug}$$

$$= 2/3 \cdot 248$$

$$= 165.12 \text{ N./mm}^2$$

Bearing Area $[Ba]$:

$$= L_{pl} \cdot W_{fb}$$

$$= 550.0 \cdot 250.0$$

$$= 1375.00 \text{ cm}^2$$

Bending Stress in bottom Plate (Unif. Load) Per Bednar p.156 $[Spl2]$:

$$= \beta_1 \cdot F_{lug} / Ba \cdot W_{fb}^2 / T_{pl}^2 \text{ per Roark \& Young 5th Ed.}$$

$$= 1.109 \cdot 37857.3 / 1375.0 \cdot 250.0^2 / 50.0^2$$

$$= 74.86 \text{ N./mm}^2$$

Bottom Plate Required Thickness (Uniform Load) 33.6668 mm.

Bottom Plate Required Thickness based on ADM S 3/4 $[trAD]$:

$$= 0.71 \cdot D_{gp} \cdot \sqrt{(F_{lug} / (L_{pl} \cdot W_{fb}) / S_{pa})}$$

$$= 0.71 \cdot 300.0 \cdot \sqrt{(37857 / (550.0 \cdot 250.0) / 165.119)}$$

$$= 27.238 \text{ mm.}$$

If using the AD Code recommendations, the force bearing width (W_{fb}) must be greater than or equal to 1/3 of the bottom plate radial width (W_{pl}) plus the pad thickness (Pad_{thk}), if there is a pad.

Bottom Support Plate Allowable Stress $[Spa]$:

$$= 2/3 \cdot Y_{lug}$$

$$= 2/3 \cdot 248$$



$$= 165.12 \text{ N./mm}^2$$

Gusset Plate Axial Stress (Force / Gusset Plate Area) [Sgp]:

$$= (\text{Flug}/2) / (\text{Wgp} \cdot \text{Tgp})$$

$$= (37857/2) / (250.0 \cdot 50.0)$$

$$= 14.85 \text{ N./mm}^2$$

Check required Gusset Plate Thickness:

Required Thickness of Gussets per AISC: 13.5616 mm.

Gusset Plate Allowable Stress [Sga]:

$$= (1 - (Klr)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr^3) / (8 \cdot Cc^3))$$

$$= (1 - (36.0)^2 / (2 \cdot 122.18^2)) 248 / (5/3 + 3 \cdot (36.0) / (8 \cdot 122.18) - (36.0^3) / (8 \cdot 122.18^3))$$

$$= 133.56 \text{ N./mm}^2$$

Dimension [Gtp]:

$$= 2 \cdot \text{Wgp} - \text{Wpl}$$

$$= 2 \cdot 250.0 - 400.0$$

$$= 100.0 \text{ mm.}$$

Angle [αG]:

$$= \pi / 2 - \text{atan} ((\text{Wpl} - \text{Gtp}) / \text{Hgp})$$

$$= \pi / 2 - \text{atan} ((400.0 - 100.0) / 450.0)$$

$$= 56.3 \text{ deg}$$

Maximum Compressive Gusset Plate Stress per Bednar [SgpB]:

$$= \text{Flug} (3 \cdot \text{Dlug} - \text{Wpl}) / (\text{Tgp} \cdot \text{Wpl}^2 \cdot (\text{sind}(\alpha G))^2)$$

$$= 37857 (3 \cdot 300.0 - 400.0) / (50.0 \cdot 400.0^2 \cdot (\text{sind}(56.31))^2)$$

$$= 33.52 \text{ N./mm}^2$$

Gusset Plate Allowable Compressive Stress [SgaB]:

$$= 18000 / (1 + (1 / 18000) \cdot (\text{Hgp} / \text{sind}(\alpha G) / (0.289 \cdot \text{Tgp}))^2)$$

$$= 18000 / (1 + (1 / 18000) \cdot (450.0 / \text{sind}(56.31) / (0.289 \cdot 50.0))^2)$$

$$= 115.15 \text{ N./mm}^2$$

There was no uplift. Please choose an appropriate bolt size for this support design.

Review notes about nozzle loadings and supports in the Warnings report.

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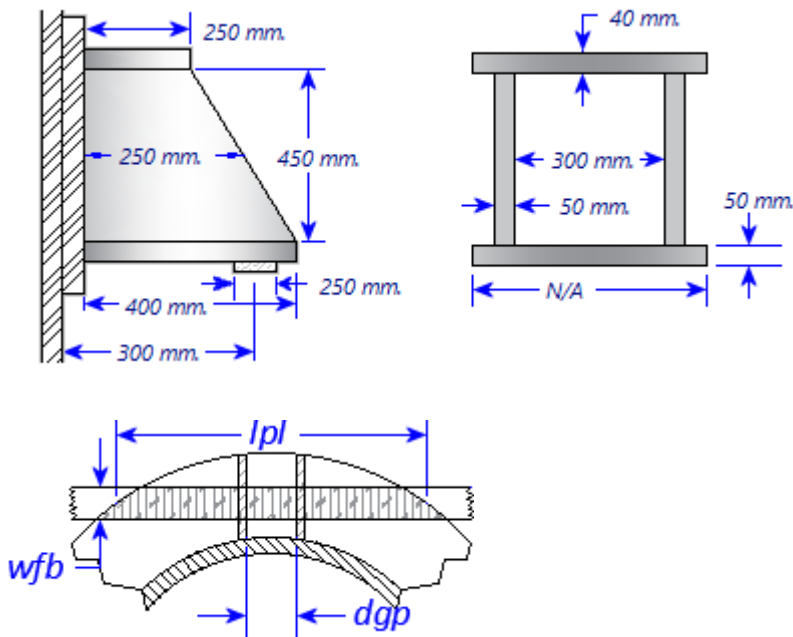
Support Lug Calculations: Test Case

Input Data: Support Lugs

Type of Geometry : Gussets with Continuous Top Ring

Number of Support Lugs	Nlug	4
Distance from Vessel OD to Lug Contact Point	Dlug	300.0000 mm.
Lug Support Force Bearing Width	Wfb	250.0000 mm.
Lug Material		SA-387 11
Lug Yield Stress		309.99 N./mm ²
Radial Width of bottom Support Lug Plate	Wpl	400.0000 mm.
Effective Force Bearing Length	Lpl	550.0000 mm.
Thickness of bottom Support Lug Plate	Tpl	50.0000 mm.
Distance between Gussets	Dgp	300.0000 mm.
Mean Width of Gusset Plate	Wgp	250.0000 mm.
Height of Gusset Plate	Hgp	450.0000 mm.
Thickness of Gusset Plate	Tgp	50.0000 mm.
Radial Width of Reinforcing Ring	Wtp	250.0000 mm.
Thickness of Top Plate	Ttp	40.0000 mm.

Bolt Material	SA-193 B7
Bolt Allowable Stress at Design Temperature	172.00 N./mm ²
Thread Series	TEMA Metric
Bolt Diameter	42.00 mm.



Results for Support Lugs:

Description: Support Lugs

Overturning Moment at Support Lug	330.2 Kg-m.
Weight Load at the top of one Lug	34269.5 Kgf

Force on one Lug, Test Condition [Flug]:

$$= (W/Nlug + Mlug / (Rlug \cdot Nlug / 2))$$

$$= (137078 / 4 + 330 / (1418.0 \cdot 4 / 2))$$

$$= 34385.89 \text{ Kgf}$$

Ring Girder Calculations Per Bednar pp.157-159:

Force P acting in the plane of the ring [P]:

$$= Flug \cdot Dlug / (Hgp + Tpl)$$



$$= 34386 \cdot 300.0 / (450.0 + 50.0)$$

$$= 20631.54 \text{ Kg}$$

Angle [α]:

$$= \pi / N_{lug}$$

$$= \pi / 4$$

$$= 45.0 \text{ deg}$$

Maximum bending moment (at the load points) [M1]:

$$= 0.5 \cdot P(OD/2 + W_{tp}/2) \cdot (1/\alpha - \cos(\alpha)/\sin(\alpha))$$

$$= 0.5 \cdot 20631.54 \cdot 1243.0(1/0.79 - \cos(45.0)/\sin(45.0))$$

$$= 3503.68 \text{ Kg-m.}$$

Tangential (axial) thrust induced in the ring [T1]:

(Top ring in compression, bottom ring in tension)

$$= 0.5 \cdot P \cdot \cos(\alpha) / \sin(\alpha)$$

$$= 0.5 \cdot 20631.54 \cdot \cos(45.0) / \sin(45.0)$$

$$= 10315.77 \text{ Kg}$$

Resulting combined stress at the load point [S1]:

$$= T1 / (T_{tp} \cdot W_{tp}) + M1 / (T_{tp} \cdot W_{tp}^2 / 6)$$

$$= 10316 / (40.0 \cdot 250.0) + 3503.68 / (40.0 \cdot 250.0^2 / 6)$$

$$= 92.58 \text{ N./mm}^2$$

Bending moment at the midpoints between the loads [M2]:

$$= 0.5 \cdot P(OD/2 + W_{tp}/2) \cdot (1/\sin(\alpha) - 1/\alpha)$$

$$= 0.5 \cdot 20631.54 \cdot 1243.0(1/\sin(45.0) - 1/0.79)$$

$$= 1807.68 \text{ Kg-m.}$$

Axial thrust [T2]:

$$= 0.5 \cdot P / \sin(\alpha)$$

$$= 0.5 \cdot 20631.54 / \sin(45.0)$$

$$= 14588.70 \text{ Kg}$$

Combined maximum stress [S2]:

$$= T2 / (W_{tp} \cdot T_{tp}) + M2 / (T_{tp} \cdot W_{tp}^2 / 6)$$

$$= 14589 / (250.0 \cdot 40.0) + 1807.68 / (40.0 \cdot 250.0^2 / 6)$$

$$= 56.85 \text{ N./mm}^2$$

Ring Allowable Stress [Sa]:

$$= 2/3 \cdot Y_{lug}$$

$$= 2/3 \cdot 310$$

$$= 206.66 \text{ N./mm}^2$$

Bearing Area [Ba]:

$$= L_{pl} \cdot W_{fb}$$

$$= 550.0 \cdot 250.0$$

$$= 1375.00 \text{ cm}^2$$

Bending Stress in bottom Plate (Unif. Load) Per Bednar p.156 [Spl2]:

$$= \beta_1 \cdot F_{lug} / B_a \cdot W_{fb}^2 / T_{pl}^2 \text{ per Roark \& Young 5th Ed.}$$

$$= 1.109 \cdot 34385.9 / 1375.0 \cdot 250.0^2 / 50.0^2$$

$$= 68.00 \text{ N./mm}^2$$

Bottom Plate Required Thickness (Uniform Load) 28.6803 mm.

Bottom Plate Required Thickness based on ADM S 3/4 [trAD]:

$$= 0.71 \cdot D_{gp} \cdot \sqrt{(F_{lug} / (L_{pl} \cdot W_{fb}) / S_{pa})}$$

$$= 0.71 \cdot 300.0 \cdot \sqrt{(34386 / (550.0 \cdot 250.0) / 206.663)}$$

$$= 23.203 \text{ mm.}$$

If using the AD Code recommendations, the force bearing width (Wfb) must be greater than or equal to 1/3 of the bottom plate radial width (Wpl) plus the pad thickness (Padthk), if there is a pad.

Bottom Support Plate Allowable Stress [Spa]:

$$= 2/3 \cdot Y_{lug}$$

$$= 2/3 \cdot 310$$



$$= 206.66 \text{ N./mm}^2$$

Gusset Plate Axial Stress (Force / Gusset Plate Area) [Sgp]:

$$= (\text{Flug}/2) / (\text{Wgp} \cdot \text{Tgp})$$

$$= (34386/2) / (250.0 \cdot 50.0)$$

$$= 13.49 \text{ N./mm}^2$$

Check required Gusset Plate Thickness:

Required Thickness of Gussets per AISC: 12.7889 mm.

Gusset Plate Allowable Stress [Sga]:

$$= (1 - (Klr)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr^3) / (8 \cdot Cc^3))$$

$$= (1 - (36.0)^2 / (2 \cdot 114.0^2)) 310 / (5/3 + 3 \cdot (36.0) / (8 \cdot 114.0) - (36.0^3) / (8 \cdot 114.0^3))$$

$$= 165.36 \text{ N./mm}^2$$

Dimension [Gtp]:

$$= 2 \cdot \text{Wgp} - \text{Wpl}$$

$$= 2 \cdot 250.0 - 400.0$$

$$= 100.0 \text{ mm.}$$

Angle [αG]:

$$= \pi / 2 - \text{atan} ((\text{Wpl} - \text{Gtp}) / \text{Hgp})$$

$$= \pi / 2 - \text{atan} ((400.0 - 100.0) / 450.0)$$

$$= 56.3 \text{ deg}$$

Maximum Compressive Gusset Plate Stress per Bednar [SgpB]:

$$= \text{Flug} (3 \cdot \text{Dlug} - \text{Wpl}) / (\text{Tgp} \cdot \text{Wpl}^2 \cdot (\text{sind}(\alpha G))^2)$$

$$= 34386 (3 \cdot 300.0 - 400.0) / (50.0 \cdot 400.0^2 \cdot (\text{sind}(56.31))^2)$$

$$= 30.44 \text{ N./mm}^2$$

Gusset Plate Allowable Compressive Stress [SgaB]:

$$= 18000 / (1 + (1 / 18000) \cdot (\text{Hgp} / \text{sind}(\alpha G) / (0.289 \cdot \text{Tgp}))^2)$$

$$= 18000 / (1 + (1 / 18000) \cdot (450.0 / \text{sind}(56.31) / (0.289 \cdot 50.0))^2)$$

$$= 115.15 \text{ N./mm}^2$$

There was no uplift. Please choose an appropriate bolt size for this support design.

Review notes about nozzle loadings and supports in the Warnings report.

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**Input, Nozzle Desc: D1-3"****From: 10**

Pressure for Reinforcement Calculations	P	17.334	bars
Temperature for Internal Pressure	Temp	286	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	286	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	165.12	N./mm ²
Parent Allowable Stress At Ambient	Sva	207.00	N./mm ²
Inside Diameter of Hemispherical Head	D	2064.00	mm.
Head Finished (Minimum) Thickness	t	18.0000	mm.
Head Internal Corrosion Allowance	cas	1.5000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

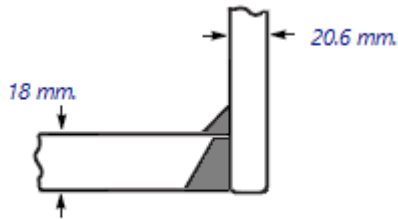
Material		SA-336 F21	
Material UNS Number		K31545	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	169.56	N./mm ²
Allowable Stress At Ambient	Sna	207.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	20.5740	mm.
Flange Material		SA-336 F21	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Outside Projection	ho	300.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
This is a Manway, Coupling or Access Opening.			

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: D1-3"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	3.000	in.
Actual Thickness Used in Calculation	0.810	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 79.2 \cdot (\exp(17.334/(169.558 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 0.4069 + 1.5 + 0.0 = 1.9069 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 169.558 \cdot 1.0 \cdot \ln((2 \cdot 19.074 + 79.2) / 79.2) - 0.0$$

$$= 666.608 - 0.0 = 666.608 \text{ bars}$$

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[169.6/165.1, 1]$$

$$= 1.000$$

Thickness of Nozzle [tn]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 20.574 - 1.5$$

$$= 19.074 \text{ mm.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2064.0/2 + 1.5$$

$$= 1033.500 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1033.5 \cdot 16.5}, 2 \cdot 39.6)$$

$$= 79.200 \text{ mm.}$$

Intermediate value [Cn]:

$$= \min(((t + t_e)/t_n)^{0.35}, 1)$$

$$= \min(((16.5 + 0.0)/19.074)^{0.35}, 1)$$

$$= \min(0.951, 1)$$

$$= 0.951$$

Intermediate value [Cp]:

$$= 9999999.000$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min(t + t_e + F_p \cdot \sqrt{R_n \cdot t_n}, L_{pr1} + t)$$

$$= \min(16.5 + 0.0 + 0.951 \cdot \sqrt{39.6 \cdot 19.074}, 300.0 + 16.5)$$

$$= 42.624 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A_5 \cdot frp) / LR)$$

$$= 16.5 + ((0.0 \cdot 1.0) / 79.2)$$

$$= 16.500 \text{ mm.}$$



Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR$$

$$= 16.5 \cdot 79.2$$

$$= 13.068 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$

$$= 19.074 \cdot 42.624$$

$$= 8.130 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$

$$= 13.068 + 1.0(8.13 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 21.698 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 19.074 / \ln[1 + 19.074/39.6]$$

$$= 48.514 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 16.5 / \ln[1 + 16.5/1033.5]$$

$$= 1041.725 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 165.119 \cdot 1.0$$

$$= 247.7 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$

$$= 17.334 \cdot 48.514 \cdot 42.624$$

$$= 365.5 \text{ Kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} \cdot (LR + t_n)/2$$

$$= 17.334 \cdot 1041.725 \cdot (79.2 + 19.074)/2$$

$$= 9048.3 \text{ Kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= (P \cdot R_{xs} \cdot R_{nc})/2$$

$$= (17.334 \cdot 1041.725 \cdot 39.6)/2$$

$$= 3646.1 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N)/P$$

$$= (9048.3 + 3646.1 + 365.5)/17.334$$

$$= 738.8 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / AT - R_{xs} / (2 \cdot t_{eff}))$$

$$= 247.678 / (2 \cdot 738.812 / 21.698 - 1041.725 / (2 \cdot 16.5))$$

$$= 67.8 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot S / [t / R_{xs}]$$

$$= 2 \cdot 165.119 [16.5 / 1041.725]$$

$$= 52.3 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:



$$\begin{aligned}
 &= \min(P_{\max 1}, P_{\max 2}) \\
 &= \min(67.794, 52.304) \\
 &= 52.304 \text{ bars}
 \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned}
 &= (f_N + f_S + f_Y) / A_T \\
 &= (365.529 + 9048.34 + 3646.074) / 21.698 \\
 &= 59.027 \text{ N./mm}^2
 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned}
 &= P \cdot R_{\text{xs}} / (2 \cdot t_{\text{eff}}) \\
 &= 17.334 \cdot 1041.725 / (2 \cdot 16.5) \\
 &= 54.7 \text{ N./mm}^2
 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned}
 &= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}}) \\
 &= \max(2 \cdot 59.027 - 54.724, 54.724) \\
 &= 63.3 \text{ N./mm}^2
 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	247.68	N./mm ²
Local Primary Membrane Stress	PL	63.33	N./mm ²
Maximum Allowable Working Pressure	Pmax	52.30	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned}
 &= (R_{\text{nc}} + t_n) / R_{\text{nc}} \\
 &= (39.6 + 19.074) / 39.6 \\
 &= 1.482 \text{ For set-in Nozzles}
 \end{aligned}$$

Weld Length of Nozzle to Shell Weld [L_T]:

$$\begin{aligned}
 &= \pi / 2 \cdot (R_n + t_n) \\
 &= \pi / 2 \cdot (39.6 + 19.074) \\
 &= 92.165 \text{ mm.}
 \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$t_c, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 19.074, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$\begin{aligned}
 &= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi / 4 \cdot P \cdot R_n^2 \cdot k_y^2) \\
 &= \min(3646 \cdot 1.48, 1.5 \cdot 169.6(8.389 + 0.0), \pi / 4 \cdot 17.3 \cdot 39.6^2 \cdot 1.48^2) \\
 &= 477.957 \text{ Kg}
 \end{aligned}$$

Weld Stress Value [τ]:

$$\begin{aligned}
 &= fwelds / (L_T(0.49 \cdot L41T + 0.6 \cdot t_{w1} + 0.49 \cdot L43T)) \\
 &= 477.957 / (92.165(0.49 \cdot 7.071 + 0.6 \cdot 16.5 + 0.49 \cdot 0.0)) \\
 &= 3.805 < \text{ or } = \text{ to } 165.119 \text{ Weld Stress Passed}
 \end{aligned}$$

Nozzle Calculations per Section 4.5: External Pressure Case:Nozzle Radius for Force Calculation [R_{xn}]:

$$\begin{aligned}
 &= t_n / \ln[1 + t_n / R_n] \\
 &= 19.074 / \ln[1 + 19.074 / 39.6] \\
 &= 48.514 \text{ mm.}
 \end{aligned}$$

Shell Radius for Force Calculation [R_{xs}]:

$$\begin{aligned}
 &= t_{\text{eff}} / \ln[1 + t_{\text{eff}} / R_{\text{eff}}] \\
 &= 16.5 / \ln[1 + 16.5 / 1033.5] \\
 &= 1041.725 \text{ mm.}
 \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned}
 &= 1.5 \cdot S \\
 &= 1.5 \cdot 165.119 \\
 &= 247.7 \text{ N./mm}^2
 \end{aligned}$$



Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$

$$= 1.034 \cdot 48.514 \cdot 42.624$$

$$= 21.8 \text{ Kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} \cdot (LR + t_n)/2$$

$$= 1.034 \cdot 1041.725 \cdot (79.2 + 19.074)/2$$

$$= 539.8 \text{ Kgf}$$

Discontinuity Force from External Pressure [fY]:

$$= (P \cdot R_{xs} \cdot R_{nc})/2$$

$$= (1.034 \cdot 1041.725 \cdot 39.6)/2$$

$$= 217.5 \text{ Kgf}$$

Area Resisting External Pressure [Ap]:

$$= (fS + fY + fN)/P$$

$$= (539.8 + 217.5 + 21.8)/1.034$$

$$= 738.8 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / AT - R_{xs} / (2 \cdot t_{eff}))$$

$$= 247.678 / (2 \cdot 738.812 / 21.698 - 1041.725 / (2 \cdot 16.5))$$

$$= 67.8 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot S / [t / R_{xs}]$$

$$= 2 \cdot 165.119 [16.5 / 1041.725]$$

$$= 52.3 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(67.794, 52.304)$$

$$= 52.304 \text{ bars}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (fN + fS + fY) / AT$$

$$= (21.808 + 539.832 + 217.528) / 21.698$$

$$= 3.522 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / (2 \cdot t_{eff})$$

$$= 1.034 \cdot 1041.725 / (2 \cdot 16.5)$$

$$= 3.3 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 3.522 - 3.265, 3.265)$$

$$= 3.8 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	247.68	N./mm ²
Local Primary Membrane Stress	PL	3.78	N./mm ²
Maximum Allowable External Pressure	Pmax	52.30	bars

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: A

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	-43 °C
User entered Minimum Design Temperature	:	-10 °C
The Curve for this Material is	:	A
From Figure 3.8M- with PWHT, the MDMT is	:	10 °C
Temperature reduction per Graph 3.13	:	53 °C
The computed stress ratio is	:	0.263

STEAM SUPERHEATER

A150NRJ10EP001

PV Elite 28 U2 Licensee: SL User

FileName : STEAM SUPERHEATER

Nozzle Calcs.: D1-3"

PV Elite®

v28



HEXAGON

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Noz1: 13 4:54pm Sep 16,2026

The computed thickness ratio [sr]:

$= (tr \cdot E^*) / (tnom - ci - cext)$

$= (4.337 \cdot 1.0) / (18.0 - 1.5 - 0.0)$

$= 0.263$

Gov. MDMT of the nozzle to shell joint welded assembly : -43 °C

The Drop for this Nozzle is : 1.6693 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 319.6693 mm.

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**Input, Nozzle Desc: A2-24 "****From: 20**

Pressure for Reinforcement Calculations	P	17.334	bars
Temperature for Internal Pressure	Temp	286	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	286	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	165.12	N./mm ²
Parent Allowable Stress At Ambient	Sva	207.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2032.00	mm.
Design Length of Section	L	5343.9702	mm.
Shell Finished (Minimum) Thickness	t	34.0000	mm.
Shell Internal Corrosion Allowance	cas	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		450.00	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

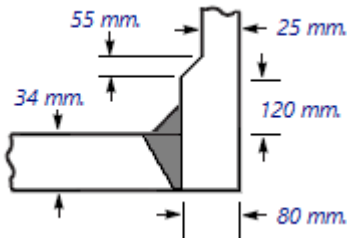
Material		SA-336 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	165.12	N./mm ²
Allowable Stress At Ambient	Sna	207.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		22.0315	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	25.0000	mm.
Hub Height of Integral Nozzle	h	120.0000	mm.
Height of Beveled Transition	L'	55.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		80.0000	mm.
Corrosion Allowance	can	1.5000	mm.
Outside Projection	ho	300.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	34.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: A2-24 "

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	22.031	in.
Actual Thickness Used in Calculation	0.984	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 562.6 \cdot (\exp(17.334/(165.119 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 2.9689 + 1.5 + 0.0 = 4.4689 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 165.119 \cdot 1.0 \cdot \ln((2 \cdot 23.5 + 562.6) / 562.6) - 0.0$$

$$= 132.474 - 0.0 = 132.4737 \text{ bars}$$

This nozzle or its diameter limit intersects weld seam # [3] on this element. The distance between the nozzle and seam is, 159.83 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 541.65 mm.

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	9.840 mm
Min. required shell thickness, internal p + c :	12.239 mm.
Min. required shell thickness, external p + c :	12.555 mm.

Nozzle neck minimum required thickness : 9.840 mm.

Minimum nozzle neck thickness 24.994 mm., is greater than required thickness 9.840 mm.

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained :	43.2,	Allowable :	165.1 N./mm ²	Passed
Expansion :	0.0,	Allowable :	422.0 N./mm ²	Passed
Occasional :	10.0,	Allowable :	219.6 N./mm ²	Passed
Shear :	18.5,	Allowable :	115.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[165.1/165.1, 1]$$

$$= 1.000$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 80.0 - 1.5$$

$$= 78.500 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$



$$= 25.0 - 1.5$$

$$= 23.500 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 2035.0/32.5$$

$$= 62.615 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2032.0/2 + 1.5$$

$$= 1017.500 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{(R_{eff} \cdot t)}, 2 \cdot R_n)$$

$$= \min(\sqrt{(1017.5 \cdot 32.5)}, 2 \cdot 281.3)$$

$$= 181.848 \text{ mm.}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)}$$

$$= \min[1.5 \cdot 32.5, 0.0] + \sqrt{(281.3 \cdot 78.5)}$$

$$= 148.600 \text{ mm.}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 300.0$$

$$= 300.000 \text{ mm.}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(32.5 + 0.0)$$

$$= 260.000 \text{ mm.}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[148.6, 300.0, 260.0] + 32.5$$

$$= 181.100 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 32.5 + ((0.0 \cdot 1.0) / 181.848)$$

$$= 32.500 \text{ mm.}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}}))$$

$$= \min(12, (2 \cdot 281.3 + 78.5) / (\sqrt{(2035.0 + 32.5) \cdot 32.5}))$$

$$= 2.473$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 32.5 \cdot 181.848 \cdot \max((2.473/5)^{0.85}, 1)$$

$$= 59.101 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A2A + A2B$$

$$= 119.713 + 18.361$$

$$= 138.074 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$

$$= 59.101 + 1 \cdot (138.074 + 0.0) + 0.5 + 0.0 + 0.0$$



$$= 197.674 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 78.5 / \ln[1 + 78.5/281.3]$$

$$= 318.941 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 32.5 / \ln[1 + 32.5/1017.5]$$

$$= 1033.664 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 165.119 \cdot 1.0$$

$$= 247.7 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 17.334 \cdot 318.941 \cdot 181.1$$

$$= 10210.3 \text{ Kg}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 17.334 \cdot 1033.664 (181.848 + 78.5)$$

$$= 47570.9 \text{ Kg}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 17.334 \cdot 1033.664 \cdot 281.3$$

$$= 51399.2 \text{ Kg}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (47570.9 + 51399.2 + 10210.3) / 17.334$$

$$= 6176.4 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 247.678 / (2 \cdot 6176.427 / 197.674 - 1033.664 / 32.5)$$

$$= 80.7 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / [t / R_{xs}]$$

$$= 165.119 [32.5 / 1033.664]$$

$$= 51.9 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(80.709, 51.913)$$

$$= 51.913 \text{ bars}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (10210.278 + 47570.898 + 51399.227) / 197.674$$

$$= 54.166 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 17.334 \cdot 1033.664 / 32.5$$

$$= 55.1 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 54.166 - 55.136, 55.136)$$

$$= 55.1 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:



Allowed Local Primary Membrane Stress	Sallow	247.68	N./mm ²
Local Primary Membrane Stress	PL	55.14	N./mm ²
Maximum Allowable Working Pressure	Pmax	51.91	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (281.3 + 78.5) / 281.3$$

$$= 1.279 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [L_T]:

$$= \pi/2 \cdot (R_n + t_n)$$

$$= \pi/2 \cdot (281.3 + 78.5)$$

$$= 565.173 \text{ mm.}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$t_c, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 78.500, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$$

$$= \min(51399 \cdot 1.28, 1.5 \cdot 165.1(138.074 + 0.0), \pi/4 \cdot 17.3 \cdot 281.3^2 \cdot 1.28^2)$$

$$= 17972.961 \text{ Kg}$$

Weld Stress Value [τ]:

$$= fwelds / (L_T(0.49 \cdot L41T + 0.6 \cdot t_{w1} + 0.49 \cdot L43T))$$

$$= 17972.961 / (565.173(0.49 \cdot 7.071 + 0.6 \cdot 32.5 + 0.49 \cdot 0.0))$$

$$= 13.580 < \text{ or } = \text{ to } 165.119 \text{ Weld Stress Passed}$$

Nozzle/Structural Discontinuity Proximity Warning:

This nozzle's diameter limit intersects a major structural discontinuity. Reduce the diameter limit in the nozzle dialog so that there is no interference or move the nozzle farther away from the intersecting weld.

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [R_{xn}]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 78.5 / \ln[1 + 78.5/281.3]$$

$$= 318.941 \text{ mm.}$$

Shell Radius for Force Calculation [R_{xs}]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 32.5 / \ln[1 + 32.5/1017.5]$$

$$= 1033.664 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S$$

$$= 1.5 \cdot 165.119$$

$$= 247.7 \text{ N./mm}^2$$

Force acting on the Nozzle [f_N]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 1.034 \cdot 318.941 \cdot 181.1$$

$$= 609.2 \text{ Kg}$$

Force acting on the Shell [f_S]:

$$= P \cdot R_{xs}(L_R + t_n)$$

$$= 1.034 \cdot 1033.664(181.848 + 78.5)$$

$$= 2838.1 \text{ Kg}$$



Discontinuity Force from External Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 1.034 \cdot 1033.664 \cdot 281.3$$

$$= 3066.5 \text{ Kgf}$$

Area Resisting External Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (2838.1 + 3066.5 + 609.2) / 1.034$$

$$= 6176.4 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 247.678 / (2 \cdot 6176.427 / 197.674 - 1033.664 / 32.5)$$

$$= 80.7 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / [t / R_{xs}]$$

$$= 165.119 [32.5 / 1033.664]$$

$$= 51.9 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(80.709, 51.913)$$

$$= 51.913 \text{ bars}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (609.154 + 2838.12 + 3066.522) / 197.674$$

$$= 3.232 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 1.034 \cdot 1033.664 / 32.5$$

$$= 3.3 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 3.232 - 3.289, 3.289)$$

$$= 3.3 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	247.68	N./mm ²
Local Primary Membrane Stress	PL	3.29	N./mm ²
Maximum Allowable External Pressure	Pmax	51.91	bars

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: A

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	-30 °C
User entered Minimum Design Temperature	:	-10 °C
The Curve for this Material is	:	A
From Figure 3.8M- with PWHT, the MDMT is	:	23 °C
Temperature reduction per Graph 3.13	:	53 °C
The computed stress ratio is	:	0.263

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (8.557 \cdot 1.0) / (34.0 - 1.5 - 0.0)$$

$$= 0.263$$

Gov. MDMT of the nozzle to shell joint welded assembly : -30 °C

The Drop for this Nozzle is : 65.8421 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 399.8421 mm.

Input Echo, WRC297 Item 1, Description: A2-24 " :



Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	2032.000	mm.
Shell Thickness	34.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-387 11	
Vessel Cold S.I. Allowable	Smc 207.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 165.12	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm.
Nozzle Diameter	559.600	mm.
Nozzle Thickness	80.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-336 F11	
Nozzle Cold S.I. Allowable	SNmc 207.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 165.12	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUstained loads.

Design Internal Pressure	Dp	17.33	bars
Radial Load	P	8810.95	Kgf
Circumferential Shear	Vc	8810.95	Kgf
Longitudinal Shear	Vl	6608.18	Kgf
Circumferential Moment	Mc	13739.75	Kg-m.
Longitudinal Moment	Ml	13739.75	Kg-m.
Torsional Moment	Mt	15853.58	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, D_n/T_n (8.995) is < 10
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 719.6 + 2 \cdot 1.65 \cdot \sqrt{1033.75(34.0 - 1.5)}$$

$$= 1324.472 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2067.500	mm.
Nozzle Outside Diameter	(d) =	719.600	mm.
Vessel Thickness used	(T) =	32.500	mm.
Nozzle Thickness used	(t) =	78.500	mm.
T / t	=	0.414	
d / t	=	10.000	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.776$$

Nr/P	=	0.061
Mr/P	=	0.111
M0/P	=	0.035
N0/P	=	0.081
MrD/Mc	=	0.268



NrDL/Mc = 0.098
 MØd/Mc = 0.081
 NØDL/Mc = 0.084
 MrD/Ml = 0.095
 NrDL/Ml = 0.055
 MØD/Ml = 0.026
 NØDL/Ml = 0.094

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses Normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	-6	-6	-6	-6
Outplane Bending (P)	-17	17	-17	17
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-16	-16	16	16
Outplane Bending (ML)	-28	28	28	-28
Normal Pressure Stress	53	55	53	55
Outplane Stress Summary	-15	77	74	54

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	-5	-5	-5	-5
Inplane Bending (P)	-54	54	-54	54
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-9	-9	9	9
Inplane Bending (ML)	-100	100	100	-100
Inplane Pressure Stress	26	26	26	26
Inplane Stress Summary	-143	167	77	-14

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	2	2	-2	-2
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	5	5	5	5
Shear Stress Summary	8	8	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	144	168	80	69

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	-6	-6	-6	-6
Outplane Bending (P)	-17	17	-17	17
Outplane Membrane (Mc)	-14	-14	14	14
Outplane Bending (Mc)	-86	86	86	-86
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	26	26	26	26

Outplane Stress Summary	-97	108	103	-33
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	-5	-5	-5	-5
Inplane Bending (P)	-54	54	-54	54
Inplane Membrane (Mc)	-17	-17	17	17
Inplane Bending (Mc)	-285	285	285	-285
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	53	55	53	55

Inplane Stress Summary	-308	372	296	-163
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	5	5	5	5

Shear Stress Summary	4	4	7	7
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	308	372	296	163

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	-6	-6	-6	-6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-16	-16	16	16
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6	8	6	8

Hoop Stress Summary	-16	-14	16	18
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	-6	6	-6	6
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-5	-5	5	5
Axial Bending (ML)	-12	12	12	-12
Axial Pressure Stress	3	3	3	3

Axial Stress Summary	-22	15	14	3
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
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(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	2	2	2	2
Shear Stress Summary	3	3	1	1

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside
Two * Max Shear Stress	23	31	17	18

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside
Hoop Membrane (P)	-6	-6	-6	-6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-14	-14	14	14
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6	8	6	8
Hoop Stress Summary	-14	-12	14	16

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	-6	6	-6	6
Axial Membrane (Mc)	-5	-5	5	5
Axial Bending (Mc)	-39	39	39	-39
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	3	3	3	3
Axial Stress Summary	-49	43	41	-24

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	2	2	2	2
Shear Stress Summary	1	1	3	3

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside
Two * Max Shear Stress	49	56	42	41

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl



Circ. Pm (SUS)	53	55	53	55	53	55	53	55
Circ. Pl (SUS)	-22	-22	10	10	-22	-22	12	12
Circ. Q (SUS)	-45	45	11	-11	-339	339	231	-231
Long. Pm (SUS)	26	26	26	26	26	26	26	26
Long. Pl (SUS)	-14	-14	4	4	-20	-20	8	8
Long. Q (SUS)	-154	154	46	-46	-103	103	69	-69
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-1	-1	1	1
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	53.0	55.0	53.0	55.0	53.0	55.0	53.0	55.0
Pm+Pl (SUS)	31.2	33.2	63.1	65.1	31.0	33.0	65.0	67.0
Pm+Pl+Q (Total)	142.4	166.6	78.2	70.3	308.1	372.1	296.2	164.3

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	55.00	165.12	Passed
Pm+Pl (SUS)	67.03	247.68	Passed
Pm+Pl+Q (TOTAL)	372.06	495.36	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	6	8	6	8	6	8	6	8
Circ. Pl (SUS)	-22	-22	10	10	-20	-20	8	8
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	3	3	3	3	3	3	3	3
Long. Pl (SUS)	-5	-5	5	5	-5	-5	5	5
Long. Q (SUS)	-18	18	6	-6	-45	45	33	-33
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	6.0	8.0	6.0	8.0	6.0	8.0	6.0	8.0
Pm+Pl (SUS)	16.0	14.0	16.0	18.0	14.0	12.0	14.0	16.0
Pm+Pl+Q (Total)	20.8	30.3	17.2	18.2	47.1	55.1	41.1	41.2

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	8.00	165.12	Passed
Pm+Pl (SUS)	18.00	247.68	Passed
Pm+Pl+Q (TOTAL)	55.15	495.36	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

**Input, Nozzle Desc: B2-24 "****From: 40**

Pressure for Reinforcement Calculations	P	17.334	bars
Temperature for Internal Pressure	Temp	286	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	286	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	165.12	N./mm ²
Parent Allowable Stress At Ambient	Sva	207.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2032.00	mm.
Design Length of Section	L	5343.9697	mm.
Shell Finished (Minimum) Thickness	t	34.0000	mm.
Shell Internal Corrosion Allowance	cas	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		4250.00	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

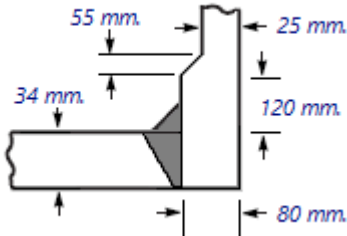
Material		SA-336 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	165.12	N./mm ²
Allowable Stress At Ambient	Sna	207.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		22.0315	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	25.0000	mm.
Hub Height of Integral Nozzle	h	120.0000	mm.
Height of Beveled Transition	L'	55.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		80.0000	mm.
Corrosion Allowance	can	1.5000	mm.
Outside Projection	ho	300.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	34.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: B2-24 "

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	22.031	in.
Actual Thickness Used in Calculation	0.984	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 562.6 \cdot (\exp(17.334/(165.119 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 2.9689 + 1.5 + 0.0 = 4.4689 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 165.119 \cdot 1.0 \cdot \ln((2 \cdot 23.5 + 562.6) / 562.6) - 0.0$$

$$= 132.474 - 0.0 = 132.4737 \text{ bars}$$

This nozzle or its diameter limit intersects weld seam # [3] on this element. The distance between the nozzle and seam is, 195.34 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 541.65 mm.

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	9.840 mm
Min. required shell thickness, internal p + c :	12.239 mm.
Min. required shell thickness, external p + c :	12.555 mm.

Nozzle neck minimum required thickness : 9.840 mm.

Minimum nozzle neck thickness 24.994 mm., is greater than required thickness 9.840 mm.

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained :	43.2,	Allowable :	165.1 N./mm ²	Passed
Expansion :	0.0,	Allowable :	422.0 N./mm ²	Passed
Occasional :	10.0,	Allowable :	219.6 N./mm ²	Passed
Shear :	18.5,	Allowable :	115.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[165.1/165.1, 1]$$

$$= 1.000$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 80.0 - 1.5$$

$$= 78.500 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$



$$= 25.0 - 1.5$$

$$= 23.500 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 2035.0/32.5$$

$$= 62.615 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2032.0/2 + 1.5$$

$$= 1017.500 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{(Reff \cdot t)}, 2 \cdot R_n)$$

$$= \min(\sqrt{(1017.5 \cdot 32.5)}, 2 \cdot 281.3)$$

$$= 181.848 \text{ mm.}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)}$$

$$= \min[1.5 \cdot 32.5, 0.0] + \sqrt{(281.3 \cdot 78.5)}$$

$$= 148.600 \text{ mm.}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 300.0$$

$$= 300.000 \text{ mm.}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(32.5 + 0.0)$$

$$= 260.000 \text{ mm.}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[148.6, 300.0, 260.0] + 32.5$$

$$= 181.100 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 32.5 + ((0.0 \cdot 1.0) / 181.848)$$

$$= 32.500 \text{ mm.}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + teff) \cdot teff}))$$

$$= \min(12, (2 \cdot 281.3 + 78.5) / (\sqrt{(2035.0 + 32.5) \cdot 32.5}))$$

$$= 2.473$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 32.5 \cdot 181.848 \cdot \max((2.473/5)^{0.85}, 1)$$

$$= 59.101 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A2A + A2B$$

$$= 119.713 + 18.361$$

$$= 138.074 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot Leg41^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + frn(A2 + A3) + A41 + A42 + A43$$

$$= 59.101 + 1 \cdot (138.074 + 0.0) + 0.5 + 0.0 + 0.0$$



$$= 197.674 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 78.5 / \ln[1 + 78.5/281.3]$$

$$= 318.941 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 32.5 / \ln[1 + 32.5/1017.5]$$

$$= 1033.664 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 165.119 \cdot 1.0$$

$$= 247.7 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 17.334 \cdot 318.941 \cdot 181.1$$

$$= 10210.3 \text{ Kg}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 17.334 \cdot 1033.664 (181.848 + 78.5)$$

$$= 47570.9 \text{ Kg}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 17.334 \cdot 1033.664 \cdot 281.3$$

$$= 51399.2 \text{ Kg}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (47570.9 + 51399.2 + 10210.3) / 17.334$$

$$= 6176.4 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 247.678 / (2 \cdot 6176.427 / 197.674 - 1033.664 / 32.5)$$

$$= 80.7 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / [t / R_{xs}]$$

$$= 165.119 [32.5 / 1033.664]$$

$$= 51.9 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(80.709, 51.913)$$

$$= 51.913 \text{ bars}$$

Average Primary Membrane Stress [σAvg]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (10210.278 + 47570.898 + 51399.227) / 197.674$$

$$= 54.166 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 17.334 \cdot 1033.664 / 32.5$$

$$= 55.1 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 54.166 - 55.136, 55.136)$$

$$= 55.1 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:



Allowed Local Primary Membrane Stress	Sallow	247.68	N./mm ²
Local Primary Membrane Stress	PL	55.14	N./mm ²
Maximum Allowable Working Pressure	Pmax	51.91	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (281.3 + 78.5) / 281.3$$

$$= 1.279 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [L_T]:

$$= \pi/2 \cdot (R_n + t_n)$$

$$= \pi/2 \cdot (281.3 + 78.5)$$

$$= 565.173 \text{ mm.}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$t_c, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 78.500, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$$

$$= \min(51399 \cdot 1.28, 1.5 \cdot 165.1(138.074 + 0.0), \pi/4 \cdot 17.3 \cdot 281.3^2 \cdot 1.28^2)$$

$$= 17972.961 \text{ Kg}$$

Weld Stress Value [τ]:

$$= fwelds / (L_T(0.49 \cdot L41T + 0.6 \cdot t_{w1} + 0.49 \cdot L43T))$$

$$= 17972.961 / (565.173(0.49 \cdot 7.071 + 0.6 \cdot 32.5 + 0.49 \cdot 0.0))$$

$$= 13.580 < \text{ or } = \text{ to } 165.119 \text{ Weld Stress Passed}$$

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [R_{xn}]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 78.5 / \ln[1 + 78.5/281.3]$$

$$= 318.941 \text{ mm.}$$

Shell Radius for Force Calculation [R_{xs}]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 32.5 / \ln[1 + 32.5/1017.5]$$

$$= 1033.664 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S$$

$$= 1.5 \cdot 165.119$$

$$= 247.7 \text{ N./mm}^2$$

Force acting on the Nozzle [f_N]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 1.034 \cdot 318.941 \cdot 181.1$$

$$= 609.2 \text{ Kg}$$

Force acting on the Shell [f_S]:

$$= P \cdot R_{xs}(L_R + t_n)$$

$$= 1.034 \cdot 1033.664(181.848 + 78.5)$$

$$= 2838.1 \text{ Kg}$$

Discontinuity Force from External Pressure [f_Y]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 1.034 \cdot 1033.664 \cdot 281.3$$

$$= 3066.5 \text{ Kg}$$

Area Resisting External Pressure [A_p]:



$$\begin{aligned}
 &= (fS + fY + fN)/P \\
 &= (2838.1 + 3066.5 + 609.2)/1.034 \\
 &= 6176.4 \text{ cm}^2
 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned}
 &= Sallow / (2 \cdot Ap/AT - Rxs/teff) \\
 &= 247.678 / (2 \cdot 6176.427/197.674 - 1033.664/32.5) \\
 &= 80.7 \text{ bars}
 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned}
 &= S[t/Rxs] \\
 &= 165.119[32.5/1033.664] \\
 &= 51.9 \text{ bars}
 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned}
 &= \min(Pmax1, Pmax2) \\
 &= \min(80.709, 51.913) \\
 &= 51.913 \text{ bars}
 \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned}
 &= (fN + fS + fY) / AT \\
 &= (609.154 + 2838.12 + 3066.522) / 197.674 \\
 &= 3.232 \text{ N./mm}^2
 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned}
 &= P \cdot Rxs / teff \\
 &= 1.034 \cdot 1033.664 / 32.5 \\
 &= 3.3 \text{ N./mm}^2
 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned}
 &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\
 &= \max(2 \cdot 3.232 - 3.289, 3.289) \\
 &= 3.3 \text{ N./mm}^2
 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	247.68	N./mm ²
Local Primary Membrane Stress	PL	3.29	N./mm ²
Maximum Allowable External Pressure	Pmax	51.91	bars

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: A**

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	-30 °C
User entered Minimum Design Temperature	:	-10 °C
The Curve for this Material is	:	A
From Figure 3.8M- with PWHT, the MDMT is	:	23 °C
Temperature reduction per Graph 3.13	:	53 °C
The computed stress ratio is	:	0.263

The computed thickness ratio [sr]:

$$\begin{aligned}
 &= (tr \cdot E^*) / (tnom - ci - cext) \\
 &= (8.557 \cdot 1.0) / (34.0 - 1.5 - 0.0) \\
 &= 0.263
 \end{aligned}$$

Gov. MDMT of the nozzle to shell joint welded assembly : -30 °C

The Drop for this Nozzle is : 65.8421 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 399.8421 mm.

Input Echo, WRC297 Item 1, Description: B2-24 " :

Diameter Basis for Cylindrical Shell	ID
Shell Internal Corrosion Allowance	1.5000 mm.
Shell External Corrosion Allowance	0.0000 mm.
Shell Diameter	2032.000 mm.
Shell Thickness	34.0000 mm.



Shell Stress Concentration Factor		1.000	
Vessel Material		SA-387 11	
Vessel Cold S.I. Allowable	Smc	207.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	165.12	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance	1.5000	mm.	
Nozzle Diameter	559.600	mm.	
Nozzle Thickness	80.0000	mm.	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-336 F11		
Nozzle Cold S.I. Allowable	SNmc	207.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	165.12	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	17.33	bars
Radial Load	P	8810.95	Kgf
Circumferential Shear	Vc	8810.95	Kgf
Longitudinal Shear	Vl	6608.18	Kgf
Circumferential Moment	Mc	13739.75	Kg-m.
Longitudinal Moment	Ml	13739.75	Kg-m.
Torsional Moment	Mt	15853.58	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (8.995) is < 10
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 719.6 + 2 \cdot 1.65 \cdot \sqrt{(1033.75(34.0 - 1.5))}$$

$$= 1324.472 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2067.500	mm.
Nozzle Outside Diameter	(d) =	719.600	mm.
Vessel Thickness used	(T) =	32.500	mm.
Nozzle Thickness used	(t) =	78.500	mm.
T / t	=	0.414	
d / t	=	10.000	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.776$$

Nr/P	=	0.061
Mr/P	=	0.111
M0/P	=	0.035
N0/P	=	0.081
MrD/Mc	=	0.268
NrDL/Mc	=	0.098
M0d/Mc	=	0.081
N0DL/Mc	=	0.084
MrD/Ml	=	0.095
NrDL/Ml	=	0.055
M0D/Ml	=	0.026

NØDL/M1 = 0.094

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Membrane (P)	-6	-6	-6	-6
Outplane Bending (P)	-17	17	-17	17
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-16	-16	16	16
Outplane Bending (ML)	-28	28	28	-28
Normal Pressure Stress	53	55	53	55
Outplane Stress Summary	-15	77	74	54

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Inplane Membrane (P)	-5	-5	-5	-5
Inplane Bending (P)	-54	54	-54	54
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-9	-9	9	9
Inplane Bending (ML)	-100	100	100	-100
Inplane Pressure Stress	26	26	26	26
Inplane Stress Summary	-143	167	77	-14

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Shear (Vc)	2	2	-2	-2
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	5	5	5	5
Shear Stress Summary	8	8	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Two * Max Shear Stress	144	168	80	69

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Outplane Membrane (P)	-6	-6	-6	-6
Outplane Bending (P)	-17	17	-17	17
Outplane Membrane (Mc)	-14	-14	14	14
Outplane Bending (Mc)	-86	86	86	-86
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	26	26	26	26
Outplane Stress Summary	-97	108	103	-33

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to	Cu Left	Cl Left	Du Right	D1 Right
--	------------	------------	-------------	-------------



circumferential plane)	Outside	Inside	Outside	Inside
Inplane Membrane (P)	-5	-5	-5	-5
Inplane Bending (P)	-54	54	-54	54
Inplane Membrane (Mc)	-17	-17	17	17
Inplane Bending (Mc)	-285	285	285	-285
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	53	55	53	55
Inplane Stress Summary	-308	372	296	-163

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	5	5	5	5
Shear Stress Summary	4	4	7	7

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	308	372	296	163

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	-6	-6	-6	-6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-16	-16	16	16
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6	8	6	8
Hoop Stress Summary	-16	-14	16	18

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	-6	6	-6	6
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-5	-5	5	5
Axial Bending (ML)	-12	12	12	-12
Axial Pressure Stress	3	3	3	3
Axial Stress Summary	-22	15	14	3

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	2	2	2	2

Shear Stress Summary 3 3 1 1

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	23	31	17	18

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	-6	-6	-6	-6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-14	-14	14	14
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6	8	6	8
Hoop Stress Summary	-14	-12	14	16

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	-6	6	-6	6
Axial Membrane (Mc)	-5	-5	5	5
Axial Bending (Mc)	-39	39	39	-39
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	3	3	3	3
Axial Stress Summary	-49	43	41	-24

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	2	2	2	2
Shear Stress Summary	1	1	3	3

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	49	56	42	41

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	53	55	53	55	53	55	53	55
Circ. Pl (SUS)	-22	-22	10	10	-22	-22	12	12
Circ. Q (SUS)	-45	45	11	-11	-339	339	231	-231
Long. Pm (SUS)	26	26	26	26	26	26	26	26
Long. Pl (SUS)	-14	-14	4	4	-20	-20	8	8



Long. Q (SUS)	-154	154	46	-46	-103	103	69	-69
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	2	2	-2	-2	-1	-1	1	1
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	53.0	55.0	53.0	55.0	53.0	55.0	53.0	55.0
Pm+P1 (SUS)	31.2	33.2	63.1	65.1	31.0	33.0	65.0	67.0
Pm+P1+Q (Total)	142.4	166.6	78.2	70.3	308.1	372.1	296.2	164.3

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	55.00	165.12	Passed
Pm+P1 (SUS)	67.03	247.68	Passed
Pm+P1+Q (TOTAL)	372.06	495.36	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	6	8	6	8	6	8	6	8
Circ. P1 (SUS)	-22	-22	10	10	-20	-20	8	8
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	3	3	3	3	3	3	3	3
Long. P1 (SUS)	-5	-5	5	5	-5	-5	5	5
Long. Q (SUS)	-18	18	6	-6	-45	45	33	-33
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	6.0	8.0	6.0	8.0	6.0	8.0	6.0	8.0
Pm+P1 (SUS)	16.0	14.0	16.0	18.0	14.0	12.0	14.0	16.0
Pm+P1+Q (Total)	20.8	30.3	17.2	18.2	47.1	55.1	41.1	41.2

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	8.00	165.12	Passed
Pm+P1 (SUS)	18.00	247.68	Passed
Pm+P1+Q (TOTAL)	55.15	495.36	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

**Input, Nozzle Desc: A1-24"****From: 50**

Pressure for Reinforcement Calculations	P	17.152	bars
Temperature for Internal Pressure	Temp	286	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	286	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	165.12	N./mm ²
Parent Allowable Stress At Ambient	Sva	207.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1964.00	mm.
Design Length of Section	L	6350.0303	mm.
Shell Finished (Minimum) Thickness	t	136.0000	mm.
Shell Internal Corrosion Allowance	cas	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		7100.00	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

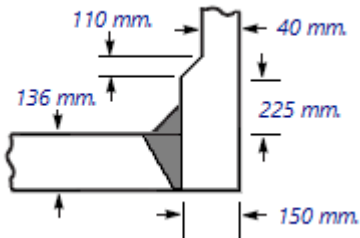
Material		SA-336 F21	
Material UNS Number		K31545	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	169.56	N./mm ²
Allowable Stress At Ambient	Sna	207.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		20.8504	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	40.0000	mm.
Hub Height of Integral Nozzle	h	225.0000	mm.
Height of Beveled Transition	L'	110.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		150.0000	mm.
Corrosion Allowance	can	1.5000	mm.
Outside Projection	ho	300.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	136.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: A1-24"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	20.850	in.
Actual Thickness Used in Calculation	1.575	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 532.6 \cdot (\exp(17.152/(169.558 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 2.7076 + 1.5 + 0.0 = 4.2076 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 169.558 \cdot 1.0 \cdot \ln((2 \cdot 38.5 + 532.6) / 532.6) - 0.0$$

$$= 228.945 - 0.0 = 228.9446 \text{ bars}$$

This nozzle or its diameter limit intersects weld seam # [3] on this element. The distance between the nozzle and seam is, 154.49 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 778.50 mm.

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	9.840 mm
Min. required shell thickness, internal p + c :	11.770 mm.
Min. required shell thickness, external p + c :	13.652 mm.

Nozzle neck minimum required thickness : 9.840 mm.

Minimum nozzle neck thickness 40.005 mm., is greater than required thickness 9.840 mm.

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained :	14.1,	Allowable :	169.6 N./mm ²	Passed
Expansion :	0.0,	Allowable :	456.6 N./mm ²	Passed
Occasional :	5.5,	Allowable :	225.5 N./mm ²	Passed
Shear :	5.5,	Allowable :	118.7 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[169.6/165.1, 1]$$

$$= 1.000$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 150.0 - 1.5$$

$$= 148.500 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$



$$= 40.0 - 1.5$$

$$= 38.500 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 1967.0/134.5$$

$$= 14.625 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 1964.0/2 + 1.5$$

$$= 983.500 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{R_{eff} \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{983.5 \cdot 134.5}, 2 \cdot 266.3)$$

$$= 363.704 \text{ mm.}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 134.5, 0.0] + \sqrt{266.3 \cdot 148.5}$$

$$= 198.861 \text{ mm.}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 300.0$$

$$= 300.000 \text{ mm.}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(134.5 + 0.0)$$

$$= 1076.000 \text{ mm.}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[198.861, 300.0, 1076.0] + 134.5$$

$$= 333.361 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 134.5 + ((0.0 \cdot 1.0) / 363.704)$$

$$= 134.500 \text{ mm.}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}}))$$

$$= \min(12, (2 \cdot 266.3 + 148.5) / (\sqrt{(1967.0 + 134.5) \cdot 134.5}))$$

$$= 1.281$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 134.5 \cdot 363.704 \cdot \max((1.281/5)^{0.85}, 1)$$

$$= 489.182 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$

$$= 148.5 \cdot 333.361$$

$$= 495.040 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$

$$= 489.182 + 1.0(495.04 + 0.0) + 0.5 + 0.0 + 0.0$$



$$= 984.723 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 148.5 / \ln[1 + 148.5/266.3]$$

$$= 335.084 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 134.5 / \ln[1 + 134.5/983.5]$$

$$= 1049.313 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 165.119 \cdot 1.0$$

$$= 247.7 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 17.152 \cdot 335.084 \cdot 333.361$$

$$= 19537.5 \text{ Kg}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 17.152 \cdot 1049.313 (363.704 + 148.5)$$

$$= 94004.8 \text{ Kg}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 17.152 \cdot 1049.313 \cdot 266.3$$

$$= 48874.0 \text{ Kg}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (94004.8 + 48874.0 + 19537.5) / 17.152$$

$$= 9286.0 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 247.678 / (2 \cdot 9285.985 / 984.723 - 1049.313 / 134.5)$$

$$= 224.0 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / [t / R_{xs}]$$

$$= 165.119 [134.5 / 1049.313]$$

$$= 211.6 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(223.958, 211.636)$$

$$= 211.636 \text{ bars}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (19537.5 + 94004.75 + 48873.996) / 984.723$$

$$= 16.175 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 17.152 \cdot 1049.313 / 134.5$$

$$= 13.4 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 16.175 - 13.382, 13.382)$$

$$= 19.0 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:



Allowed Local Primary Membrane Stress	Sallow	247.68	N./mm ²
Local Primary Membrane Stress	PL	18.97	N./mm ²
Maximum Allowable Working Pressure	Pmax	211.64	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (266.3 + 148.5) / 266.3$$

$$= 1.558 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [L_T]:

$$= \pi/2 \cdot (R_n + t_n)$$

$$= \pi/2 \cdot (266.3 + 148.5)$$

$$= 651.566 \text{ mm.}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$t_c, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 148.500, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$$

$$= \min(48874 \cdot 1.56, 1.5 \cdot 169.6(495.04 + 0.0), \pi/4 \cdot 17.2 \cdot 266.3^2 \cdot 1.56^2)$$

$$= 23635.721 \text{ Kg}$$

Weld Stress Value [τ]:

$$= fwelds / (L_T(0.49 \cdot L41T + 0.6 \cdot t_{w1} + 0.49 \cdot L43T))$$

$$= 23635.721 / (651.566(0.49 \cdot 7.071 + 0.6 \cdot 134.5 + 0.49 \cdot 0.0))$$

$$= 4.227 < \text{or to } 165.119 \text{ Weld Stress Passed}$$

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [R_{xn}]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 148.5 / \ln[1 + 148.5/266.3]$$

$$= 335.084 \text{ mm.}$$

Shell Radius for Force Calculation [R_{xs}]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 134.5 / \ln[1 + 134.5/983.5]$$

$$= 1049.313 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S$$

$$= 1.5 \cdot 165.119$$

$$= 247.7 \text{ N./mm}^2$$

Force acting on the Nozzle [f_N]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 1.034 \cdot 335.084 \cdot 333.361$$

$$= 1178.1 \text{ Kg}$$

Force acting on the Shell [f_S]:

$$= P \cdot R_{xs}(L_R + t_n)$$

$$= 1.034 \cdot 1049.313(363.704 + 148.5)$$

$$= 5668.2 \text{ Kg}$$

Discontinuity Force from External Pressure [f_Y]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 1.034 \cdot 1049.313 \cdot 266.3$$

$$= 2947.0 \text{ Kg}$$

Area Resisting External Pressure [A_p]:



$$= (fS + fY + fN) / P$$

$$= (5668.2 + 2947.0 + 1178.1) / 1.034$$

$$= 9286.0 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap / AT - Rxs / teff)$$

$$= 247.678 / (2 \cdot 9285.984 / 984.723 - 1049.313 / 134.5)$$

$$= 224.0 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t / Rxs]$$

$$= 165.119 [134.5 / 1049.313]$$

$$= 211.6 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(223.958, 211.636)$$

$$= 211.636 \text{ bars}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (fN + fS + fY) / AT$$

$$= (1178.052 + 5668.2 + 2946.953) / 984.723$$

$$= 0.975 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot Rxs / teff$$

$$= 1.034 \cdot 1049.313 / 134.5$$

$$= 0.8 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 0.975 - 0.807, 0.807)$$

$$= 1.1 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	247.68	N./mm ²
Local Primary Membrane Stress	PL	1.14	N./mm ²
Maximum Allowable External Pressure	Pmax	211.64	bars

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: A

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm.. This component must be impact tested.

The Drop for this Nozzle is : 91.9073 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 527.9073 mm.

Input Echo, WRC297 Item 1, Description: A1-24" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	1964.000	mm.
Shell Thickness	136.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-387 11	
Vessel Cold S.I. Allowable	Smc	207.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	165.12 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.



Nozzle Corrosion Allowance	1.5000	mm.
Nozzle Diameter	529.600	mm.
Nozzle Thickness	150.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-336 F21	
Nozzle Cold S.I. Allowable	SNmc 207.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 169.56	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	17.15	bars
Radial Load	P	-3915.97	Kgf
Circumferential Shear	Vc	3915.97	Kgf
Longitudinal Shear	Vl	2936.98	Kgf
Circumferential Moment	Mc	6106.56	Kg-m.
Longitudinal Moment	Ml	6106.56	Kg-m.
Torsional Moment	Mt	7046.03	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, D_v/T_v (16.441) is < 20
check the limitations of WRC bulletin 297.
Warning - The ratio, D_n/T_n (5.531) is < 10
check the limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$$

$$= 829.6 + 2 \cdot 1.65 \cdot \sqrt{(1050.75(136.0 - 1.5))}$$

$$= 2070.180 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2101.500	mm.
Nozzle Outside Diameter	(d) =	829.600	mm.
Vessel Thickness used	(T) =	134.500	mm.
Nozzle Thickness used	(t) =	148.500	mm.
T / t	=	0.906	
d / t	=	10.000	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.560$$

Nr/P	=	0.096
Mr/P	=	0.144
M0/P	=	0.047
N0/P	=	0.109
MrD/Mc	=	0.267
NrDL/Mc	=	0.136
M0d/Mc	=	0.095
N0DL/Mc	=	0.053
MrD/Ml	=	0.142
NrDL/Ml	=	0.085
M0D/Ml	=	0.051
N0DL/Ml	=	0.136

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside



Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	0	0	0	0
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	-1	1	1	-1
Normal Pressure Stress	11	13	11	13

Outplane Stress Summary	10	13	14	12
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	1	-1	1	-1
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-3	3	3	-3
Inplane Pressure Stress	5	5	5	5

Inplane Stress Summary	4	7	11	1
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0

Shear Stress Summary	0	0	0	0
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	10	13	14	12
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	0	0	0	0
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-2	2	2	-2
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	5	5	5	5

Outplane Stress Summary	4	7	9	3
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	1	-1	1	-1
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-6	6	6	-6
Inplane Membrane (ML)	0	0	0	0



Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	11	13	11	13

Inplane Stress Summary	6	17	20	5
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	0	0	0	0

Shear Stress Summary	0	0	0	0
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	6	17	20	6
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	4	2	4

Hoop Stress Summary	2	4	3	5
---------------------	---	---	---	---

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-1	-1	1	1
Axial Bending (ML)	-1	1	1	-1
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	0	1	5	0
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	0	0	0	0
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Outside	Inside	Outside	Inside

Two * Max Shear Stress	3	4	5	5
------------------------	---	---	---	---

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	4	2	4
Hoop Stress Summary	2	4	3	5

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	-1	-1	1	1
Axial Bending (Mc)	-3	3	3	-3
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1
Axial Stress Summary	-2	3	7	-2

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	5	5	7	7

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	11	13	11	13	11	13	11	13
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	-1	1	1	-1	-5	5	7	-7
Long. Pm (SUS)	5	5	5	5	5	5	5	5
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	-2	2	4	-4	-2	2	2	-2
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	11.0	13.0	11.0	13.0	11.0	13.0	11.0	13.0



Pm+Pl (SUS)	11.0	13.0	11.0	13.0	11.0	13.0	11.0	13.0
Pm+Pl+Q (Total)	10.0	14.0	12.0	12.0	6.0	18.0	18.0	6.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	13.00	165.12	Passed
Pm+Pl (SUS)	13.00	247.68	Passed
Pm+Pl+Q (TOTAL)	18.00	495.36	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	2	4	2	4	2	4	2	4
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	-1	-1	1	1	-1	-1	1	1
Long. Q (SUS)	0	0	2	-2	-2	2	4	-4
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	2.0	4.0	2.0	4.0	2.0	4.0	2.0	4.0
Pm+Pl (SUS)	2.0	4.0	2.0	4.0	2.0	4.0	2.0	4.0
Pm+Pl+Q (Total)	2.0	4.0	4.0	4.0	4.0	4.0	6.0	6.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	4.00	169.56	Passed
Pm+Pl (SUS)	4.00	254.34	Passed
Pm+Pl+Q (TOTAL)	6.00	508.67	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

Input, Nozzle Desc: B1-24"

From: 60

Pressure for Reinforcement Calculations	P	17.152	bars
Temperature for Internal Pressure	Temp	286	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	286	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	165.12	N./mm ²
Parent Allowable Stress At Ambient	Sva	207.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1964.00	mm.
Design Length of Section	L	6350.0308	mm.
Shell Finished (Minimum) Thickness	t	136.0000	mm.
Shell Internal Corrosion Allowance	cas	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		10100.00	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

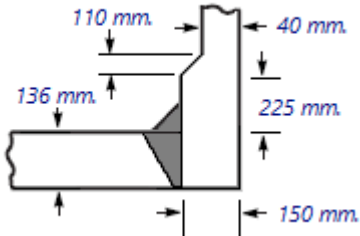
Material		SA-336 F21	
Material UNS Number		K31545	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	169.56	N./mm ²
Allowable Stress At Ambient	Sna	207.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		20.8504	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	40.0000	mm.
Hub Height of Integral Nozzle	h	225.0000	mm.
Height of Beveled Transition	L'	110.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		150.0000	mm.
Corrosion Allowance	can	1.5000	mm.
Outside Projection	ho	300.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	136.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: B1-24"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	20.850	in.
Actual Thickness Used in Calculation	1.575	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 532.6 \cdot (\exp(17.152/(169.558 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 2.7076 + 1.5 + 0.0 = 4.2076 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 169.558 \cdot 1.0 \cdot \ln((2 \cdot 38.5 + 532.6) / 532.6) - 0.0$$

$$= 228.945 - 0.0 = 228.9446 \text{ bars}$$

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 772.43 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 778.50 mm.

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	9.840 mm
Min. required shell thickness, internal p + c :	11.770 mm.
Min. required shell thickness, external p + c :	13.652 mm.

Nozzle neck minimum required thickness : 9.840 mm.

Minimum nozzle neck thickness 40.005 mm., is greater than required thickness 9.840 mm.

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained :	14.1,	Allowable :	169.6 N./mm ²	Passed
Expansion :	0.0,	Allowable :	456.6 N./mm ²	Passed
Occasional :	5.5,	Allowable :	225.5 N./mm ²	Passed
Shear :	5.5,	Allowable :	118.7 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[169.6/165.1, 1]$$

$$= 1.000$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 150.0 - 1.5$$

$$= 148.500 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$



$$= 40.0 - 1.5$$

$$= 38.500 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 1967.0/134.5$$

$$= 14.625 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 1964.0/2 + 1.5$$

$$= 983.500 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{R_{eff} \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{983.5 \cdot 134.5}, 2 \cdot 266.3)$$

$$= 363.704 \text{ mm.}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)}$$

$$= \min[1.5 \cdot 134.5, 0.0] + \sqrt{(266.3 \cdot 148.5)}$$

$$= 198.861 \text{ mm.}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 300.0$$

$$= 300.000 \text{ mm.}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(134.5 + 0.0)$$

$$= 1076.000 \text{ mm.}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[198.861, 300.0, 1076.0] + 134.5$$

$$= 333.361 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 134.5 + ((0.0 \cdot 1.0) / 363.704)$$

$$= 134.500 \text{ mm.}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}}))$$

$$= \min(12, (2 \cdot 266.3 + 148.5) / (\sqrt{(1967.0 + 134.5) \cdot 134.5}))$$

$$= 1.281$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 134.5 \cdot 363.704 \cdot \max((1.281/5)^{0.85}, 1)$$

$$= 489.182 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$

$$= 148.5 \cdot 333.361$$

$$= 495.040 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$

$$= 489.182 + 1.0(495.04 + 0.0) + 0.5 + 0.0 + 0.0$$



$$= 984.723 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 148.5 / \ln[1 + 148.5/266.3]$$

$$= 335.084 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 134.5 / \ln[1 + 134.5/983.5]$$

$$= 1049.313 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 165.119 \cdot 1.0$$

$$= 247.7 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 17.152 \cdot 335.084 \cdot 333.361$$

$$= 19537.5 \text{ Kg}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 17.152 \cdot 1049.313 (363.704 + 148.5)$$

$$= 94004.8 \text{ Kg}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 17.152 \cdot 1049.313 \cdot 266.3$$

$$= 48874.0 \text{ Kg}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (94004.8 + 48874.0 + 19537.5) / 17.152$$

$$= 9286.0 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 247.678 / (2 \cdot 9285.985 / 984.723 - 1049.313 / 134.5)$$

$$= 224.0 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / [t / R_{xs}]$$

$$= 165.119 [134.5 / 1049.313]$$

$$= 211.6 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(223.958, 211.636)$$

$$= 211.636 \text{ bars}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (19537.5 + 94004.75 + 48873.996) / 984.723$$

$$= 16.175 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 17.152 \cdot 1049.313 / 134.5$$

$$= 13.4 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 16.175 - 13.382, 13.382)$$

$$= 19.0 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:



Allowed Local Primary Membrane Stress	Sallow	247.68	N./mm ²
Local Primary Membrane Stress	PL	18.97	N./mm ²
Maximum Allowable Working Pressure	Pmax	211.64	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (266.3 + 148.5) / 266.3$$

$$= 1.558 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [L_T]:

$$= \pi/2 \cdot (R_n + t_n)$$

$$= \pi/2 \cdot (266.3 + 148.5)$$

$$= 651.566 \text{ mm.}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$t_c, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 148.500, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$$

$$= \min(48874 \cdot 1.56, 1.5 \cdot 169.6(495.04 + 0.0), \pi/4 \cdot 17.2 \cdot 266.3^2 \cdot 1.56^2)$$

$$= 23635.721 \text{ Kg}$$

Weld Stress Value [τ]:

$$= fwelds / (L_T(0.49 \cdot L41T + 0.6 \cdot t_{w1} + 0.49 \cdot L43T))$$

$$= 23635.721 / (651.566(0.49 \cdot 7.071 + 0.6 \cdot 134.5 + 0.49 \cdot 0.0))$$

$$= 4.227 < \text{or to } 165.119 \text{ Weld Stress Passed}$$

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [R_{xn}]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 148.5 / \ln[1 + 148.5/266.3]$$

$$= 335.084 \text{ mm.}$$

Shell Radius for Force Calculation [R_{xs}]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 134.5 / \ln[1 + 134.5/983.5]$$

$$= 1049.313 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S$$

$$= 1.5 \cdot 165.119$$

$$= 247.7 \text{ N./mm}^2$$

Force acting on the Nozzle [f_N]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 1.034 \cdot 335.084 \cdot 333.361$$

$$= 1178.1 \text{ Kg}$$

Force acting on the Shell [f_S]:

$$= P \cdot R_{xs}(L_R + t_n)$$

$$= 1.034 \cdot 1049.313(363.704 + 148.5)$$

$$= 5668.2 \text{ Kg}$$

Discontinuity Force from External Pressure [f_Y]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 1.034 \cdot 1049.313 \cdot 266.3$$

$$= 2947.0 \text{ Kg}$$

Area Resisting External Pressure [A_p]:



$$= (fS + fY + fN) / P$$

$$= (5668.2 + 2947.0 + 1178.1) / 1.034$$

$$= 9286.0 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap / AT - Rxs / teff)$$

$$= 247.678 / (2 \cdot 9285.984 / 984.723 - 1049.313 / 134.5)$$

$$= 224.0 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t / Rxs]$$

$$= 165.119 [134.5 / 1049.313]$$

$$= 211.6 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(223.958, 211.636)$$

$$= 211.636 \text{ bars}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (fN + fS + fY) / AT$$

$$= (1178.052 + 5668.2 + 2946.953) / 984.723$$

$$= 0.975 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot Rxs / teff$$

$$= 1.034 \cdot 1049.313 / 134.5$$

$$= 0.8 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 0.975 - 0.807, 0.807)$$

$$= 1.1 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	247.68	N./mm ²
Local Primary Membrane Stress	PL	1.14	N./mm ²
Maximum Allowable External Pressure	Pmax	211.64	bars

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: A

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm..
This component must be impact tested.

The Drop for this Nozzle is : 91.9073 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 527.9073 mm.

Input Echo, WRC297 Item 1, Description: B1-24" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	1964.000	mm.
Shell Thickness	136.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-387 11	
Vessel Cold S.I. Allowable	Smc	207.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	165.12 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.



Nozzle Corrosion Allowance	1.5000	mm.
Nozzle Diameter	529.600	mm.
Nozzle Thickness	150.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-336 F21	
Nozzle Cold S.I. Allowable	SNmc 207.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 169.56	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	17.15	bars
Radial Load	P	-3915.97	Kgf
Circumferential Shear	Vc	3915.97	Kgf
Longitudinal Shear	Vl	2936.98	Kgf
Circumferential Moment	Mc	6106.56	Kg-m.
Longitudinal Moment	Ml	6106.56	Kg-m.
Torsional Moment	Mt	7046.03	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, D_v/T_v (16.441) is < 20
check the limitations of WRC bulletin 297.
Warning - The ratio, D_n/T_n (5.531) is < 10
check the limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$$

$$= 829.6 + 2 \cdot 1.65 \cdot \sqrt{(1050.75(136.0 - 1.5))}$$

$$= 2070.180 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2101.500	mm.
Nozzle Outside Diameter	(d) =	829.600	mm.
Vessel Thickness used	(T) =	134.500	mm.
Nozzle Thickness used	(t) =	148.500	mm.
T / t	=	0.906	
d / t	=	10.000	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.560$$

Nr/P	=	0.096
Mr/P	=	0.144
M0/P	=	0.047
N0/P	=	0.109
MrD/Mc	=	0.267
NrDL/Mc	=	0.136
M0d/Mc	=	0.095
N0DL/Mc	=	0.053
MrD/Ml	=	0.142
NrDL/Ml	=	0.085
M0D/Ml	=	0.051
N0DL/Ml	=	0.136

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside

Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	0	0	0	0
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	-1	1	1	-1
Normal Pressure Stress	11	13	11	13

Outplane Stress Summary	10	13	14	12
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	1	-1	1	-1
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-3	3	3	-3
Inplane Pressure Stress	5	5	5	5

Inplane Stress Summary	4	7	11	1
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0

Shear Stress Summary	0	0	0	0
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	10	13	14	12

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	0	0	0	0
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-2	2	2	-2
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	5	5	5	5

Outplane Stress Summary	4	7	9	3
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	1	-1	1	-1
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-6	6	6	-6
Inplane Membrane (ML)	0	0	0	0



Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	11	13	11	13

Inplane Stress Summary	6	17	20	5
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	0	0	0	0

Shear Stress Summary	0	0	0	0
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	6	17	20	6
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	4	2	4

Hoop Stress Summary	2	4	3	5
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-1	-1	1	1
Axial Bending (ML)	-1	1	1	-1
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	0	1	5	0
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	0	0	0	0
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Outside	Inside	Outside	Inside

Two * Max Shear Stress	3	4	5	5
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	4	2	4
Hoop Stress Summary	2	4	3	5

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	-1	-1	1	1
Axial Bending (Mc)	-3	3	3	-3
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1
Axial Stress Summary	-2	3	7	-2

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	5	5	7	7

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	11	13	11	13	11	13	11	13
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	-1	1	1	-1	-5	5	7	-7
Long. Pm (SUS)	5	5	5	5	5	5	5	5
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	-2	2	4	-4	-2	2	2	-2
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	11.0	13.0	11.0	13.0	11.0	13.0	11.0	13.0



Pm+Pl (SUS)	11.0	13.0	11.0	13.0	11.0	13.0	11.0	13.0
Pm+Pl+Q (Total)	10.0	14.0	12.0	12.0	6.0	18.0	18.0	6.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	13.00	165.12	Passed
Pm+Pl (SUS)	13.00	247.68	Passed
Pm+Pl+Q (TOTAL)	18.00	495.36	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	2	4	2	4	2	4	2	4
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	-1	-1	1	1	-1	-1	1	1
Long. Q (SUS)	0	0	2	-2	-2	2	4	-4
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	2.0	4.0	2.0	4.0	2.0	4.0	2.0	4.0
Pm+Pl (SUS)	2.0	4.0	2.0	4.0	2.0	4.0	2.0	4.0
Pm+Pl+Q (Total)	2.0	4.0	4.0	4.0	4.0	4.0	6.0	6.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	4.00	169.56	Passed
Pm+Pl (SUS)	4.00	254.34	Passed
Pm+Pl+Q (TOTAL)	6.00	508.67	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

**Input, Nozzle Desc: M1-24"****From: 70**

Pressure for Reinforcement Calculations	P	17.152	bars
Temperature for Internal Pressure	Temp	286	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	286	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	165.12	N./mm ²
Parent Allowable Stress At Ambient	Sva	207.00	N./mm ²
Inside Diameter of Hemispherical Head	D	2100.00	mm.
Head Finished (Minimum) Thickness	t	68.0000	mm.
Head Internal Corrosion Allowance	cas	1.5000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

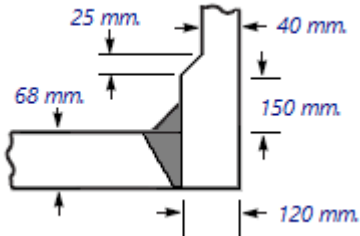
Material		SA-336 F21	
Material UNS Number		K31545	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	169.56	N./mm ²
Allowable Stress At Ambient	Sna	207.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		20.8504	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	40.0000	mm.
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	150.0000	mm.
Height of Beveled Transition	L'	25.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		120.0000	mm.
Corrosion Allowance	can	1.5000	mm.
Outside Projection	ho	350.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		68.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
This is a Manway, Coupling or Access Opening.			
Flange Class		1500	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: M1-24"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	20.850	in.
Actual Thickness Used in Calculation	1.575	in.

**Note:**

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 532.6 \cdot (\exp(17.152/(169.558 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 2.7076 + 1.5 + 0.0 = 4.2076 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 169.558 \cdot 1.0 \cdot \ln((2 \cdot 38.5 + 532.6) / 532.6) - 0.0$$

$$= 228.945 - 0.0 = 228.9446 \text{ bars}$$

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[169.6/165.1, 1]$$

$$= 1.000$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 120.0 - 1.5$$

$$= 118.500 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 40.0 - 1.5$$

$$= 38.500 \text{ mm.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2100.0/2 + 1.5$$

$$= 1051.500 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{(Reff \cdot t)}, 2 \cdot R_n)$$

$$= \min(\sqrt{(1051.5 \cdot 66.5)}, 2 \cdot 266.3)$$

$$= 264.433 \text{ mm.}$$

Intermediate value [Cn]:

$$= \min(((t + t_e)/t_n)^{0.35}, 1)$$

$$= \min(((66.5 + 0.0)/118.5)^{0.35}, 1)$$

$$= \min(0.817, 1)$$

$$= 0.817$$

Intermediate value [Cp]:

$$= 999999.000$$

Effective Nozzle Wall Length Outside the Vessel [LH]:



$$\begin{aligned}
 &= \min(t + t_e + F_p \cdot \sqrt{(R_n \cdot t_n)}, L_{pr1} + t) \\
 &= \min(66.5 + 0.0 + 0.817 \cdot \sqrt{(266.3 \cdot 118.5)}, 350.0 + 66.5) \\
 &= 211.621 \text{ mm.}
 \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned}
 &= t + ((A_5 \cdot frp) / LR) \\
 &= 66.5 + ((0.0 \cdot 1.0) / 264.433) \\
 &= 66.500 \text{ mm.}
 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned}
 &= t \cdot LR \\
 &= 66.5 \cdot 264.433 \\
 &= 175.848 \text{ cm}^2
 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned}
 &= t_n \cdot LH \\
 &= 118.5 \cdot 211.621 \\
 &= 250.771 \text{ cm}^2
 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned}
 &= 0.5 \cdot Leg^2 \\
 &= 0.5 \cdot 10.0^2 \\
 &= 0.500 \text{ cm}^2
 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned}
 &= A1 + frn(A2 + A3) + A41 + A42 + A43 \\
 &= 175.848 + 1.0(250.771 + 0.0) + 0.5 + 0.0 + 0.0 \\
 &= 427.119 \text{ cm}^2
 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned}
 &= t_n / \ln[1 + t_n/R_n] \\
 &= 118.5 / \ln[1 + 118.5/266.3] \\
 &= 321.923 \text{ mm.}
 \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned}
 &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\
 &= 66.5 / \ln[1 + 66.5/1051.5] \\
 &= 1084.409 \text{ mm.}
 \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned}
 &= 1.5 \cdot S \cdot E \\
 &= 1.5 \cdot 165.119 \cdot 1.0 \\
 &= 247.7 \text{ N./mm}^2
 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned}
 &= P \cdot R_{xn} \cdot LH \\
 &= 17.152 \cdot 321.923 \cdot 211.621 \\
 &= 11915.5 \text{ Kg}
 \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned}
 &= P \cdot R_{xs} \cdot (LR + t_n)/2 \\
 &= 17.152 \cdot 1084.409 \cdot (264.433 + 118.5)/2 \\
 &= 36315.1 \text{ Kg}
 \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned}
 &= (P \cdot R_{xs} \cdot R_{nc})/2 \\
 &= (17.152 \cdot 1084.409 \cdot 266.3)/2 \\
 &= 25254.3 \text{ Kg}
 \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned}
 &= (f_S + f_Y + f_N)/P \\
 &= (36315.1 + 25254.3 + 11915.5)/17.152 \\
 &= 4201.4 \text{ cm}^2
 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot A_p/AT - R_{xs}/(2 \cdot t_{eff}))$$



$$= 247.678 / (2 \cdot 4201.429 / 427.119 - 1084.409 / (2 \cdot 66.5))$$

$$= 215.0 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot S[t/Rxs]$$

$$= 2 \cdot 165.119[66.5/1084.409]$$

$$= 202.5 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(214.988, 202.502)$$

$$= 202.502 \text{ bars}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (fN + fS + fY) / AT$$

$$= (11915.521 + 36315.121 + 25254.34) / 427.119$$

$$= 16.872 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot Rxs / (2 \cdot teff)$$

$$= 17.152 \cdot 1084.409 / (2 \cdot 66.5)$$

$$= 14.0 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 16.872 - 13.985, 13.985)$$

$$= 19.8 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	247.68	N./mm ²
Local Primary Membrane Stress	PL	19.76	N./mm ²
Maximum Allowable Working Pressure	Pmax	202.50	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$

$$= (266.3 + 118.5) / 266.3$$

$$= 1.445 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [LT]:

$$= \pi/2 \cdot (Rn + tn)$$

$$= \pi/2 \cdot (266.3 + 118.5)$$

$$= 604.442 \text{ mm.}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 118.500, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2)$$

$$= \min(25254 \cdot 1.44, 1.5 \cdot 169.6(289.308 + 0.0), \pi/4 \cdot 17.2 \cdot 266.3^2 \cdot 1.44^2)$$

$$= 20340.500 \text{ Kg}$$

Weld Stress Value [τ]:

$$= fwelds / (LT(0.49 \cdot L41T + 0.6 \cdot tw1 + 0.49 \cdot L43T))$$

$$= 20340.5 / (604.442(0.49 \cdot 7.071 + 0.6 \cdot 66.5 + 0.49 \cdot 0.0))$$

$$= 7.610 < \text{or} = \text{to } 165.119 \text{ Weld Stress Passed}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

$$= tn / \ln[1 + tn/Rn]$$

$$= 118.5 / \ln[1 + 118.5/266.3]$$

$$= 321.923 \text{ mm.}$$



Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 66.5 / \ln[1 + 66.5/1051.5]$$

$$= 1084.409 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S$$

$$= 1.5 \cdot 165.119$$

$$= 247.7 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 1.034 \cdot 321.923 \cdot 211.621$$

$$= 718.5 \text{ Kg}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} \cdot (L_R + t_n)/2$$

$$= 1.034 \cdot 1084.409 \cdot (264.433 + 118.5)/2$$

$$= 2189.7 \text{ Kg}$$

Discontinuity Force from External Pressure [fY]:

$$= (P \cdot R_{xs} \cdot R_{nc})/2$$

$$= (1.034 \cdot 1084.409 \cdot 266.3)/2$$

$$= 1522.8 \text{ Kg}$$

Area Resisting External Pressure [Ap]:

$$= (f_S + f_Y + f_N)/P$$

$$= (2189.7 + 1522.8 + 718.5)/1.034$$

$$= 4201.4 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / (2 \cdot t_{eff}))$$

$$= 247.678 / (2 \cdot 4201.429 / 427.119 - 1084.409 / (2 \cdot 66.5))$$

$$= 215.0 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot S[t/R_{xs}]$$

$$= 2 \cdot 165.119[66.5/1084.409]$$

$$= 202.5 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(214.988, 202.502)$$

$$= 202.502 \text{ bars}$$

Average Primary Membrane Stress [σAvg]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (718.469 + 2189.691 + 1522.76) / 427.119$$

$$= 1.017 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot R_{xs} / (2 \cdot t_{eff})$$

$$= 1.034 \cdot 1084.409 / (2 \cdot 66.5)$$

$$= 0.8 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 1.017 - 0.843, 0.843)$$

$$= 1.2 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	247.68	N./mm ²
Local Primary Membrane Stress	PL	1.19	N./mm ²
Maximum Allowable External Pressure	Pmax	202.50	bars

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)



This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : -104 °C
User entered Minimum Design Temperature : -10 °C
The Curve for this Material is : A

As the Stress Ratio < 0.240, the MDMT is : -104 °C
The computed stress ratio is : 0.058

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$
$$= (2.216 \cdot 1.0) / (40.0 - 1.5 - 0.0)$$
$$= 0.058$$

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: A

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : -104 °C
User entered Minimum Design Temperature : -10 °C
The Curve for this Material is : A

As the Stress Ratio < 0.240, the MDMT is : -104 °C
The computed stress ratio is : 0.066

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$
$$= (4.366 \cdot 1.0) / (68.0 - 1.5 - 0.0)$$
$$= 0.066$$

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction : -104 °C

The Drop for this Nozzle is : 73.0513 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 491.0513 mm.

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**Input Echo, Tubesheet Number 1, Description: Tube sheet****Shell Data:****Main Shell Description: sheel 4**

Shell Maximum Design Pressure	Psd,max	51.00 bars
Shell Maximum Operating Pressure	Psox,max	32.50 bars
Shell Minimum Operating Pressure	Psox,min	0.00 bars
Shell Thickness	ts	34.0000 mm.
Shell Internal Corrosion Allowance	cas	1.5000 mm.
Shell External Corrosion Allowance	casext	0.0000 mm.
Inside Diameter of Shell	Ds	2032.000 mm.
Shell Temperature for Internal Pressure	Ts	286.00 °C
Shell Material		SA-387 11

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Shell Material UNS Number		K11789
Shell Allowable Stress at Temperature	Ss	165.12 N./mm ²
Shell Allowable Stress at Ambient		207.00 N./mm ²

Channel Data:**Channel Description: Channel-1**

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	191.00 bars
Channel Maximum Operating Pressure	Ptox,max	25.00 bars
Channel Minimum Operating Pressure	Ptox,min	0.00 bars
Channel Thickness	tc	136.0000 mm.
Channel Internal Corrosion Allowance	cac	1.5000 mm.
Channel External Corrosion Allowance	cacext	0.0000 mm.
Inside Diameter of Channel	Dc	1964.000 mm.
Channel Design Temperature	TEMPC	310.00 °C
Channel Flange Matl		SA-387 11

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Channel Material UNS Number		K11789
Channel Allowable Stress at Temperature	Sc	162.80 N./mm ²
Channel Allowable Stress at Ambient		207.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	559
Tube Wall Thickness	et	2.7000 mm.
Tube Outside Diameter	D	38.1000 mm.
Total Straight Tube Length	Lt	4500.00 mm.
Straight Tube Length (bet. inner tubsht faces) L		4400.00 mm.
Design Temperature of the Tubes		465.00 °C
Tube Material		SB-163
Tube Material UNS Number		N08825
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		157.40 N./mm ²
Tube Allowable Stress At Ambient		161.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	175.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	75.0000 mm.
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.1100 mm.
Groove Weld Leg	ag	2.1100 mm.
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		4.21.2
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	921.615 mm.
Largest Center-to-Center Tube Distance	U1	800.0000 mm.
Length of Expanded Portion of Tube	ltx	0.0000 mm.
Tube-side pass partition groove depth	hg	0.0000 mm.
Area of Tube Layout	Ap	25623.5 cm ²



Perimeter of Tube Layout Cp 5952.6 mm.

Tubesheet Data:

Tubesheet TYPE: U-tube, Integral both sides, Conf. a

Tubesheet Design Metal Temperature T 465.00 °C
Tubesheet Material SA-336 F21

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

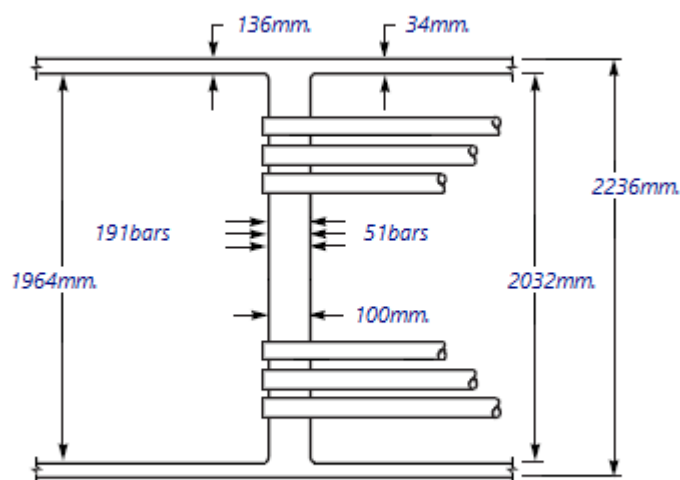
Tubesheet Material UNS Number K31545
Tubesheet Allowable Stress at Temperature S 120.60 N./mm²
Tubesheet Allowable Stress at Ambient Tt 207.00 N./mm²
Thickness of Tubesheet h 100.0000 mm.
Tubesheet Corr. Allowance (Shell side) Cats 1.5000 mm.
Tubesheet Corr. Allowance (Channel side) Catc 1.5000 mm.
Tubesheet Outside Diameter A 2236.000 mm.

Area of the Untubed Lanes AL 0.0 cm²

Junction Stress Reduction option Perform Plastic Calculation
Perform Differential Pressure Design No
Run Multiple Load Cases Yes

Tubesheet Integral with Both
Tubesheet Extended as Flange No

Tubesheet Integral w/Shell and Channel



Configuration a:

ASME Division 2 Tubesheet Results per Part 4.18, 2025**Elasticity/Expansion Material Properties :**

Shell - TM-1 Material Group C

Elastic Mod. at Design Temperature 286.0 °C 0.18730E+09 KPa.
Elastic Mod. at Ambient Temperature 21.1 °C 0.20408E+09 KPa.

Channel - TM-1 Material Group C

Elastic Mod. at Design Temperature 310.0 °C 0.18581E+09 KPa.



Elastic Mod. at Ambient Temperature 21.1 °C 0.20408E+09 KPa.

Tubes - TM-4 N08825

Elastic Mod. at Tubsht. Design Temp. 465.0 °C 0.16930E+09 KPa.

Elastic Mod. at Ambient Temperature 21.1 °C 0.19305E+09 KPa.

TubeSheet - TM-1 Material Group D

Elastic Mod. at Design Temperature 465.0 °C 0.17800E+09 KPa.

Elastic Mod. at Ambient Temperature 21.1 °C 0.21097E+09 KPa.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Computed Minimum Required Thickness [t]:

= $0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i$ Equation 4.3.1

= $0.5 \cdot 32.7 \cdot (\exp(191.0/(157.4 \cdot 1.0)) - 1) + 0$

= 2.1096 + 0

= 2.1096 mm.

Tube Required Thickness under External Pressure (Shellside pressure) :

Determine Parameter [Mx]:

= $L / \sqrt{(R_o \cdot t)}$

= $5321.615 / \sqrt{(19.05 \cdot 2.7)}$

= 742.017

Compute the Factor [Ch]:

Since $M_x > 2 \cdot (D_o/t)^{0.94}$, $Ch = 0.55 / (D_o/t)$

= $0.55 / (38.1/2.7)$

= 0.0390

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

= $1.6 \cdot Ch \cdot E_y(t / D_o)$

= $1.6 \cdot 0.039 \cdot 0.16930E+09(2.7/38.1)$

= 748.239 N./mm²

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

= Fhe / F_y

= $748.2/175.0$

= 4.276

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

= $\sigma_{ys} / \sigma_{uts}$

= $175.0/565.8$

= 0.309

Ratio [AE]:

= Fhe / E

= $748.2/0.16930E+09$

= 0.004419

Factor [m1]:

= $(\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$

= $(\ln(0.309) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$

= 0.255

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

= $\sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$

= $175.0(1 + 0.002) / \ln(1 + 0.002)^{0.255301}$

= 857.151 N./mm²

Curve Fitting Exponent for, Stainless Steel and Nickel Base Alloys [m2]:

= $m2f(m2f2 - R)$

= $0.75(1.0 - 0.309)$

= 0.518



Curve Fitting Constant [A2]:

$$= (\text{outs} \cdot \exp(m2)) / (m2 \cdot m2)$$

$$= (565.8 \cdot \exp(0.518)) / (0.518 \cdot 0.518)$$

$$= 1335.403 \text{ N./mm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.3091.5 - 0.5 \cdot 0.3092.5 - 0.3093.5$$

$$= 0.215$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	282	0.569	0.901E-05	-0.160E-04	0.171E-04	0.525E-04	0.158E+07	0.174E+00
2	141	-2.799	0.119E-05	0.117E-05	0.895E-05	-0.866E-05	0.308E+08	0.175E-03
3	70	-4.482	0.158E-06	0.158E-06	0.470E-05	-0.469E-05	0.110E+09	0.378E-02
4	106	-3.641	0.515E-06	0.514E-06	0.685E-05	-0.682E-05	0.605E+08	0.267E-02
5	123	-3.220	0.808E-06	0.801E-06	0.791E-05	-0.780E-05	0.434E+08	0.157E-02
6	132	-3.009	0.988E-06	0.975E-06	0.843E-05	-0.826E-05	0.366E+08	0.801E-03
7	137	-2.904	0.109E-05	0.107E-05	0.869E-05	-0.846E-05	0.336E+08	0.343E-03
8	139	-2.851	0.114E-05	0.112E-05	0.882E-05	-0.856E-05	0.322E+08	0.920E-04
9	140	-2.825	0.117E-05	0.114E-05	0.889E-05	-0.861E-05	0.315E+08	0.393E-04
10	139	-2.838	0.115E-05	0.113E-05	0.885E-05	-0.859E-05	0.318E+08	0.269E-04
11	140	-2.832	0.116E-05	0.114E-05	0.887E-05	-0.860E-05	0.316E+08	0.605E-05
12	139	-2.835	0.116E-05	0.113E-05	0.886E-05	-0.859E-05	0.317E+08	0.105E-04
13	139	-2.833	0.116E-05	0.114E-05	0.887E-05	-0.859E-05	0.317E+08	0.222E-05
14	140	-2.832	0.116E-05	0.114E-05	0.887E-05	-0.860E-05	0.317E+08	0.191E-05
15	140	-2.833	0.116E-05	0.114E-05	0.887E-05	-0.860E-05	0.317E+08	0.152E-06
16	140	-2.833	0.116E-05	0.114E-05	0.887E-05	-0.860E-05	0.317E+08	0.881E-06
17	140	-2.833	0.116E-05	0.114E-05	0.887E-05	-0.860E-05	0.317E+08	0.364E-06

Factor of Safety [FS]:

$$\text{Since } F_{ic} > 0.55 F_y \text{ and } F_{ic} < F_y$$

$$= 2.407 - 0.741 \cdot F_{ic} / F_y$$

$$= 2.407 - 0.741 \cdot 140.02/175.0$$

$$= 1.814$$

Allowable Circumferential Compressive Stress [Fha]:

$$= F_{ic}/FS$$

$$= 140.02/1.814$$

$$= 77.185 \text{ N./mm}^2$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot F_{ha} (t / D_o)$$

$$= 2 \cdot 77.1802 (2.7/38.1)$$

$$= 109.389 \text{ bars}$$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	2.1096 mm.
Allowable Internal Pressure at Corroded thickness	240.55 bars
Required Internal Design Pressure	191.00 bars
Allowable External Pressure at Corroded thickness	109.39 bars
Required External Design Pressure	51.00 bars
Required Thickness due to Shell Side pressure	1.7507 mm.

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:



Equivalent outer tube limit circle diameter [Do]:

$$\begin{aligned}
 &= 2 \cdot r_o + dt \\
 &= 2 \cdot 921.6149 + 38.1 \\
 &= 1881.330 \text{ mm.}
 \end{aligned}$$

Basic ligament efficiency for shear [μ]:

$$\begin{aligned}
 &= (p - dt) / p \\
 &= (75.0 - 38.1) / 75.0 \\
 &= 0.492
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ρ_s]:

$$\begin{aligned}
 &= D_s / D_o \text{ (Configurations a, b, c)} \\
 &= 2032.0 / 1881.33 \\
 &= 1.0801
 \end{aligned}$$

Ratio [ρ_c]:

$$\begin{aligned}
 &= D_c / D_o \text{ (Configurations a, e, f)} \\
 &= 1964.0 / 1881.33 \\
 &= 1.0439
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [ρ]:

$$\begin{aligned}
 &= l_{tx} / h \\
 &= 0 / 100.0 \\
 &= 0.000 \text{ (must be } 0 \leq \rho \leq 1 \text{)}
 \end{aligned}$$

Effective tube hole Diameter [d^*]:

$$\begin{aligned}
 &= \max(dt - 2t_t(E_t T / E) \cdot (S_t T / S) \cdot (\rho); dt - 2t_t) \\
 &= \max(38.1 - 2 \cdot 2.7 (0.16930E+09 / 1.7800E+09) \cdot \\
 &\quad (157 / 120) \cdot (0); 38.1 - 2 \cdot 2.7) \\
 &= 38.1000 \text{ mm.}
 \end{aligned}$$

Effective tube pitch [p^*]:

$$\begin{aligned}
 &= p / \sqrt{ (1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)) } \\
 &= 75.0 / \sqrt{ (1 - 4 \cdot \min(0; 4 \cdot 1881.33 \cdot 75.0) / (\pi \cdot 1881.33^2)) } \\
 &= 75.0000 \text{ mm.}
 \end{aligned}$$

Effective ligament efficiency for bending [μ^*]:

$$\begin{aligned}
 &= (p^* - d^*) / p^* \\
 &= (75.0 - 38.1) / 75.0 \\
 &= 0.4920
 \end{aligned}$$

$h/p = 1.333333$; $\mu^* = 0.492000$; $E^* = 99944016 \text{ KPa}$.

$E^*/E = 0.561479$; $v^* = 0.292221$

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$\begin{aligned}
 &= ((12(1 - v_s^2))^{1/4}) / \sqrt{ ((D_s + t_s) t_s) } \\
 &= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ ((2032.0 + 34.0) 34.0) } \\
 &= 0.0069 \text{ 1/mm.}
 \end{aligned}$$

Shell coefficient [k_s]:

$$\begin{aligned}
 &= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - v_s^2)) \\
 &= 0.007 \cdot 1873002 \cdot 34.0^3 / (6(1 - 0.3^2)) \\
 &= 92476920.0000 \text{ bars} \cdot \text{mm.}^2
 \end{aligned}$$

Shell coefficient [λ_s]:

$$\begin{aligned}
 &= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2) \\
 &= 6(2032.0 + 34.0) \cdot 92476920 / 100.0^3 \cdot (1 + 100.0 \cdot 0.007 + (100.0 \cdot 0.007)^2 / 2) \\
 &= 2202245.2 \text{ bars}
 \end{aligned}$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot v_s / (2(D_s + t_s)))$$



$$= 2032.0^2 / (4 \cdot 0.18730E+09 \cdot 34.0) \cdot (1 - 2032.0 \cdot 0.3 / (2(2032.0 + 34.0)))$$

$$= 0.13817E+00 \text{ mm./N./mm}^2$$

Parameter [ws]:

$$= \rho_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.0801 \cdot 92476920 \cdot 0.0069 \cdot 0.138173 (1 + 100.0 \cdot 0.0069)$$

$$= 15959.5137 \text{ mm.}^2$$

Channel coefficient [β_c]:

$$= ((12(1 - v_c^2))^{1/4} / \sqrt{(D_c + t_c)t_c})$$

$$= ((12(1 - 0.3^2))^{1/4} / \sqrt{(1964.0 + 136.0)136.0})$$

$$= 0.0034 \text{ 1/mm.}$$

Channel coefficient [kc]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - v_s^2))$$

$$= 0.003 \cdot 1858110 \cdot 136.0^3 / (6(1 - 0.3^2))$$

$$= 2911871744.0000 \text{ bars}\cdot\text{mm.}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot kc / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot Cq\Delta)$$

$$= 6(1964.0 + 136.0 / \cos(0)) \cdot 0.29119E+1 / 100.0^3 \cdot (1.0 + 100.0 \cdot 0.003 \cdot (1.0 + 1.0) / 2 + (100.0 \cdot 0.003)^2 / 2 \cdot 1.0)$$

$$= 51292308.0 \text{ bars}$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c)))$$

$$= 1964.0^2 / (4 \cdot 0.18581E+09 \cdot 136.0) \cdot (1 - 1964.0 \cdot 0.3 / (2(1964.0 + 136.0)))$$

$$= 0.32805E-01 \text{ mm./N./mm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

Parameter [wc]:

$$= \rho_c \cdot kc [\delta_c (CmTheta \cdot \beta_c + Cq\Delta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$

$$= 1.044 \cdot 0.29119E+1 [0.032805 (1.0 + 0.003 \cdot 1.0 \cdot 100.0) - 0 (1.0 / 100.0 + 1.0 \cdot 0.003 / 2)]$$

$$= 45461.8164 \text{ mm.}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 2236.0 / 1881.33$$

$$= 1.1885$$

Coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.29) / 99944016 \cdot (2202245 + 51292308 + 0.17800E+09 \cdot \ln(1.19))$$

$$= 38.1012$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= D_o^2 / 16 \cdot [(\rho_s - 1) \cdot (\rho_s^2 + 1) \cdot P_s - (\rho_c - 1) \cdot (\rho_c^2 + 1) \cdot P_t]$$

$$= 1881.33^2 / 16 \cdot [(1.08 - 1) \cdot (1.08^2 + 1) \cdot 51.0 - (1.044 - 1) \cdot (1.044^2 + 1) \cdot 191.0]$$

$$= -1922447.0000 \text{ bars}\cdot\text{mm.}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= Mts + wc \cdot Pt - ws \cdot Ps$$

$$= -1922447.0 + 45461.816 \cdot 191.0 - 15959.514 \cdot 51.0$$

$$= 5946825.5000 \text{ bars}\cdot\text{mm.}^2$$

**ASME Section VIII-2 Part 4.18.7.4 Step 7:**

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= (M \cdot Do^2/32 \cdot F(P_s - P_t)) / (1 + F)$$

$$= (5946825 - 1881.33^2/32 \cdot 38.101 (51.0 - 191.0)) / (1 + 38.1)$$

$$= 15240953.0000 \text{ bars} \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= M_p + Do^2/64 \cdot (3 + \nu) \cdot (P_s - P_t)$$

$$= 15240953 + 1881.33^2/64 \cdot (3 + 0.292) \cdot (51.0 - 191.0)$$

$$= -10248878.0000 \text{ bars} \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(|M_p|; |M_o|)$$

$$= \max(|15240953|; |-10248878|)$$

$$= 15240953.0000 \text{ bars} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu \cdot (h - h_g')^2)$$

$$= 6 \cdot 15240953 / (0.492 (100.0 - 0)^2)$$

$$= 1858.7606 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot S$$

$$= 2 \cdot 120.6$$

$$= 241.21 \text{ N./mm}^2$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu \cdot (h - h_g')^2)$$

$$= 6 \cdot 14125981 / (0.492 (267.2685 - 0)^2)$$

$$= 241.1762 \text{ N./mm}^2$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + C_{ats} + C_{atc}$$

$$= 267.2685 + 0 + 0$$

$$= 267.2685 \text{ mm.}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(267.2685; 126.961)$$

$$= 267.2685 \text{ mm.}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:Absolute pressure differential [$|P_s - P_t|$]:

$$= |P_s - P_t|$$

$$= |51.0 - 191.0|$$

$$= 140.000 \text{ bars}$$

Shear Stress check [τ_{limit}]:

$$= 4 \cdot \mu \cdot h / Do \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)$$

$$= 4 \cdot 0.492 \cdot 100.0 / 1881.33 \cdot \min(0.8 \cdot 120.604; 0.533 \cdot 227.807)$$

$$= 10.093 \text{ N./mm}^2$$

Note: As $F_s - P_t > \tau_{limit}$ & A_p, C_p are given, the more accurate formula for τ is used.

Average shear stress at the outer edge of perforated region [τ]:

$$= (1 / (4 \cdot \mu)) \cdot (1 / h \{ (4 \cdot A_p) / C_p \}) \cdot |P_s - P_t|$$

$$= (1 / (4 \cdot 0.492)) \cdot (1 / 100.0 \{ (4 \cdot 25623.5) / 5952.6 \}) \cdot |51.0 - 191.0|$$

$$= 122.4960 \text{ N./mm}^2$$

Minimum required thickness for Shear [HreqS]:

$$= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot |P_s - P_t| + C_{ats} + C_{atc}$$

$$= 1 / (4 \cdot 0.492 \cdot 96.483) \cdot (4 \cdot 25623.5 / 5952.57) \cdot |51.0 - 191.0| + 0$$

$$= 126.9610 \text{ mm.}$$

**ASME Section VIII-2 Part 4.18.7.4 Step 10:**

Note: For a given Shell thickness of: 34.000mm.

Min. Shell len. adjacent to the tubesheet is: 473.123 mm.

Axial Shell Membrane Stress [σ_s, m]:

$$= D_s^2 / (4 \cdot t_s (D_s + t_s)) \cdot P_s$$

$$= 2032.0^2 / (4 \cdot 34.0 (2032.0 + 34.0)) \cdot 51.0$$

$$= 74.9503 \text{ N./mm}^2$$

Axial Shell Bending Stress [σ_s, b]:

$$= 6 \cdot k_s / t_s^2 \cdot [\beta S \cdot P_s + 6(1 - \nu^*) / E^* \cdot$$

$$Do / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (M_p + Do^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 2079011 / 34.0^2 \cdot [0.007 \cdot 0.7027 + 6(1 - 0.292) / 99944016 \cdot$$

$$1881.33 / 100.0^3 \cdot (1 + 100.0 \cdot 0.007 / 2) \cdot (15240953 + 1881.33^2 / 32 \cdot$$

$$(51.0 - 191.0))]$$

$$= 106.318 \text{ N./mm}^2$$

Shell Membrane + Bending Stress [σ_s]:

$$= | \sigma_{s,m} | + | \sigma_{s,b} | \text{ (Should be } \leq 1.5 \cdot S_s)$$

$$= 181.268 \text{ N./mm}^2 \text{ (Should be } \leq 247.68)$$

Note: For a given Channel thickness of: 136.000 mm.

Min. Channel len. adjacent to the tubesheet is: 930.278mm.

Axial Channel Membrane Stress [σ_c, m]:

$$= D_c^2 / (4 \cdot t_c (D_c + t_c) \cdot \cos(\theta)) \cdot P_t$$

$$= 1964.0^2 / (4 \cdot 136.0 (1964.0 + 136.0) \cdot \cos(0)) \cdot 191.0$$

$$= 64.495 \text{ N./mm}^2$$

Axial Channel Bending Stress [σ_c, b]:

$$= 6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta \cdot \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t -$$

$$6((1 - \nu^*) / E^*) \cdot Do / h^3 \cdot (C_m \theta +$$

$$h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot (M_p + Do^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 65462980 / 136.0^2 \cdot [(0.032805 \cdot 1.0 \cdot 0.003 - 0 \cdot 1.0) \cdot 191.0 -$$

$$6((1 - 0.292) / 99944016) \cdot 1881.33 / 100.0^3 \cdot (1.0 +$$

$$0.003 \cdot 100.0 / 2 \cdot 1.0) \cdot (15240953 + 1881.33^2 / 32 \cdot (51.0 - 191.0))]$$

$$= 416.880 \text{ N./mm}^2$$

Channel Membrane + Bending Stress [σ_c]:

$$= | \sigma_{c,m} | + | \sigma_{c,b} | \text{ (Should be } \leq 1.5 \cdot S_c)$$

$$= 481.374 \text{ N./mm}^2 \text{ (Should be } \leq 244.20)$$

Step 11, The Cylinder-to-Tubesheet Junction is Overstressed.

Performing a simplified Elastic-Plastic calculation

(option 3) to reduce the overstress condition.

Modify E_s and/or E_c and recompute from Step 4 onwards.

Modified Elasticity [E_c]:

$$= E_c \cdot \sqrt{1.5 \cdot S_c / \sigma_c}$$

$$= 0.18581 \text{E}+09 \cdot \sqrt{1.5 \cdot 162.8 / 481.37}$$

$$= 132343680.0 \text{ KPa.}$$

Tubesheet Bending Stress at Original Thk., after the Elastic-Plastic iteration:

$$= 6 \cdot M / (\mu^* (h - h'g)^2)$$

$$= 6 \cdot 15151956 / (0.492 (100.0 - 0)^2)$$

$$= 1847.9 \text{ N./mm}^2 \text{ (Should be } \leq 241.2)$$

Note: Tubesheet is overstressed after the Elastic-Plastic iteration

Recomputing the tubesheet required thickness.

Required Tubesheet Thk., for Bending Stress after the Elastic-Plastic iteration [H_{reqB}]:

$$= h + C_{ats} + C_{atc} = 267.2685 + 0 + 0 = 267.2685 \text{ mm.}$$

Required Tubesheet Thk. after the Elastic-Plastic iteration (includes CA) [H_{req}]:

$$= \max(H_{reqB}, H_{reqS}) = \max(267.2685; 126.961) = 267.2685 \text{ mm.}$$



Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (38.1 - 2.7) 2.7$$

$$= 3.003 \text{ cm}^2$$

Tube Weld Size Results per VIII-2 4.21.2:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 2.7 (38.1 - 2.7) 157.399$$

$$= 4819.360 \text{ Kg}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 2.11 (38.1 + 0.67 \cdot 2.11) 120.6$$

$$= 2737.9912 \text{ Kg}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 157.399 / 120.604$$

$$= 1.305$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 2.11 (38.1 + 0.67 \cdot 2.11) 120.6$$

$$= 1771.6415 \text{ Kg}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{(0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d} - 0.75 \cdot d_o \right)$$

$$= 2 \left(\sqrt{(0.75 \cdot 38.1)^2 + 1.07 \cdot 2.7 \cdot (38.1 - 2.7) 1.305 \cdot 1.0} - 0.75 \cdot 38.1 \right)$$

$$= 4.494 \text{ mm.}$$

Minimum Required Fillet Weld Leg [afr]: 2.2471 mm.

Minimum Required Groove Weld Leg [agr]: 2.2471 mm.

The Tube-Tubesheet Joint weld is not adequate.

Warning: This tube-tubesheet joint can not be qualified as a Full strength Jt.

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual		Allowable	Pass/Fail
<i>Tubesheet bend. stress</i>	1847.9	<=	241.2 N./mm ²	FAIL
<i>Tubesheet shear stress</i>	122.5	<=	96.5 N./mm ²	FAIL
Stress in Shell at Tubesheet	181.3	<=	247.7 N./mm ²	Ok
Stress in Channel at Tubesheet	481.4	<=	488.8 N./mm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm.)	Required	Actual	P/F
<i>Tubesheet Thickness :</i>	267.268	100.000	Fail
<i>Tube-Tubesheet Fillet Weld Leg :</i>	2.247	2.110	Fail
<i>Tube-Tubesheet Groove Weld Leg :</i>	2.247	2.110	Fail

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 6 Load Cases:

--Reqd. Thk. + CA ---- Tubesheet Stresses Case Pass/



Case#	Tbsht	Extnsn	Bend	Allwd	Shear	Allwd	Type	Fail
D1uc	302.417	...	2526	241	167	96	Ps+Pt D1	Fail
D2uc	161.046	...	667	241	45	96	Ps+Pt D2	Fail
D3uc	267.268	...	1848	241	122	96	Ps+Pt D3	Fail
D1c	304.572	...	2686	241	172	96	Ps+Pt-c D1	Fail
D2c	163.788	...	709	241	46	96	Ps+Pt-c D2	Fail
D3c	269.771	...	1976	241	126	96	Ps+Pt-c D3	Fail
Max:	304.5724	...	mm.	11.135		1.786	(Str. Ratio)	

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

-- Integral Cyl. Reqd. Thk.	Shell	Stress	Channel	Stress	Pass
Case#	Shell	Channel	Stress	Allwd	Stress Allwd Fail
D1uc	...	...	215	248	635 244 Fail
D2uc	...	...	396	495	153 244 Ok
D3uc	...	...	181	248	481 489 Ok
D1c	...	...	222	248	656 244 Fail
D2c	...	...	414	495	160 244 Ok
D3c	...	...	192	248	496 244 Fail
Max	...	...	0.896		2.686

Note:

Because there was no net pressure differential across the tubesheet due to the specified pressures psd,min and ptd,min; load case D4 was not processed as the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 6 Load Cases:

Thickness (mm.)	Required	Actual	P/F
Tubesheet thickness :	304.572	100.000	Fail
Tube thickness :	2.110	2.700	Ok
Tube-tubesheet fillet weld leg :	2.247	2.110	Fail
Tube-tubesheet groove weld leg :	2.247	2.110	Fail

Given Shell Thickness: 34.0000 mm.

Given Channel Thickness: 136.0000 mm.

Min Shell length of thk, (34.000) adj. to tubesheet: 473.123 mm.

Min Channel length of thk, (136.000) adj. to tubesheet: 930.278 mm.

Warning: This tube-tubesheet joint can not be qualified as a Full strength Jt.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force	Tubeside	0 shellside	Shellside	0 tubeside
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Condition	MAWP	Stress Rat.	MAWP	Stress Rat.
Tubesheet Bending Stress	17.152	1.000	17.337	1.000
Tubesheet Shear Stress	106.961	1.000	106.961	1.000
Shell Stress (Axial, Junction)	106.961	6.236	17.337	0.568
Tubesheet-Channel Junction	106.961	6.236	17.337	0.223
Tube Pressure Stress	240.552	1.000	109.388	1.000
Minimum MAWP	17.152		17.337	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	31.357	1.000	31.741	1.000
Tubesheet Shear Stress	188.836	1.000	188.836	1.000
Shell Stress (Axial, Junction)	188.836	0.678	31.741	0.793
Tubesheet-Channel Junction	31.357	0.335	31.741	0.307
Tube Pressure Stress	246.051	1.000	148.677	1.000
Minimum MAPnc	31.357		31.741	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Tubesheet MDMT Calculations:

Shell Side MDMT calculation:

Governing thickness on the shell side per figure 3.10:

$$\begin{aligned}
 &= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{shell thickness})) \\
 &= \max(100.0/4, \min(100.0, 34.0)) \\
 &= 34.000 \text{ mm.}
 \end{aligned}$$

MDMT reduction ratio:

$$\begin{aligned}
 &= \min(\text{ps}/\text{Shellside MAPnc}, 1) \\
 &= \min(51.0/31.74, 1) \\
 &= 1.000
 \end{aligned}$$

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : 23 °C
 User entered Minimum Design Temperature : -10 °C
 The Curve for this Material is : A
 From Figure 3.8M- with PWHT, the MDMT is : 23 °C
 Temperature reduction per Graph 3.13 : 0 °C
 The computed stress ratio is : 1.000

Warning: Impact Testing Is Mandatory!

Channel Side MDMT calculation:

Governing thickness on the shell side per figure 3.10:

$$\begin{aligned}
 &= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{channel thickness})) \\
 &= \max(100.0/4, \min(100.0, 136.0)) \\
 &= 100.000 \text{ mm.}
 \end{aligned}$$

MDMT reduction ratio:

$$\begin{aligned}
 &= \min(\text{pt}/\text{Tubeside MAPnc}, 1) \\
 &= \min(191.0/31.36, 1) \\
 &= 1.000
 \end{aligned}$$

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : 42 °C
 User entered Minimum Design Temperature : -10 °C
 The Curve for this Material is : A
 From Figure 3.8M- with PWHT, the MDMT is : 42 °C
 Temperature reduction per Graph 3.13 : 0 °C



The computed stress ratio is : 1.000

Warning: Impact Testing Is Mandatory!

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